

Impact of Maximal Overexpression of a Non-toxic Protein on Yeast Cell Physiology

Reviewed Preprint

v2 • July 17, 2025

Revised by authors

Reviewed Preprint

v1 • July 26, 2024

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This **convincing** study advances our understanding of the physiological consequences of the strong overexpression of non-toxic proteins in baker's yeast. The findings suggest that a massive protein burden results in nitrogen starvation and a shift in metabolism likely regulated via the TORC1 pathway, as well as defects in ribosome biogenesis in the nucleolus. The study presents findings and tools that are **important** for the cell biology and protein homeostasis fields.

<https://doi.org/10.7554/eLife.99572.2.sa4>

Abstract

Summary

While it is recognized that excess expression of non-essential proteins burdens cell growth, the physiological state of cells under such stress is largely unknown. This is because it is challenging to distinguish between adverse effects arising from the properties of the expressed excess protein (cytotoxicity) and those caused solely by protein overexpression. In this study, we attempted to identify the model protein with the lowest cytotoxicity in yeast cells by introducing a new neutrality index. We found that a non-fluorescent fluorescent protein (mox-YG) and an inactive glycolytic enzyme (Gpm1- CCmut) showed the lowest cytotoxicity. These proteins can be expressed at levels exceeding 40% of total protein while maintaining yeast growth. The transcriptome of cells expressing mox-YG to the limit indicated that the cells were in a nitrogen source requirement state. Proteome analysis revealed increased mitochondrial protein and decreased ribosome abundance, similar to the inactivated state of the TORC1 pathway. The decrease in ribosome abundance was presumably due to defective nucleolus formation, partially rescued by a mutation in the nuclear exosome. These findings suggest that massive overexpression of excess protein, termed protein burden, causes nitrogen source starvation, a metabolic shift toward more

energy-efficient respiration, and a ribosomal biosynthesis defect due to an imbalance between ribosomal protein and rRNA synthesis in the nucleolus.

Introduction

Expression levels of proteins are optimized to maximize the fitness of organisms, and the evolutionary principle that determines those levels is considered to be demand and constraint (Bruggeman et al., 2020) (Figure 1A). Demand is the evolutionary pressure that works to increase the expression of a protein to meet the requirements of its function, and constraint is the evolutionary pressure that works to decrease the expression of a protein to avoid the negative effects produced by excess protein. Constraints appear as disturbances in cellular function when the protein is overexpressed (Figure 1A, the red arrow). The mechanisms of constraining expression levels, or defects due to overexpression, can be roughly classified into four major categories (Moriya, 2015) (Figure 1B). “Resource overload” (also called “burden”) (Kastberg et al., 2022) is a disorder in which intracellular resources used for protein processing, such as synthesis, localization, degradation, and modification of proteins, are depleted or monopolized by the processing of that protein, resulting in a disruption of processing of other proteins (Dong et al., 1995; Eguchi et al., 2018; Kintaka et al., 2020, 2016; Stoebel et al., 2008). This phenomenon occurs in proteins with high expression levels. “Stoichiometry imbalance” is a disorder triggered by an imbalance of amounts between constituents within complexes, leading to untimely activation or inactivation (pathway modulation), degradation (resource overload), aggregation (promiscuous interaction), etc. (Kaizu et al., 2010; Makanae et al., 2013; Papp et al., 2003). This occurs due to the overexpression of proteins that form complexes. “Pathway modulation” is a disorder in which pathways are untimely activated or inactivated. This occurs especially with overexpression of proteins involved in pathway regulation, such as metabolic enzymes, kinases and phosphatases, and transcription factors (Prelich, 2012; Youn et al., 2017). “Promiscuous interaction” is a disorder in which weak interactions are enhanced by mass action to form complexes or aggregates that are not normally formed. This damages cellular function by inactivating essential proteins or overloading degradation resources, and is caused by overexpression of intrinsically disordered proteins or liquid-liquid phase-separating proteins (Bolognesi et al., 2016; Ma et al., 2010; Vavouri et al., 2009).

As the expression level of a protein increases, it eventually encounters one of the barriers created by either these specified constraints or others that are unknown, all of which are related to the function and physicochemical properties of the protein, and damage cellular function. The expression level at which this occurs is referred to as the “expression limit” in this paper (see Figure 1A). If the constraints could be removed from the protein, the expression limit would increase. For example, if components of a complex are simultaneously expressed to resolve a stoichiometry imbalance, the expression limits of those proteins will be increased (Kaizu et al., 2010; Makanae et al., 2013). In this manner, when (one) constraint is removed from a protein, the expression limit would increase until the next constraint barrier is hit (Figure 1A, dotted lines). The amount of expression limit created by a constraint barrier depends on the type of constraint. Stoichiometry imbalances, pathway modulations, and promiscuous interactions would create different expression limits for each protein. This is because they are determined by the activity, regulatory mechanism, and physicochemical properties of each protein. On the other hand, resource overload is presumed to be determined to some extent by the capacities of the processes in the cell (Kintaka et al., 2020) (Figure 1C). Essentially, the synthesis, in which all proteins are processed, should have the largest capacity, while processes such as transport, folding, and degradation, in which only a fraction of proteins are processed, should have smaller capacities. In fact, adding transport or degradation signals to a cytosolic protein lowers the expression limit (Kintaka et al., 2016). In that light, the final constraint barrier that emerges when all possible constraints are removed is the overloading of the synthesis process.

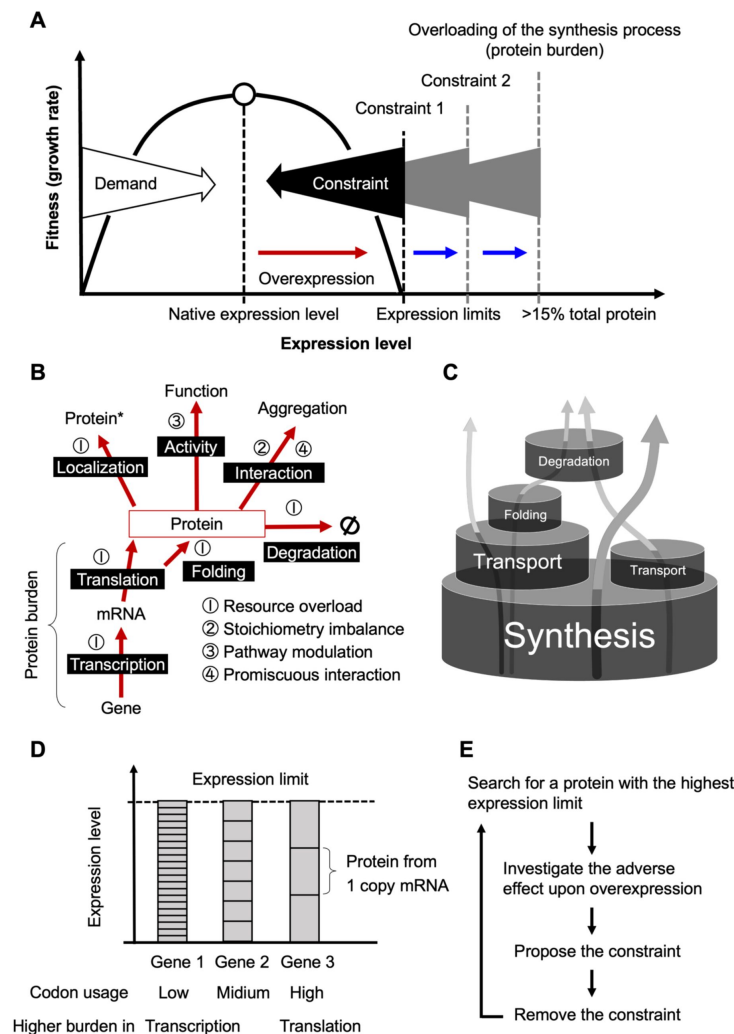


Figure 1.

Schematic diagrams illustrating the constraints on protein expression levels.

A) Protein expression levels are determined by demand and constraints (created by the authors, inspired by the work by Keren et al. (Keren et al., 2016 [\[1\]](#)). The existence of this evolutionary principle is revealed by the relationship between fitness (growth rate) and expression level when the expression level is altered, and in general, the native expression level provides the highest level of fitness for the organism (Bruggeman et al., 2020 [\[2\]](#)). The existence of constraints is revealed by a decrease in fitness upon overexpression (red arrow). Constraints determine the expression limit of the protein, i.e., the expression level at which growth inhibition occurs (dotted lines), but there can be multiple constraints for a single protein. The constraint that exists at the highest expression limit is protein burden. B) Four major mechanisms that constrain protein expression levels. The figure outlines the fate of proteins and shows what adverse effects occur upon overexpression (red arrows). The resource overload that occurs within the synthesis process (transcription and translation) is specifically referred to as protein burden. This diagram is a more concise redrawing of the author's earlier one (Moriya, 2015 [\[3\]](#)). That asterisk means differentially localized proteins. C) A barrel model for resource overload (modified from the author's earlier one) (Kintaka et al., 2020 [\[4\]](#)). The size of the barrels represents the capacity of each process, and the arrows represent the fate of the proteins processed there. The expression limit of a protein is defined by the lowest-capacity process that processes the protein. Among these processes, synthesis, which processes all proteins, is considered to have the largest capacity. Thus, proteins processed only by synthesis would have the highest expression limits, and their overexpression would overload synthesis (i.e., cause protein burden). Previous studies by the authors, conducted in this context, estimated that protein burden occurs at more than 15% of total protein (Eguchi et al., 2018 [\[5\]](#)). D) Difference in burden due to gene structure (codon optimization). The amount of transcription required to achieve the same protein expression limit depends on the degree of codon optimization. This may change whether transcription or translation results in a higher burden. An ideal framework for protein burden studies. See text for details.

Overloading of the synthesis process is specifically referred to as “protein burden” or “protein cost” (Eguchi et al., 2018 [↗](#); Kafri et al., 2016 [↗](#)). It has long been the subject of study, mainly as an effect on growth produced by excess (gratuitous) proteins that are not necessary for cellular function (Bruggeman et al., 2020 [↗](#); Scott et al., 2010 [↗](#)). This idea was first proposed in studies of the cost of unnecessary induction of the *lac* operon in *Escherichia coli* (Koch, 1983 [↗](#)). In recent years, much attention has been paid to the actual mechanisms by which massive expression of excess protein inhibits growth and the response of cells (Farkas et al., 2018 [↗](#); Kafri et al., 2016 [↗](#); Metzl-Raz et al., 2020 [↗](#)). During protein burden, it is presumed that depletion of the materials, machinery, and factors necessary for transcription and translation occurs, along with competition from these monopolies. In particular, amino acids and aminoacyl-tRNAs are major candidates for depletion (Kastberg et al., 2022 [↗](#)). In addition, ribosomes, which are crucial for protein synthesis, themselves require significant resources for their synthesis and have therefore been considered major candidates for depletion during protein burden, as they compete for these limited resources (Metzl-Raz et al., 2017 [↗](#)). Mechanistic analysis of protein burden in budding yeast, a model eukaryotic cell, has been intensively conducted by Barkai and colleagues, who reported that transcription is rate-limiting in phosphate starvation and translation in nitrogen source starvation (Kafri et al., 2016 [↗](#)), and that basic transcription factors are limiting resources (Metzl-Raz et al., 2020 [↗](#)). In addition, a cellular response in the form of increased cell size and enhanced protein synthesis capacity has been observed (Kafri et al., 2016 [↗](#)). Thus, the previously expected decrease in ribosomes has not been detected.

As mentioned above, to investigate protein burden, we need to overexpress proteins with minimal constraints. This is because overexpressing proteins with additional constraints cannot overload the synthetic process, as the additional constraints reduce the expression limits (**Figure 1A** [↗](#)). Also, the cellular damage and cellular response generated by the constraint mask the protein burden response. However, it remains unclear which proteins are not subject to constraints other than protein burden. Proteins that would induce protein burden have been determined by assuming that they would be minimally constrained based on their properties, and this assumption has been confirmed experimentally. Model proteins selected in this manner in yeast include fluorescent proteins (FPs) and certain glycolytic enzymes (Eguchi et al., 2018 [↗](#); Farkas et al., 2018 [↗](#); Kafri et al., 2016 [↗](#); Kintaka et al., 2020 [↗](#), 2016 [↗](#)). FPs do not have a specific function or interaction partner in yeast cells and are not actively sorted into cellular compartments other than the cytosol. On the other hand, glycolytic enzymes are naturally expressed at very high levels and are thus considered to be less constrained. Notably, these proteins exhibit an expression limit of approximately 15% of total protein in yeast cells (Eguchi et al., 2018 [↗](#)). However, it is not clear whether these proteins are free from constraints other than protein burden. This is because there may be unknown constraints, and there is no evidence that the measured expression limit is the “true limit” at which protein burden occurs.

In addition to the protein (amino acid sequence) itself, there is another aspect to be considered when selecting a model for protein burden. That is the nucleotide sequence of the gene used for expression, i.e., transcriptional, and translational regulatory sequences, codon usage, and so on (**Figure 1D** [↗](#)). For example, if a protein is to be expressed to the limit by a less codon-optimized and thus less translationally efficient mRNA, a large amount of mRNA would be required, which would induce a greater burden on transcription. Conversely, a similar attempt by an mRNA with high translation efficiency would induce a greater burden on translation. These are the constraints inherent in protein synthesis itself. Although there is some evidence that protein synthesis in nature is economically designed to optimally load transcription and translation (Frumkin et al., 2017 [↗](#); Hausser et al., 2019 [↗](#); Mahima and Sharma, 2023 [↗](#)), it still needs to be determined what the genetic design should be to minimize the burden.

Considering the above circumstance, a realistic approach to protein burden requires more than just drawing straightforward conclusions from the overexpression of a single model protein (or gene). Rather, it requires experimentally identifying proteins with the highest possible expression

limits, investigating the negative impacts of their overexpression, and, if any constraints are revealed, examining whether their removal can increase expression limits. Through successive iterations, this process gradually deepens our understanding of protein burden (**Figure 1E**). In each iteration, the least constrained protein (and its gene) will be generated and utilized as a benchmark for further investigation into proteins with high (or low) expression limits and the adverse effects of their overexpression. This will (unintentionally) elucidate a new mechanism of constraint. Indeed, the enhanced green fluorescent protein (EGFP), which we have used as a model for protein burden (Kintaka et al., 2020; 2016; Makanae et al., 2013), was found to form a chaperone-entrapped aggregate upon overexpression and trigger the heat shock response due to its low folding ability (Namba et al., 2022). Its overexpression also causes abnormal yeast cell morphology through proteasomal stress as it contains cysteines. The moxGFP (mox), which improved folding properties and lacks cysteine residues (Costantini et al., 2015), showed a higher expression limit and no longer induced abnormal morphologies. However, even when overexpressing mox, the anticipated depletion and competition for factors, or cellular responses to them have not been observed (Namba et al., 2022).

In this study, we found that a mutation causing loss of fluorescence in mox, and other fluorescent proteins increased the expression limit up to 3-fold, to more than 40% of total protein. Introducing a neutrality index (NI) to assess protein constraint, we found that non-fluorescent mox (mox-YG) and enzymatically inactive Gpm1 (Gpm1-CCmut) are currently the least constrained. Upon overexpression of mox-YG, depletion of amino acid or nitrogen sources, decreased ribosomal expression, and a metabolic shift from glycolysis to oxygen respiration were observed. Thus, physiological states, as anticipated with such protein burdens, are only generated by massive overexpression of proteins with significantly lower constraints. Some of the responses found appear to be due to inactivation of the TORC1 pathway, suggesting that eukaryotic cells use this regulatory pathway to buffer protein burden. In the burdened cells, the nucleolus was dysplastic, and mutations in the nuclear exosome partially restored this. Thus, the abnormal formation of liquid-liquid phase-separated organelles due to dilution may be one of the defects caused by the strong protein burden.

Results

Fluorescent property is a constraint on the expression of fluorescent proteins

To evaluate the expression limits of FPs, we overexpressed FPs to levels that induced growth inhibition. Gene constructs, in which the *TDH3* promoter controls the target protein genes, were incorporated into plasmids designed for the genetic-tug-of-war (gTOW) method (**Figures 2A** and **Figure 2—figure supplement 1A**) (Moriya et al., 2006). Subsequently, yeast cells were transformed and cultured in a medium depleted of leucine. The maximum growth rate (MGR) of cells was calculated from serial measurements of optical density at 660 nm (OD660). Protein levels were determined through the analysis of proteins from cells in the logarithmic growth phase (**Figure 2—figure supplement 1B**). For each protein, we also analyzed a mutant in which the Tyr residue forming the fluorophore was replaced with Gly (YG mutation), resulting in a loss of fluorescence. This is because we unexpectedly found that this mutation substantially increased the expression limits of FPs (see below).

Overexpression of the FPs and mutants all caused growth inhibition (**Figure 2B**, C). EGFP and mox were expressed at about 15% of the total protein as previously reported (Kintaka et al., 2016), but mCherry (mChe) and YG mutants were expressed at higher levels. Mox-YG expressed 44% of the total protein (**Figures 2D**, E). Our previous work had evaluated only expression limits (Eguchi et al., 2018), but we realized that we could not properly assess the constraints on each protein if the growth rates at the expression limit were different. For example, the mChe's

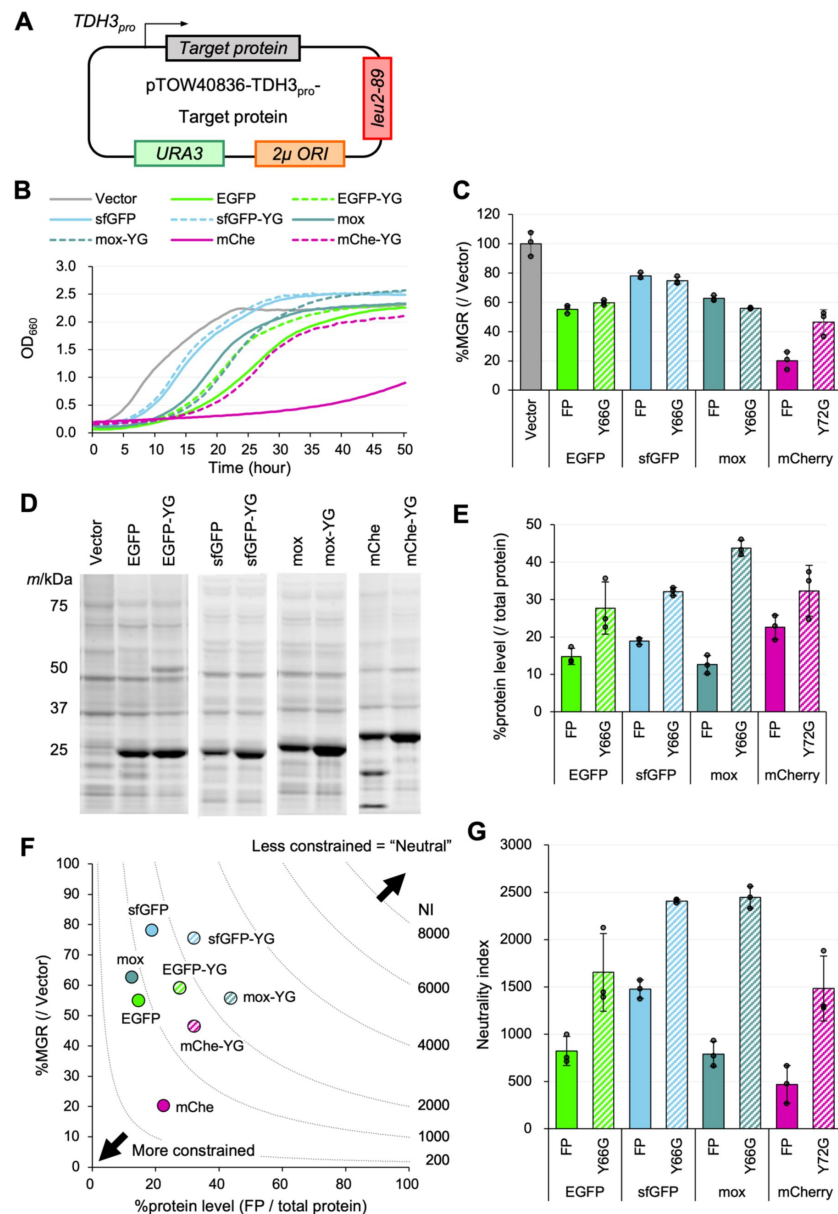


Figure 2.

Evaluation of expression constraint (or neutrality) of fluorescent proteins.

A) Experimental setup of the analysis. Target proteins were expressed under the control of the $TDH3_{pro}$ promoter ($TDH3_{pro}$) using the multicopy plasmid $pTOW40836$. The cells were pre-cultured under $-Ura$ conditions, and then cultured in $-LeuUra$ conditions. B) Growth curves of the vector control and cells overexpressing FPs and their mutants in synthetic medium ($-LeuUra$). The solid or dotted lines show the average calculated from three biological replicates. Growth curves with the standard deviations (SD) of replicates are shown in Figure 2—figure supplement 2. C) MGR of cells overexpressing FPs and their mutants (percent over the vector control). D) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing FPs. All proteins were separated after staining with a fluorescent dye. E) Protein level of the target protein (percent over the total protein). The amount was calculated from the intensity of the target bands separated by SDS-PAGE of D. See Figure 2—figure supplement 1B for the method of FP and total protein quantification. Note that EGFP-YG and mCherry may not be accurately measured due to dimerization and cleavage of the protein, respectively. F) Relationship between MGR and the protein level. MGR data are the same as in C, and the protein level data are the same as in E. The neutrality indexes (NIs), calculated from the products of MGR and protein levels, are indicated by the lines. If the investigated proteins have the same NI, they are located on the same dotted line segment. The neutrality index for the FPs and mutants. The neutrality index is the product of $\%MGR / (Vector)$ and $\%protein\ level / (total\ protein)$.

expression limit (20%) is higher than that of sfGFP (17%), but the relative MGRs are 20% and 80%, respectively, which is much lower for mChe. The relationship between %MGR and %protein level when the control is set at 100% is shown in the scatter plot in **Figure 2F**. The lower left indicates more constraint as lower expression levels lead to growth inhibition, and the upper right indicates less constraint as higher expression levels do not cause growth inhibition. Here, we describe “unconstrained” as “neutral” and calculate the neutrality index (NI) as the product of %MGR and %protein level.

The bars and error bars in **C**, **E**, and **G** show the means and SDs calculated from three biological replicates. The raw data is shown with dot plots. Note that the regime of NI in **Figure 2F** shows a theoretical neutrality distribution but does not show the relationship between growth rate and the expression level when the expression level of a protein with a particular neutrality is increased or decreased. For example, 20% of the growth rate would be maintained when the expression of mox-YG (NI = 2,400) is increased to 100%, but this is not realistic. We also note that in this study, we calculated NI using the percentage of the target protein among total cellular proteins separated by SDS-PAGE. Therefore, to calculate NI, the expression limit of the target protein must be sufficiently high to be detectable by this method. On the other hand, if detection sensitivity is improved, for example by using Western blotting, it becomes possible to estimate the NI even for proteins under strong constraints—i.e., those that exhibit cytotoxicity. In fact, when we recalculated the NI for EGFPs fused with various localization signals from our previous study (Kintaka et al. 2016), all values were found to be below 100 (**Figure 2—figure supplement 3**).

The NI of each protein indicates that introducing a non-fluorescent mutation in FPs uniformly increases neutrality (**Figure 2G**), suggesting that “having fluorescence” is a constraint on the expression of FPs. The most neutral proteins were sfGFP-YG and mox-YG, with NIs of about 2,400. Notably, despite significant differences in the degree of growth inhibition and expression limits between the two proteins, their NIs are nearly identical (**Figures 2F**, **G**). They differ only in two amino acid residues (Cys - Ser) (**Figure 2—figure supplement 4**), which seem to generate differences in expression efficiency. However, this does not appear to affect their NIs.

We also investigated the impact of expression efficiency on NI with another FP, mChe. The mChe used in this study is codon optimized (the codon adaptation index (CAI) = 0.584) (Keppler-Ross et al., 2008), while the mChe used by Kafri et al. for protein cost analysis (here referred to as mChe-Kafri) is not codon optimized (CAI = 0.236) (Kafri et al., 2016). When overexpressed, mChe-Kafri (and mChe-Kafri-YG) had lower expression and growth inhibition than mChe (and mChe-YG) (**Figures 2—figure supplement 5A-D**). This may be due to the lower translation rate of mChe-Kafri mRNA. However, their NIs were not significantly different, especially for the YG mutation (**Figures 2—figure supplement 5E, F**). Thus, the constraints indicated by NI are apparently not affected by translation efficiency. In the study using this mCherry-Kafri, it was also reported that the more mChe-Kafri was expressed, the more the cells enlarged (Kafri et al., 2016). However, no cell enlargement was observed with mox-YG, which can be expressed more than mCherry. Interestingly, the mChe-YG cells did not show enlargement (**Figures 2—figure supplement 6A, B**), indicating that cell enlargement is a phenomenon that occurs specifically when fluorescent mChe is overexpressed.

It should be noted that the growth inhibition caused by the overexpression of mox-YG in this study is a burden on translation rather than on transcription. This is because overexpression of a frameshift mutant (moxFS), in which four nucleotides were inserted directly after the ATG of the mox-YG gene, abolished the notable growth delay (**Figures 2—figure supplement 6C, D**).

Constraint on fluorescent protein expression arises from fluorophore properties

Next, we further analyzed the possibility that fluorescence constrains the expression of FPs. When yeast was grown in the dark, the degree of growth inhibition of *mox* was the same (**Figure 2—figure supplement 7A**) and the expression limit of *mox* was rather increased (**Figure 2—figure supplement 7B**). Phototoxicity does indeed exist when expressing *mox* with fluorescence. This is because, when exposed to strong light, cells overexpressing *mox* show a reduced growth rate (**Figure 2—figure supplement 7C-E**) and abnormalities in their internal structures (**Figure 2—figure supplement 7F**). However, the expression limits of *mox* and *mox*-YG were almost unchanged by the light (**Figure 2—figure supplement 7G, H**). Thus, the constraint was not due to phototoxicity itself.

The number of Tyr residues was also irrelevant because NI was like that of *mox*-YG when a single Tyr was added to the C-terminus of *mox*-YG (**Figure 2—figure supplement 8A-E**). Reactive oxygen species (H_2O_2) are generated during fluorophore formation. When H_2O_2 was added to the medium, stronger growth inhibition occurred only in *mox* (**Figure 2—figure supplement 8F-H**). This suggests that the reactive oxygen species generated during fluorophore formation are responsible for the constraint. However, this possibility is not supported by the omics analysis described later (see Discussion). *Mox*-YG, like *mox*, was present throughout the cytoplasm and did not cause any characteristic localization or aggregation (**Figure 2—figure supplement 9**).

During this study, we examined a *mox*GFP (*mox*-T203I) in which Thr203 was accidentally replaced by Ile (**Figure 2—figure supplement 10**). The fluorescence of this mutant is reduced to 78% of *mox* (**Figures 2—figure supplement 10A-C**). The growth rate of the cells expressing this *mox*-T203I was like that of *mox* and *mox*-YG (**Figure 2—figure supplement 10E**). Still, the expression limit was intermediate between *mox* and *mox*-YG (**Figure 2—figure supplement 10F**), resulting in the NI intermediate between *mox* and *mox*-YG (**Figure 2—figure supplement 10G**). This result further supports that fluorescence is related to constraints. However, as mentioned above, fluorescence emission itself should not be toxic (**Figure 2—figure supplement 7**). We next focused on the formation process of the fluorophore: the tyrosine of the GFP fluorophore undergoes phenol and phenolate state changes, with Thr203 stabilizing the phenol form and substitution to Ile reducing the phenolate form (Kummer et al., 2000). Furthermore, in enhanced GFPs such as *mox*, substitution of Ser65 before Tyr66 for Thr anchors the phenolate form. To investigate the possibility that this is a factor determining the limit of *mox*, we produced *mox* with Thr65 replaced with Ser and examined its expression limit. The *mox*-T65S mutant was expressed 12% more than *mox* (**Figures 2—figure supplement 10H, I**), suggesting that the phenol-type fluorophore can be a limiting factor for the expression level of the fluorescent protein, while the concrete mechanism is still unclear.

Overexpression of *mox*-YG causes an amino acid starvation response

To understand the cell physiology under protein burden, we next performed transcriptome (RNA-seq) analysis of the cells overexpressing *mox*-YG, the least constrained protein. Although the expression limit of *mox*-YG is about 3-fold higher than that of *mox* (**Figure 2E**), the growth rates at the expression limits of both are not largely different (**Figure 2C**). Thus, by comparing transcriptomes upon their overexpression, we may be able to distinguish and extract the general transcriptional response due to reduced growth rates from the (protein burden) response associated with the massive expression of a less constrained protein.

The expression changes under *mox* and *mox-YG* overexpression were calculated as fold change (FC) over the vector control (**Figures 3—figure supplement 1A, B**). The overall trend of expression changes was similar ($r = 0.64$, **Figure 3A**). This similar variation may include responses associated with the expression system and reduced growth rate. On the other hand, there were also different transcriptional responses between the two. Gene categories that showed significant expression changes in either *mox* or *mox-YG* overexpression are shown in **Figure 3B**. In addition to the enhanced ribosome synthesis and decreased glycolysis observed in previous *mox* overexpression (Namba et al., 2022), enhanced oxidative phosphorylation was detected. Interestingly, “Ribosome biogenesis”, which was elevated in *mox* overexpression, showed a rather decreasing trend in *mox-YG* overexpression (**Figure 3B**). Indeed, *IFH1*, which is involved in ribosomal gene transcription (Martin et al., 2004), was upregulated in *mox*, but downregulated in *mox-YG* (**Figure 3A**). Different trends were also observed for “amino acid metabolism” (**Figure 3B**). The amino acid transporter genes *AGP1* and *GAP1* were upregulated only in *mox-YG* overexpression (**Figure 3A**, C). Expressions of specific amino acid and nitrogen source transporter genes were strongly induced in *mox-YG* (**Figure 3C**). The expression of glucose transporter genes was also very different (**Figure 3C**). These results suggest that *mox* and *mox-YG* induce similar but distinct transcriptional responses, particularly that *mox-YG* overexpression may result in depletion of amino acid and nitrogen sources and decreased ribosome synthesis.

The transcriptional responses in *mox-YG* overexpression obtained by RNA-seq were confirmed by transcriptional reporter analysis (**Figure 3D** and **Figure 3—figure supplement 2**). In this reporter analysis, we also tested overexpression of a frameshift mutant (*moxFS*) that imposes a burden on transcription but not on translation (**Figure 2—figure supplement 6C, D**). As a result, most of the responses observed under *mox-YG* overexpression were not observed under *moxFS* overexpression, suggesting that these transcriptional responses are likely caused by a burden on translation. The only exception was *TDH3pro*, whose activity decreased even under *moxFS* overexpression. Since *TDH3pro* is used for driving overexpression in this system, this result suggests that transcriptional competition may occur as a result of overexpression. Notably, in our gTOW experiments, leucine depletion is employed to induce overexpression; however, we consider it unlikely that leucine depletion *per se* is responsible for the induction of *GAP1*, because *GAP1* expression remained unchanged in the vector control even under gradual leucine depletion (**Figure 3—figure supplement 3**).

Overexpression of *mox-YG* causes a proteomic response that partially overlaps with TORC1 inactivation

Protein burden is an overload on synthetic resources, and thus, when this overload occurs strongly, it is expected that there would be a reduction in the amount of proteins other than those being overexpressed due to competition for synthetic resources. Indeed, in cells overexpressing *mox-YG*, the amounts of proteins other than *mox-YG* were reduced (**Figure 4A**). Interestingly, the total amount of protein per cell remained almost constant, with the amount of other proteins decreasing as *mox-YG* was overexpressed. This trend was generally true for the overexpression of other FP (**Figure 4—figure supplement 1**). This suggests that there is a mechanism (or constraint) that maintains the overall cellular protein levels.

The maintenance of total cellular protein levels under overexpression conditions may be due to constraints such as intracellular space limitations or limitations in the availability of nutrients in the medium. To investigate whether nutrient limitation contributes to this phenomenon, we attempted overexpression of *mox* and *mox-YG* in YPD medium, which is considered to be more nutrient-rich than the SC medium. Since leucine depletion could not be used to drive overexpression in this context, we employed an alternative strategy using the aureobasidin A resistance marker (*AUR1d*) to increase plasmid copy number (**Figure 4—figure supplement 2A**). In this experiment, *mox-YG* was expressed at approximately 20% of the total cellular

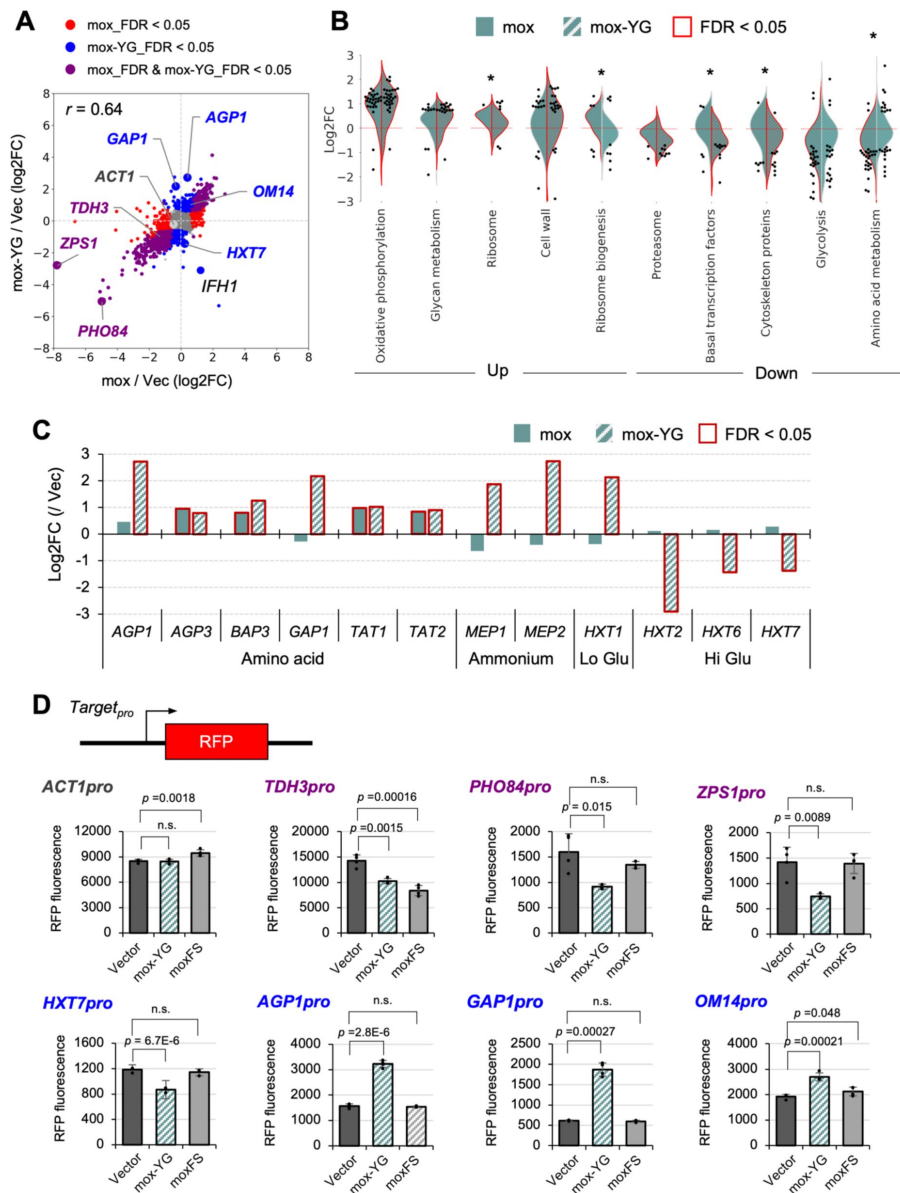


Figure 3.

Transcriptional response of mox and mox-YG overexpression.

A) Comparison of transcriptional responses of mox and mox-YG overexpression. Genes that showed common or different transcriptional responses in mox and mox-YG overexpression are shown in indicated colors. Genes that showed characteristic responses are also shown. FDR: false discovery rate. r : Pearson correlation coefficient. B) Gene groups that showed significant transcriptional changes. Among the KEGG orthology level 3 categories, gene groups in the category that were significantly up-regulated or down-regulated (FDR < 0.05) in mox or mox-YG overexpression are shown in the violin plots. Genes with significant expression changes (FDR < 0.05) within the same category are shown by swarm plots. Asterisks indicate significant differences (FDR < 0.05) between mox and mox-YG comparisons in gene groups belonging to the same category. Comparisons for all categories are shown in Figure 3—figure supplement 1C. C) Expression changes in representative amino acid, ammonium, and glucose transporters. Verification of promoter activity by the reporter assay. Constructs used for promoter analysis with transcription reporters and quantitative results of RFP fluorescence values (arbitrary unit) for each promoter. “moxF5” means the frameshift mutant of mox-YG is overexpressed. Bars and error bars indicate the mean and SD of maximum fluorescence values for RFP calculated from four biological replicates. Raw data are shown as dot plots. The p-values were calculated by performing Welch’s t-test and applying Bonferroni correction. “n.s.” means $p > 0.05$. Detailed constructs and time series data for promoters other than those shown here are in Figure 3—figure supplement 2.

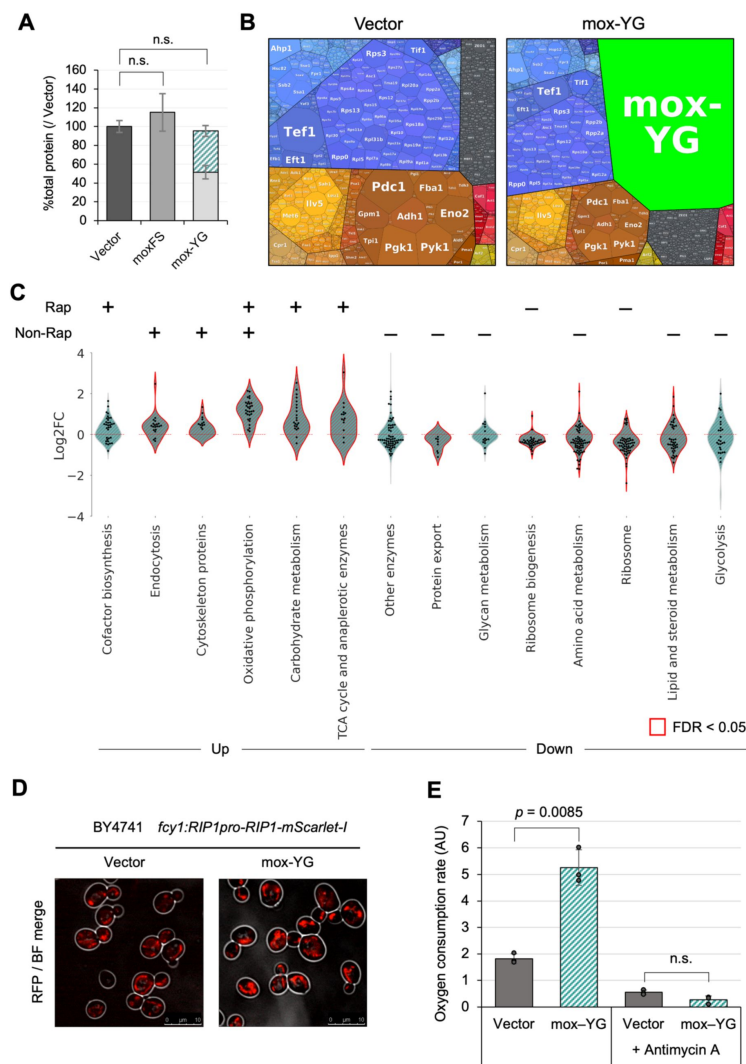


Figure 4.

Overexpression of mox-YG causes a proteome response like that of TORC1 inactivation.

A) Ratio of the total protein level over the vector control. The colored area indicates the amount of overexpressed protein. The amount was calculated from the intensity of the target bands separated by SDS-PAGE. Bars and error bars indicate the mean and SD of protein levels calculated from three biological replicates. The p-values were calculated by performing Welch's t-test and applying Bonferroni correction. "n.s." means $p > 0.05$. B) Visualization of the proteome of the vector control and mox-YG overexpressing cells using Proteomaps (Liebermeister et al., 2014). Each polygon represents the mass fraction of the protein within the proteome. Similar colors indicate similar or closely related pathways and proteins. C) Violin plots showing groups of genes in the KEGG orthology level 3 category whose expression was increased or decreased by mox-YG overexpression. Gene groups with significantly altered expression ($FDR < 0.05$) in mox-YG overexpression are shown in red. Genes with significant expression changes ($FDR < 0.05$) within the same category are shown by swarm plots. "+" and "-" indicate the categories with significantly increased or decreased expression, respectively, when separated into Rapamycin-responsive (Rap) and Rapamycin-non-responsive (Non-Rap) gene populations. Comparisons for all categories are shown in Figures 4—figure supplement 6A, B. Published data (Gowans et al., 2018) was used for comparison with the rapamycin response genes. D) Fluorescence microscopy image showing the localization of Rip1-mScarlet-I. The vector control and mox-YG overexpressing cells were observed in the log phase. Stationary phase cells, bright-field images, quantitative results, and statistical analyses are shown in Figure 4—figure supplement 8. E) Oxygen consumption rate in vector control and mox-YG overexpressing cells (arbitrary units, AU). Antimycin A was added as a control for respiratory inhibition. Measurements were conducted with three biological replicates. Bars and error bars indicate the mean and SD from three biological replicates. Raw data are shown as dot plots. The p-values were calculated by performing Welch's t-test and applying Bonferroni correction. "n.s." indicates $p > 0.05$. The measurement data are shown in Figure 4—figure supplement 9.

protein, with a corresponding increase in total protein content per cell (**Figure 4—figure supplement 2C, D**). This result suggests that in nutrient-rich YPD medium, cells may possess the capacity to buffer excess protein expression by increasing the total proteome. However, under these expression conditions, no growth inhibition was observed (**Figure 4—figure supplement 2B**), and thus the induction of *mox-YG* was not sufficiently strong. We therefore consider it necessary to investigate the regime under conditions of stronger overexpression in future studies.

To examine changes in protein expression, proteomic analysis of *mox-YG* overexpressing cells (in the SC medium) was performed. The distribution of intensities obtained from the analysis is shown in the Proteomaps in **Figure 4B**. In *mox-YG* overexpressing cells, *mox-YG* accounted for 40% of the total protein expression. **Figure 4A** suggests that in *mox-YG* overexpressing cells, proteins other than *mox-YG* are overall reduced to about 60% compared to the vector. Therefore, we considered it necessary to normalize the data overall to accurately represent changes in the per-cell proteome. However, in this study, we used data without such correction to focus on how the composition of the proteome, other than *mox-YG*, changed (see Materials and Methods and **Figures 4—figure supplement 3A–C** for more details).

Figure 4C illustrates the categories that showed significant changes upon *mox-YG* overexpression (all categories are shown in **Figure 4—figure supplement 4**). Elevated mitochondrial proteins, such as oxidative phosphorylation and TCA cycle, and decreased ribosomal protein and ribosomal synthesis were observed. Interestingly, the proteomic changes observed in *mox-YG* overexpressing cells showed little correlation with the corresponding transcriptomic changes (**Figure 4—figure supplement 3D**). The only notable exception was oxidative phosphorylation, which was upregulated at both the transcript and protein levels (**Figure 3B** and **4C**). In contrast, for example, ribosomal proteins were upregulated at the transcript level but markedly downregulated at the protein level (**Figure 3B** and **4C**). This discrepancy may indicate a cellular state in which transcriptional responses are uncoupled from protein expression as a result of protein burden.

As mentioned above, RNA-seq results suggested nitrogen source starvation in *mox-YG* overexpression. Therefore, we investigated whether this response is influenced by the regulation through the TORC1 pathway, a primary mechanism responsive to nitrogen sources. By comparing expression changes due to rapamycin (a TORC1 inhibitor) treatment (Gowans et al., 2018) and *mox-YG* overexpression, we found that about 40% of the proteins with altered expression were congruent with the group of proteins regulated by TORC1 (**Figures 4—figure supplement 5A, B**). Thus, the proteome expression changes due to *mox-YG* overexpression may include responses due to inactivation of the TORC1 pathway as well as other responses. **Figure 4C** shows which categories had significant expression changes when separated into rapamycin-responsive (Rap) and rapamycin-non-responsive (Non-Rap) gene populations. Carbohydrate metabolism, elevated TCA cycle, and decreased ribosome synthesis were suggested to be responses due to TORC1 inactivation, while endocytosis, cytoskeleton, protein transport and lipid synthesis, and glycolysis were suggested to be due to responses other than TORC1 inactivation (**Figure 4—figure supplement 6**). Oxidative phosphorylation included both Rap responses (12/32) and non-responses (20/32) (**Figure 4—figure supplement 6C**). Indeed, *mox-YG* overexpression reduced the sensitivity to rapamycin compared to the vector control (**Figure 4—figure supplement 7A–E**), supporting the notion that TORC1 activity was diminished. However, we were unable to obtain direct evidence demonstrating TORC1 inactivation. We examined the phosphorylation status of Atg13 (Kamada et al. 2000), a known TORC1 target, by Western blotting. In *mox-YG* overexpressing cells, however, the total expression level of Atg13 was markedly reduced (**Figure 4—figure supplement 7F**). Therefore, it remains inconclusive how the decreased expression of TORC1 targets affects downstream signaling under these conditions.

Above transcriptome and proteome analyses suggested that *mox*-YG overexpression induces mitochondrial development and a metabolic shift from fermentation to oxidative respiration. To verify mitochondrial development, we performed fluorescence microscopy. Indeed, mitochondria labeled with Rip1-mScarlet-I, a component of the oxidative phosphorylation pathway, were significantly more developed in *mox*-YG overexpressing cells (**Figure 4D** [↗](#), 4—figure supplement 8). Furthermore, oxygen consumption was increased in these cells (**Figure 4E** [↗](#), 4—figure supplement 9), supporting the notion that they had undergone a metabolic shift toward oxidative respiration.

Overexpression of *mox*-YG causes a dysplastic nucleolus, which is alleviated by mutations in the nuclear exosome

Ribosomes are a crucial factor for protein burden because they are themselves protein synthesizers while being synthesized in the highest amounts in the cell. Therefore, we further focused on the behavior of ribosomes in *mox*-YG cells and expression changes were investigated in detail for each ribosomal protein (**Figure 5—figure supplement 1** [↗](#)). While the majority of the 84 ribosomal proteins detected (45 of them significantly) decreased, six (Rps12, Rps31, Ppp0, Rpp2a, Rpl5, and Rpl12) showed significant increases. Interestingly, all decreased ribosomal proteins are assembled into ribosomes in the nucleolus, while all increased ribosomal proteins are assembled into ribosomes in the cytoplasm (Lafontaine, 2015 [↗](#)). They constitute the “beak” of the 40S subunit and the “P-stalk” and the “central protuberance (pc)” of the 60S subunit (**Figure 5A** [↗](#)). Electron micrographs showed an abnormal nucleolar formation of *mox*-YG overexpressing cells (**Figures 5B** [↗](#) and **Figure 5—figure supplement 3** [↗](#)), in addition to reduced ribosome density in the cytoplasm (**Figure 5—figure supplement 2A** [↗](#)). This abnormal nucleolar formation was also confirmed by fluorescence microscopy images of Nsr1 (Nucleolin), the major protein of the nucleolus (**Figure 5—figure supplement 2B-D** [↗](#)).

Based on these results, we hypothesized that overexpression of *mox*-YG creates an imbalance between the amount of ribosomal protein synthesized in the cytoplasm and the amount of rRNA synthesized in the nucleolus (**Figure 5C** [↗](#), D). The imbalance may lead to the creation of misassembled ribosomes, which are eventually degraded. We speculated that the primary enzyme degrading the misassembled ribosome should be the nuclear exosome. This is because our previous genome-wide mutation analysis identified exosome mutants (*dis3-1*, *mtr3-ts*, *mtr4-1*, *rpn42-ph*, *rpn45-ph*, and *ski6-ph*) as primary mitigators of protein burden (Kintaka et al., 2020 [↗](#)). We confirmed that the *mtr4-1* mutant does not show growth retardation upon *mox*-YG overexpression (**Figure 5E** [↗](#)). In fact, in *mtr4-1* cells, there was no difference in growth rate between the vector control and *mox*-YG overexpression (**Figure 5—figure supplement 5B** [↗](#)), but rather a shorter lag time for the *mox*-YG overexpression than the vector control (**Figure 5E** [↗](#)). We next quantified protein upon *mox*-YG overexpression in *mtr4-1* cells (**Figure 5F** [↗](#)). Compared to the wild-type (Vector), the total protein content in *mtr4-1* (Vector) cells was reduced to 78%, which was likely related to the decreased growth. In the *mtr4-1* *mox*-YG overexpressing cells, the amount of other proteins decreased (to 54%), overwhelmed by the 40% expression of *mox*-YG.

However, the overall protein content increased compared to the *mtr4-1* vector control, with a composition like the wild-type *mox*-YG overexpressing cells. Therefore, in *mtr4-1* cells, even though *mox*-YG overexpression reduced the amount of other proteins (to 54%) compared to the vector control (78%), the growth rate was not reduced (**Figure 5—figure supplement 5B** [↗](#)). This suggests that *mox*-YG overexpression restores the effect of growth inhibition caused by *mtr4-1*. Next, we observed the composition of the nucleolus in *mtr4-1* (**Figures 5G** [↗](#) and **Figure 5—figure supplement 5D, E** [↗](#)). The failure of nucleolus formation observed upon *mox*-YG overexpression was partially, but significantly, rescued in the *mtr4-1* strain. This supports our hypothesis that misassembled ribosomes are degraded by the exosome (**Figure 5D** [↗](#)), and with low exosome activity, nucleolus formation (and efficient ribosome assembly) would be possible even with overexpression of *mox*-YG (**Figure 5—figure supplement 4D** [↗](#)).

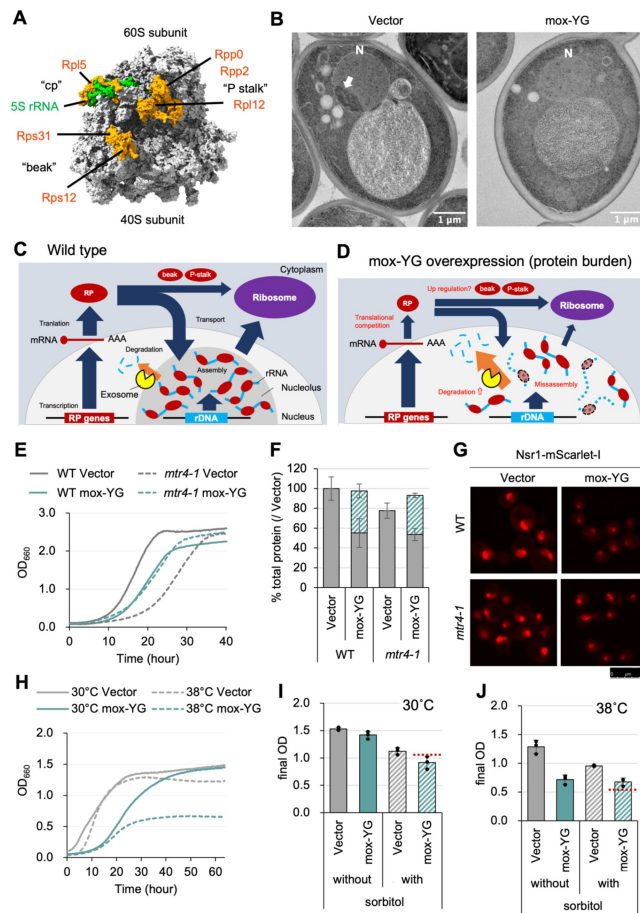


Figure 5.

mox-YG overexpression causes abnormal nucleolus formation.

A. 3D structure of the ribosome. PDB model 6GQV (Pellegrino et al., 2018) is used for the base model and colored using ChimeraX software (Meng et al., 2023). Ribosomal proteins with increased expression upon mox-YG overexpression are shown with the named structures containing them. See also Figure 5—figure supplement 1 for the quantitative data. B) Electron microscope images of the vector control and mox-YG overexpressing cells. The arrow in the image points to the nucleolus structure. N: nucleus. Images of other observed cells are shown in Figure 5—figure supplement 3. C, D) Model diagrams showing the hypothetical situation of the wild type (C) and mox-YG overexpressing cells (D). In WT cells, enough ribosomal proteins (RPs) are produced, resulting in RP- assembled rRNAs and the formation of nucleolus with normal morphology. On the other hand, in mox-YG overexpressing cells, the amount of RP is reduced due to translation competition, increasing misassembled rRNA. As a result, degradation of rRNA by exosomes may be accelerated, resulting in abnormal nucleolus morphology. E) Growth curves of WT and *mtr4-1* mutant cells with the vector control or upon overexpression of mox-YG at 30°C. The solid or dotted lines show the average calculated from three biological replicates. Growth curves with SDs of replicates are shown in Figure 5—figure supplement 5A. F) Ratio of total protein levels of WT and *mtr4-1* strains with the vector or upon mox-YG overexpression, calculated the total protein level of WT cells with the vector as 100%. Gray bars indicate expression levels of proteins other than mox-YG; green shaded bars indicate mox-YG expression levels. Error bars were SDs calculated from three biological replicates. The p-values were calculated by performing Welch's t-test. "n.s." means $p > 0.05$. G) Fluorescence microscopy image of nucleolus-localized protein Nsr1-mScarlet-I of the WT and *mtr4-1* mutant cells with the vector or mox-YG overexpression. Bright field and merged images, and quantification of the nucleolus size are shown in Figures 5—figure supplement 5D and 5E. H) Growth curves of the cells with the vector or under mox-YG overexpression cultivated at 30°C or 38°C. The solid or dotted lines show the average calculated from three biological replicates. Growth curves with SDs of replicates are shown in Figure 5—figure supplement 6A. I, J) Final OD of the cell culture with or without 1M sorbitol at 30°C (I) or 38°C (J). The bars and error bars show the means and SDs calculated from three biological replicates. The red dotted line indicates the final OD estimated from the product of (mox-YG without sorbitol) / (Vector without sorbitol) and (Vector with sorbitol) / (Vector without sorbitol). Growth curves with SDs of replicates are shown in Figures 5—figure supplement 6C, D.

While experimenting with *mox*-YG overexpressing cells, we found that these cells stop growth earlier (lower final density) at higher temperatures (38°C) than the vector controls (**Figure 5H** and **Figure 5—figure supplement 6A**). Cells at this time did not show cell death such as cell bursting (**Figure 5—figure supplement 6B**). Recently, it was reported that intracellular osmotic pressure increases at high temperatures, resulting in less formation of liquid-liquid phase-separated structures such as nucleoli, which are recovered when cells are placed in hyperosmotic conditions (Watson et al., 2023). We hypothesized that the growth inhibition of *mox*-YG overexpressing cells at high temperatures may be due to the enhanced failure of nucleolus formation, and thus cultured *mox*-YG overexpressing cells under hyperosmotic pressure. At 30°C, hyperosmolarity caused the same decrease in growth for the vector control and *mox*-YG overexpression (**Figures 5I** and **Figure 5—figure supplement 6C**). On the other hand, at 38°C, hyperosmolarity caused a decrease in growth only in the vector control and no further decrease in *mox*-YG overexpression cells (**Figures 5J** and **Figure 5—figure supplement 6D**). The same trend was observed regarding the size of the nucleolus (**Figures 5—figure supplement 6E, F**). Under the conditions with added sorbitol and under *mox*-YG overexpression, the size of the nucleoli was smaller than the control at 30°C, whereas, at 38°C, it was rather larger than the control. This suggests that *mox*-YG overexpression and hyperosmolarity may mitigate each other's growth-inhibitory effects, and one explanation may be that hyperosmolarity suppresses the abnormal organization of the nucleolus caused by *mox*-YG overexpression.

The Gpm1 mutant shows a similar NI to *mox*-YG, yet yields distinct phenotypes upon overexpression

Thus far, our investigation has focused on heterologous fluorescent proteins. Finally, we turned to endogenous yeast proteins to identify those with the lowest cytotoxicity and to examine the phenotypic consequences of their overexpression. We previously investigated the expression limits of a group of glycolytic enzymes in *S. cerevisiae*, with Gpm1 and its catalytic center mutant (CCmut) having the highest expression limits (Eguchi et al., 2018). Here, we evaluated the neutrality of the wild types and their CC-mutants of Tdh3 and Gpm1 (**Figure 6A** and **Figure 6—figure supplement 1**). The growth rates at the expression limits were higher in the CCmut for both Tdh3 and Gpm1, and in Gpm1, the CCmut grew two-fold faster (**Figures 6—figure supplement 1A, C**). The protein levels of wild types and mutants of both proteins were not significantly different (**Figures 6—figure supplement 1D, E**). The calculated NIs were highest for Gpm1-CCmut, which was equivalent to that of *mox*-YG (**Figures 6A**). We previously concluded that the activities of glycolytic enzymes generally do not affect their expression limits (Eguchi 2018), but their neutralities suggest that their activities can become constraints in some way. This may be due to a disturbance in metabolism or a property of the active enzyme, such as an effect of reactive amino acids (His or Cys, **Figure 6—figure supplement 2**).

Although Gpm1 and Gpm1-CCmut exhibited NIs comparable to that of *mox*-YG, suggesting that they are similarly unconstrained, notable differences in cell morphology were observed (**Figure 6B**, **6—figure supplement 3**). Cells overexpressing Gpm1 appeared enlarged and spherical, while those overexpressing Gpm1-CCmut did not show such hypertrophy but were still rounder than vector controls or *mox*-YG overexpressing cells. The transcriptional responses observed in *mox*-YG overexpressing cells, including the induction of *GAP1*, a marker of the amino acid starvation response, were not detected in Gpm1-CCmut overexpression (**Figure 6C** and **Figure 6—figure supplement 4**). In Gpm1-CCmut overexpressing cells, similar to *mox*-YG overexpression, the levels of proteins other than Gpm1-CCmut were reduced, despite a slight, statistically non-significant increase in total protein amount (**Figure 6—figure supplement 1F**). While mitochondrial development was observed in Gpm1-CCmut cells using Rip1-mScarlet-I labeling (**Figure 6—figure supplement 5**), no increase in oxygen consumption was detected (**Figure 6D** and **Figure 6—figure supplement 6**). The nucleolus, visualized with Nsr1-mScarlet-I, appeared similarly shrunken in Gpm1-CCmut overexpressing cells as in *mox*-YG overexpressing cells (**Figure 6E** and **Figure 6—figure supplement 7**). These results suggest

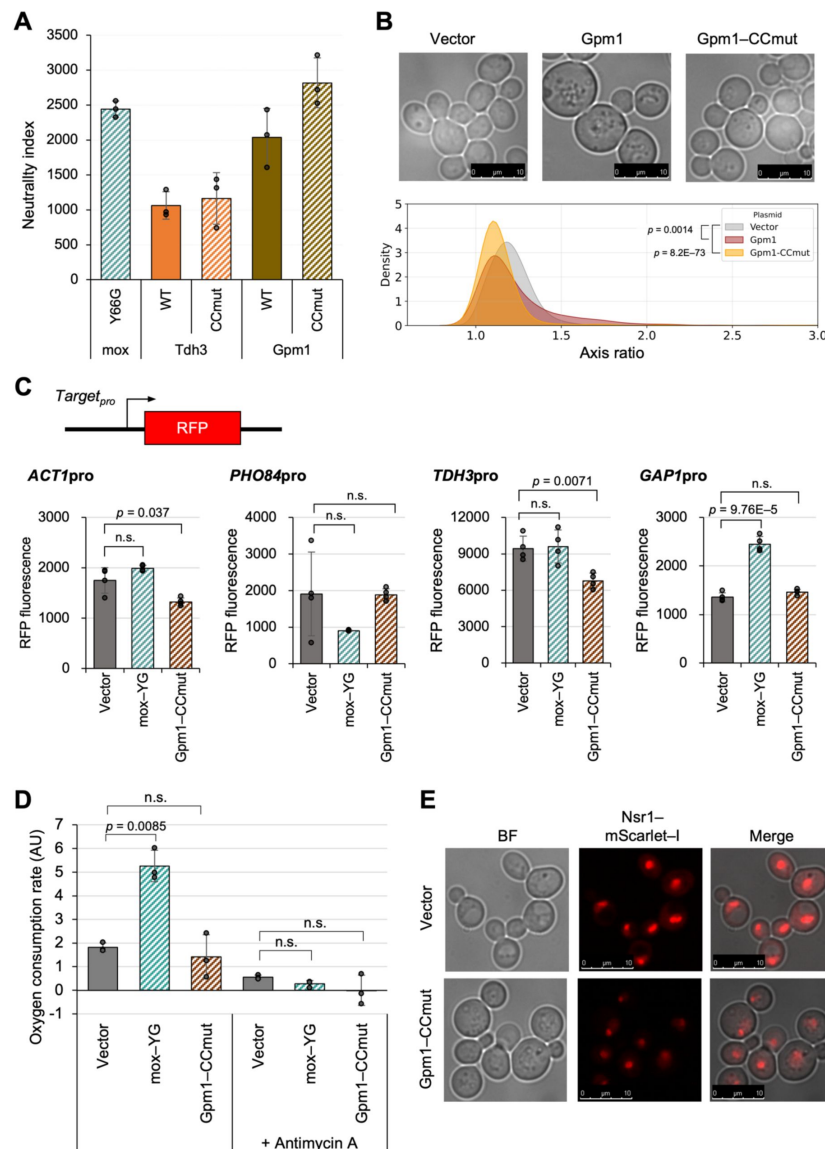


Figure 6.

The Gpm1 mutant shows a similar NI to max-YG, yet yields distinct phenotypes upon overexpression.

A) The neutrality index for Gpm1, Tdh3, and their CCmut. Data on growth rates and protein expression levels used for the neutrality index calculation are presented in Figure 6—figure supplement 1. B) Top images: representative microscopic images of cells overexpressing Gpm1 or Gpm1-CCmut. Bottom graph: Distribution of cell axis ratios upon overexpression of Gpm1 and Gpm1-CCmut, compared with the vector control. The p-values were calculated by the Mann-Whitney U-test. Statistical analysis of cell size and comparison with max-YG overexpressing cells are shown in Figure 6—figure supplement 3. C) Promoter activity reporter assay. Constructs used for promoter analysis with transcription reporters and quantitative results of RFP fluorescence values (arbitrary unit) for each promoter. Time series data are in Figure 6—figure supplement 4. D) Oxygen consumption rate in vector control, max-YG, and Gpm1-CCmut overexpressing cells (arbitrary units, AU). Antimycin A was added as a control for respiratory inhibition. The measurement data are shown in Figure 6—figure supplement 6. E) Fluorescence microscopy images of nucleolus with Nsr1-mScarlet-I. Log phase cells (OD660 = 1.0) with the control vector or under Gpm1 or Gpm1-CCmut overexpression were observed. Statistical analysis of nucleolar size is shown in Figure 6—figure supplement 7. The bars and error bars in A, C and D show the means and SDs calculated from three biological replicates. The raw data is shown with dot plots. The p-values were calculated by performing Welch's t-test and applying Bonferroni correction. "n.s." means $p > 0.05$.

that the phenotypes observed in *mox*-YG overexpressing cells include both *mox*-YG-specific effects and responses commonly associated with protein burden. Alternatively, as indicated by the abnormal cell morphology, *Gpm1*-CCmut may possess unknown constraints or cytotoxic properties not captured by the neutrality index, which could obscure the detection of phenotypes typically associated with protein burden.

Discussion

In this study, we found that the fluorescent properties of FPs constrain their expression limits. This fluorescence-induced cytotoxicity appears as different phenotypes depending on the FPs; an excess of fluorescent mCherry causes cell enlargement, but such a phenotype was not seen with green fluorescent proteins. On the other hand, the cause of the cytotoxicity is not clear currently. Fluorescence itself does not seem to constrain the expression limit, because the *mox* expression limit remained the same when cultured in the dark or under strong light (**Figure 2—figure supplement 7** [↗](#)). We also believe that H₂O₂ generated during fluorophore maturation (Craggs, 2009 [↗](#)) is unlikely. In the presence of H₂O₂, *mox* overexpressing cells grew slightly slower than *mox*-YG overexpressing cells (**Figure 2—figure supplement 8H** [↗](#)). However, no transcriptional response to oxidative stress, including the catalase gene (*CTT1*), was seen with *mox* overexpression (FDR < 0.187, over the vector control, Data S1). Because *CTT1* expression was elevated in *mox*-YG overexpression (FDR < 0.006, over the vector control, Data S1), presumably due to enhanced respiration, *mox*-YG overexpression might rather increase oxidative stress tolerance. Conversely, "inactive" proteins might accumulate more without interfering with intracellular functions because both inactive fluorescent proteins and inactive glycolytic enzymes showed higher neutrality (**Figures 2** [↗](#) and **6** [↗](#)). However, our current data did not clarify the mechanism, and investigating more proteins and their variants would be required.

The currently least cytotoxic proteins, non-fluorescent *mox*-YG and inactive *Gpm1* express more than 40% of the total protein, whereas yeast cells can maintain growth under these conditions (**Figures 2** [↗](#) and **6** [↗](#)). This is higher than the prediction in *E. coli*, which stops growing when 37% of the excess protein is expressed (Bruggeman et al., 2020 [↗](#)). This may be due to the very low cytotoxicity of the proteins expressed in this study, or it may be because eukaryotic cells have greater capacity than prokaryotic cells. In the *mox*-YG overexpressing cells, we observed a previously expected but unobserved cellular physiology of the protein burden: depletion of protein synthesis resources (i.e., nitrogen source; **Figure 3** [↗](#)) and a decrease in other proteins (glycolytic enzymes and ribosomes; **Figure 4** [↗](#)), possibly due to competition for synthesis. In addition, metabolic conversion to mitochondria and failure of nucleolus formation were observed (**Figures 4** [↗](#) and **5** [↗](#)). As a matter of fact, the nitrogen source depletion response (**Figure 3** [↗](#)) itself is not self-evident. Because the total protein content in the *mox*-YG overexpressing cells is the same as that in the vector control (**Figure 4A** [↗](#)), the intracellular nitrogen source used for protein synthesis should not be depleted. Then, why does the nitrogen source depletion response occur? Excess *mox*-YG may promote wasteful consumption of amino acids and nitrogen, or there may be pathways that sense the amount of "useful proteins" and their ability to be synthesized, or there may be depletion of specific amino acids in *mox*-YG excess. Another possibility is that there may be some mechanism that maintains the amount of total protein in the cell even when the nitrogen source is depleted. Depletion of nitrogen sources probably results in the inactivation of the TORC1 pathway (**Figure 4C** [↗](#)), metabolic conversion from fermentation to respiration, and ribosome depression to an economic mode. In *E. coli*, the ppGpp pathway plays this role (Shachrai et al., 2010 [↗](#)). While the above is thought to be a physiological response to protein burden, a defect in ribosome synthesis in the nucleolus was also suggested (**Figure 5B** [↗](#)). This could be due to a conflict between nuclear rRNA synthesis and cytoplasmic protein synthesis or to the negative effects of the inability to create a liquid-liquid phase-separated structure due to dilution effects (see below). The partial restoration of nucleolus formation by a mutation of the exosome (**Figure 5G** [↗](#)) supports this possibility.

When one type of protein is expressed as much as 40% of the total, as in the present study, the cytoplasmic crowding, viscosity, pH, etc. may change due to the nature of the protein and the effect of the reduced protein by competition. These may alter the movement of substances. It is known that when ribosomes are reduced, the fluidity of large proteins changes (Delarue et al., 2018 [DOI](#)). Water activity- osmotic pressure may change (Watson et al., 2023 [DOI](#)), and pH may also be affected. These may affect the formation of liquid-liquid phase-separated structures (Delarue et al., 2018 [DOI](#); Watson et al., 2023 [DOI](#)). The failure of nucleolus formation observed in *mox-YG* overexpression (**Figure 5B** [DOI](#)) may be due to such a condition change. In fact, the growth of *mox-YG* overexpressing cells is significantly impaired at high temperatures (**Figure 5H** [DOI](#)), where liquid-liquid phase separation is less likely to occur (Watson et al., 2023 [DOI](#)). In contrast, the effect appears to be reduced at high osmolarity (**Figure 5J** [DOI](#)), where it is more likely to occur (Watson et al., 2023 [DOI](#)). A mutation of the nuclear exosome restored nucleolus formation

(**Figure 5G** [DOI](#)), which might be due to the reformation of the liquid-liquid phase-separated structure (i.e., nucleolus) by increasing the concentration of rRNA. These have not been considered phenomena that occur with protein overexpression. The engineering of physicochemical properties of the cytoplasm thus may be possible by mass expression of proteins with specific properties.

In this study, in addition to *mox-YG*, inactive Gpm1 was also expressed in large amounts as well (**Figure 6** [DOI](#)). Both are globular proteins with relatively small molecular weight, but it is not yet clear what characteristics are necessary for them not to be cytotoxic. Further investigation of the neutrality of various proteins will clarify these features. In addition, while the overexpression of Gpm1-CCmut also caused nucleolar shrinkage similar to that observed in *mox-YG* overexpressing cells, it did not induce other phenotypes such as nitrogen starvation or enhanced respiration (**Figure 6** [DOI](#)). Therefore, it remains unclear which of the phenotypes observed in *mox-YG* overexpressing cells are general consequences of “protein burden” and which are specific to *mox-YG*. As noted in the Introduction (**Figure 1E** [DOI](#)), it is essential to continue investigating multiple proteins that appear to be “unconstrained” in order to address this issue. In this study, expression limits were investigated, but no experiments were performed to gradually increase expression levels. For example, it would be meaningful to investigate at what expression level of *mox-YG* a shift to the economy mode or failure of nucleolus formation occurs. Also, due to the limitations of the experimental system, it may not be possible to express *mox-YG* to its true limit. The expression of one specific protein cannot be 100% of the total protein, and there is always a limit. As the number of ribosomes continues to decrease due to competition for synthesis, the excess proteins cannot be synthesized either. Thus, there must be a final equilibrium point. A more powerful and controllable expression system would be needed to investigate this. Although the translation system appears to be the primary target of burden in *mox-YG* overexpression (**Figure 2—figure supplement 6C, D** [DOI](#)), the bottleneck in the expression system may also differ depending on gene design (**Figure 1D** [DOI](#)) (Hausser et al., 2019 [DOI](#)).

Materials and Methods

Strains, growth conditions, and yeast transformation

BY4741 (*MATa his3Δ1 leu2Δ0 met15Δ0 ura3Δ0*) (Brachmann et al., 1998 [DOI](#)) was used as the host strain for the experiments. Yeast culture and transformation were performed as previously described (Amberg et al., 2005 [DOI](#)). A synthetic complete (SC) medium without uracil (Ura) or leucine (Leu), as indicated, was used for yeast culture. Other strains used in this study are listed in the Key Resource Table.

Plasmid construction

The plasmids were constructed by homologous recombination activity of yeast cells (Oldenburg 1997 [↗](#)), and their sequences were verified by DNA sequencing. Plasmids used in this study are listed in the Key Resource Table.

Genetic tug-of-war

To overexpress a target protein to a level that causes growth inhibition (in this study this is referred to as “expression limit”), we used gTOW (**Figure 2—figure supplement 1A** [↗](#)) (Moriya *et al.*, 2012 [↗](#), 2006 [↗](#)). The gene of the target protein is incorporated into the gTOW plasmid (here, pTOW40836) and introduced into yeast cells lacking *ura3* and *leu2* genes. The selection marker in this case is uracil (– Ura conditions). Since this plasmid has the replication origin of 2 μ plasmid (2 μ *ORI*), this plasmid becomes multicopy in the cell (about 30 in the case of the vector). When yeast cells carrying the plasmid are transferred to a medium without leucine and uracil (–Leu/Ura conditions), the copy number of the plasmid in the cells increases (to about 120 in the case of the vector). This is because *leu2-89* on the plasmid is a *LEU2* allele with a large deletion in its promoter, so cells with higher plasmid copy numbers will grow faster in –Leu/Ura conditions. In other words, *leu2-89* acts as a bias to raise the plasmid copy number in –Leu/Ura conditions. If overexpression of the target protein inhibits growth, the plasmid copy number will be lower than the copy number that gives rise to the critical expression of that protein.

In other words, the target gene acts as a bias to lower the plasmid copy number. The resulting tug-of-war between the selection biases created by the two genes leads to a target protein being expressed at close to the critical expression level in –Leu/Ura conditions (the green-colored area in **Figure 2—figure supplement 1A** [↗](#)). By using a version of the aureobasidin A resistance gene (*AUR1*) with a deletion in its promoter (*AUR1d*) as a bias to increase plasmid copy number, experiments based on the same concept as gTOW can also be conducted in rich medium (YPD + 500 ng/mL aureobasidin A). In this case, the plasmid pTOW-*AUR1d* is used (**Figure 4—figure supplement 2A** [↗](#)).

Measuring growth rate

Cellular growth was measured by monitoring OD660, every 10 min using a compact rocking incubator (TVS062CA, ADVANTEC). The max growth rate (MGR) was calculated as described previously (Moriya *et al.* 2006 [↗](#)). Average values, SD, and *p*-values of Welch’s *t*-test were calculated from at least three biological replicates. The growth data of **Figures 2—figure supplement 10A** [↗](#) was measured by monitoring OD595, respectively, every 10 min using a microplate reader (Infinite F200, TECAN).

Protein analysis

Yeast cells overexpressing a target protein were cultivated in SC–Leu/Ura medium. The total protein was extracted from log-phase cells (OD660 = 1.0) with a NuPAGE LDS sample buffer (Thermo Fisher Scientific) after 0.2 mol/l NaOH treatment (Kushnirov 2000). For each analysis, the total protein extracted from 1 optical density (OD) unit of cells with OD660 was used. For total protein visualization, the extracted total protein was labeled with Ezlabel FluoroNeo (ATTO), as described in the manufacturer’s protocol, and separated by 4–12% sodium-dodecyl sulfate acrylamide gel electrophoresis (SDS-PAGE). Proteins were detected and measured using the LAS-4000 image analyzer (GE Healthcare) in SYBR–green fluorescence detection mode and Image Quant TL software (GE Healthcare). Protein quantification was performed as shown in **Figure 2—figure supplement 1B** [↗](#). Average values, SD, and *p*-values of Welch’s *t*-test were calculated from three biological replicates.

Western blotting of Atg13

Cells overexpressing FPs were collected at OD₆₆₀ = 1 or 2 and fixed with 100 µL of ice-cold alkaline solution (0.2 N NaOH and 0.5% β-mercaptoethanol). After 5 min of incubation on ice, 1 mL of ice-cold acetone was added to the sample and incubated at -20°C to precipitate proteins. The protein samples were precipitated with a microfuge (15,000 rpm, 5 min), air-dried, suspended in 100 µL of SDS-PAGE sample buffer, and incubated at 65°C for 15 min. The samples were thoroughly dissolved by sonication and subjected to SDS-PAGE (40 mA, 1h). After ponceau S staining, proteins are transferred to the PVDF membrane (Millipore #IPVH304F0) and blocked in 5% skim milk/TBST. Atg13 was detected using a rabbit anti-Atg13 antibody (1:3000) and peroxidase-conjugated goat anti-rabbit secondary antibodies (Jackson ImmunoResearch, #111-035-003, 1:10,000). Chemiluminescent substrates (Millipore, #WBLUF0100) and Light-Capture II (ATTO) were used for signal detection.

Measuring GFP fluorescence

GFP fluorescence (Ex485nm/Em535nm) was measured every 10 min using a microplate reader (Infinite F200, TECAN). Average values, SD, and *p*-values of Welch's *t*-test were calculated from eight biological replicates.

RNA-seq analysis

BY4741 cells overexpressing a target protein were cultured in SC-Leu/Ura medium and harvested at the logarithmic growth phase. RNA extraction was performed according to (Köhrer and Domdey, 1991 [DOI](#)).

The purified RNA was quality checked by BioAnalyzer (Agilent) or MultiNA (Shimazu), and the concentration was measured by Qubit (Thermo Fisher Scientific). Purified RNA was stored at -80°C until subsequent experiments. cDNA libraries were prepared using the TruSeq Stranded Total RNA kit (Illumina), using half the protocol of the TruSeq RNA library prep kit. Three biological replicates were analyzed for all strains. The sequences were checked for read quality by FastP (Chen et al., 2018 [DOI](#)) and then aligned using Hisat2 (Kim et al., 2019 [DOI](#)). The aligned data were formatted into bam files by Samtools (Li et al., 2009 [DOI](#)) and quantified by StringTie (Pertea et al., 2015 [DOI](#)). Finally, expression level variation analysis was performed by EdgeR (Robinson et al., 2010 [DOI](#)). The raw data were deposited into DDBJ (accession number: PRJDB18064). The processed data (Data_S1) included the average expression levels (log₁₀CPM) of 6685 genes in the vector control, the changes in expression (log₂FC) upon overexpression of *mox* or *mox*-YG compared to the vector control, and the significance values of expression change (*mox*_FDR, *mox*-YG_FDR).

Figures 3—figure supplement 1A, B [DOI](#) is the visualization of this data using volcano plots. This data is used in subsequent data analyses below.

Reporter assay

The strains used for the reporter assay were created by introducing the promoter region of each gene and the sequence of mScarlet-I, a type of RFP, into the *FCY1* locus of the genome as shown in **Figure 3—figure supplement 2A** [DOI](#). Correct integration was checked by histidine prototrophy and 5-FC resistance of the strain. The cell density (OD₅₉₅) and RFP fluorescence (Ex535nm/Em590nm) of the created strains were measured every 10 min using a microplate reader (Infinite F200PRO, TECAN). Average and SD of max RFP fluorescence, and *p*-values of Welch's *t*-test were calculated from eight biological replicates.

Proteome analysis

The vector control and mox-YG overexpressing cells (three biological replicates) were cultured in 5 mL of SC–LeuUra medium with shaking and cells were collected at OD₆₆₀ = 1. Cells were washed twice with PBS, frozen on dry ice, and transported to Kazusa DNA Research Institute. The samples were prepared as described in the previous study (Hughes et al., 2019 [↗](#); Kawashima et al., 2022 [↗](#)). The cells were dissolved in 100 mM Tris-HCl (pH 8.0) containing 4% SDS and 20 mM NaCl using BIORUPTOR BR-II (SONIC BIO Co., Kanagawa, Japan). The 20 µg of extracted proteins were quantified using Pierce™ BCA Protein Assay Kit (Thermo Fisher Scientific, WA, USA) at 500 ng/µL. The protein extracts were reduced with 20 mM tris (2-carboxyethyl) phosphine for 10 min at 80°C followed by alkylation with 30 mM iodoacetamide for 30 min at room temperature in the dark. Protein purification and digestion were performed using the sample preparation (SP3) method (Kawashima et al., 2022 [↗](#)). The tryptic digestion was performed using 500 ng/µL Trypsin/Lys-C Mix (Promega, Madison, WI, USA) for overnight at 37°C. The digests were purified using GL-Tip SDB (GL Sciences, Tokyo, Japan) according to the manufacturer's protocol. The peptides were dissolved again in 2% ACN containing 0.1% TFA and quantified using BCA assay at 150 ng/µL. The digested peptides were loaded directly using a 75 µm × 12 cm nanoLC nano-capillary column (Nikkoy Technos Co., Ltd., Tokyo, Japan) at 40°C and then separated with a 30-minute gradient (mobile phase A = 0.1% FA in water, B = 0.1% FA in 80% ACN) consisting of 0 min 8% B, 30 min 70% B at a flow rate of 200 nL/min using an UltiMate 3000 RSLCnano LC system (Thermo Fisher Scientific). The eluted peptides were detected using a quadrupole Orbitrap Exploris 480 hybrid mass spectrometer (Thermo Fisher Scientific) with normal window DIA. The MS1 scan range was set as a full scan with m/z 490-745 at a mass resolution of 15,000 to set an Auto Gain Control (AGC) target for MS1 as 3×10^6 and a maximum injection time of 23 msec. The MS2 were collected at more than m/z 200 at 30,000 resolutions to set an AGC target of 3×10^6 , maximum injection time of “auto”, and fixed normalized collision energy of 28%. The isolation width for MS2 was set to 4 m/z, and for the 500-740 m/z window pattern, an optimized window arrangement was used in Scaffold DIA (Proteome Software, Inc., Portland, OR, USA). The raw data were searched against an *in silico* predicted spectral library using DIA-NN (Demichev et al., 2020 [↗](#)). (version:1.8.1, <https://github.com/vdemichev/DiaNN> [↗](#)) First, an *in-silico* predicted spectral library was generated from the human protein sequence database (UniProt id UP000005640, reviewed, canonical, 20381 entries) using DIA-NN. The DIA-NN search parameters were as follows: protease, trypsin; missed cleavages, 1; peptide length range, 7-45; precursor charge range, 2-4; precursor mass range, 495-745; fragment ion m/z range, 200-1800; mass accuracy, 10 ppm; static modification, cysteine carbamidomethylation; enabled “Heuristic protein interferences,” “Use isotopologues,” “MBR,” and “No shared spectra.” Additional commands were set as follows: “mass acc cal 10,” “peak translation,” and “matrix spec q”. The protein identification threshold was set at < 1% for both peptide and protein false discovery rates (FDRs).

The signal intensity data for 4241 proteins (Data_S2) was used for further data analysis below.

Data analysis

The analysis and visualization of RNA-seq data and proteome data were conducted using custom Python code, and coding and execution were carried out in the Collaboratory. In the RNAseq data, 6,685 protein-coding genes were detected (Data_S1). Among these, 4,291 genes assigned to various categories of KEGG Orthology level 3 (Kanehisa et al., 2023 [↗](#)) were placed into 77 categories. Analysis was performed on 47 categories that contained more than 10 gene elements. We investigated whether the expression of genes in each category, as a group, was significantly higher or lower upon overexpression of mox and mox-YG compared to the vector control. Specifically, mox/Vector (log₂FC) and mox-YG/Vector (log₂FC) were calculated along with their FDR values, and each gene was assigned to a KEGG orthology level 3 category. Next, whether the gene groups within each category showed significantly higher or lower expression changes compared to gene groups outside of their category was tested using the Mann-Whitney *U*-test. Significant differences

were indicated in the graphs with a red border and red stripes. The graphs showed the overall distribution of genes within each category using violin plots, and only genes showing significant differences in *mox*/Vector ($\log_2\text{FC}$) and *mox*-YG/Vector ($\log_2\text{FC}$) were highlighted in swarm plots. An asterisk indicated significant differences on the graphs when the expression changes ($\log_2\text{FC}$) between *mox*/Vector and *mox*-YG/Vector were significant by the Mann-Whitney *U*-test. The significance threshold was set at $\text{FDR} < 0.05$ for all. Only categories where either *mox* or *mox*-YG overexpression showed significant expression changes were extracted and presented as **Figure 3B**, and all 47 categories were shown in **Figure 3—figure supplement 1C**.

The proteome analysis was conducted using the signal intensity data (Data_S2). This data includes proteins that were detected only in some samples under certain conditions or replicates, especially those with low signals or large expression variations. Additionally, the total protein signal sum excluding *mox*-YG was almost the same between the vector control and *mox*-YG overexpression (**Figure 4—figure supplement 3A**). This result contradicts the SDS-PAGE findings, which indicate that the total protein mass per cell, including *mox*-YG, is the same between the vector control and *mox*-YG overexpression (**Figure 4A**). Therefore, it is considered necessary to adjust the total protein signal sum, including *mox*-YG, to match that of the vector control to understand the proteome change per cell. In this study, however, we used the most conservative approach by removing proteins with zero values and not applying any overall corrections (**Figure 4—figure supplement 3B**), which allowed the analysis of 3,588 proteins. Although this may not correctly analyze "per cell" proteome change, it enables us to capture the change in proteome composition (excluding *mox*-YG), thus providing information about cellular responses. If the total correction is applied, only 138 proteins were increased (**Figure 4—figure supplement 3C**), most of which are mitochondrial proteins.

For the detected 3,588 proteins above, we compared the increase or decrease of protein groups within each category of KEGG Orthology Level 3 with those outside the category. Analysis was performed on 45 categories, each containing more than 10 proteins, to investigate whether the expression of protein groups in each category was significantly higher or lower under *mox*-YG overexpression compared to the vector control. Specifically, the average $\log_2\text{FC}$ and its FDR values were calculated using data from three replicates for *mox*-YG/Vector, and each protein was assigned to a KEGG orthology level 3 category. Next, whether the protein groups within each category showed significantly higher or lower expression changes compared to protein groups outside of their category was tested using the Mann-Whitney *U*-test. Significant differences were indicated in the graphs with a red border and red stripes. The graphs showed the overall distribution of proteins within each category using violin plots, and only proteins showing significant differences in *mox*-YG/Vector ($\log_2\text{FC}$) were highlighted in swarm plots. The significance threshold was set at $\text{FDR} < 0.05$ for all. Only categories where *mox*-YG overexpression showed significant expression changes were extracted and presented in **Figure 4C** and all 47 categories were shown in **Figure 4—figure supplement 6**. Published data (Gowans et al., 2018) was used to separate protein groups into rapamycin-responsive and non-responsive, and similar analysis and visualization were conducted (**Figures 4—figure supplement 6A, B**).

Fluorescence microscopic observation

Cells overexpressing target proteins were cultured in SC-LeuUra medium and observed under an inverted microscope (DMI 6000 B, Leica) at log phase ($\text{OD}_{660} = 1.0$). Images were acquired with Leica Application Suite and processed with Leica Application Suite X software (Leica Microsystems). Cells in brightfield images were separated from the images using *yeast_segmentation-master_v3* (Lu et al., 2019). Cell morphological traits were assessed using parameters obtained when analysed using CellProfiler (Ver. 4.2.6) (Carpenter et al., 2006). "AreaShape_Area" was used to analyse FP-overexpressing cell size (arbitrary unit; AU) in **Figure 2—figure supplement 6B**, Nsr1-GFP in **Figure 5—figure supplement 2C** and Nsr1-mScarlet-I in **Figure 5—figure supplement 5E, 6F**, and **Figure 6—figure supplement 7C**. For analysis of the fluorescence intensity of Rip1-mScarlet-I in **Figure 4—figure supplement 8B** and **Figure**

6—figure supplement 5A [↗](#). “Intensity_IntegratedIntensity” was used. Intensity_MeanIntensity” was used to analyze the fluorescence intensity of Nsr1-GFP in **Figure 5—figure supplement 2D** [↗](#) and **Figure 6—figure supplement 7C** [↗](#). Ten images per experiment were used for analysis. The axis ratio in **Figure 6—figure supplement 3** [↗](#) was calculated as the ratio of “AreaShape_MajorAxisLength” to “AreaShape_MinorAxisLength” in the data

Measurement of oxygen consumption

Cells overexpressing the target protein were collected at OD₆₆₀ = 1. The cells (0.05 OD unit) were treated with oxygen probe in SC–LeuUra and 50μM antimycin A in SC–LeuUra. Oxygen consumption was measured according to the instruction manual of the Extracellular OCR Plate Assay Kit (Dojindo, #E297). The fluorescence intensity (Ex500nm/Em650 nm) was recorded at 10 min intervals using a microplate reader (Infinite F200, TECAN). The oxygen consumption rate (AU) was calculated as the slope of a linear approximation of the kinetic data.

Electron microscopic observation

Cells grown to log phase (OD₆₀₀ = 1.0) at 30°C in SC–LeuUra medium were collected and transported to Tokai Electron Microscope Co. Transported samples were sandwiched between copper plates, flash-frozen with liquefied propane, and then dehydrated with anhydrous ethanol. cells were fixed with a 5:5 mixture of propylene oxide and resin (Quetol-651, Nissin EM Co.). Fixed cells were cut with an ultramicrotome (Ultracut UCT, Leica) to prepare 80 nm thick sections, stained with 2% uranyl acetate and lead stain solution (Sigma-Aldrich Co.), and observed with a transmission electron microscope (JEM-1400 Plus, JEOL Ltd.).

Data availability

All raw data, analysis code, and omics analysis pipelines used for generating the figures have been deposited in GitHub (https://github.com/hisaomlab/Fujita_eLife_2025 [↗](#)).

Figure supplements

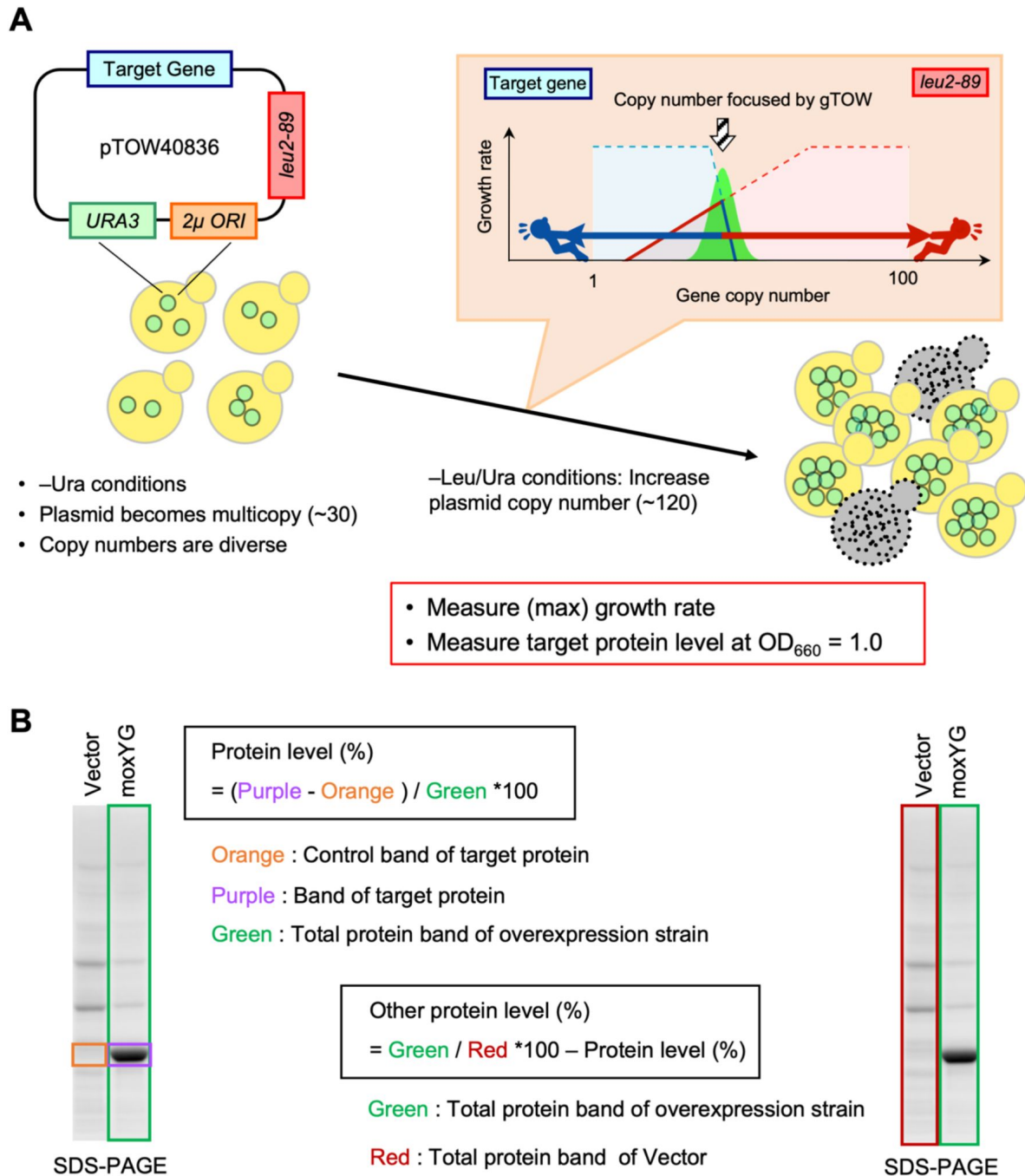


Figure 2—figure supplement 1.

Methods used for overexpression and protein expression quantification.

A) The genetic tug-of-war (gTOW) method. See Materials and Methods for details. B) Protein quantification method. All proteins present in the cells were labeled with a fluorescent dye (Ezlabel FluoroNeo) and separated by SDS-PAGE. The protein level of the target protein (Protein level) was calculated by subtracting the band at the same position of the vector control (Orange) from the band of the target protein on the gel (Purple), which was estimated from the molecular weight, and dividing the result by the total protein (Green). Other protein (Other protein level) was calculated by subtracting Protein level (%) from the value obtained by normalizing the total protein (Green) of the overexpression strain by the total protein (Red) of the vector control.

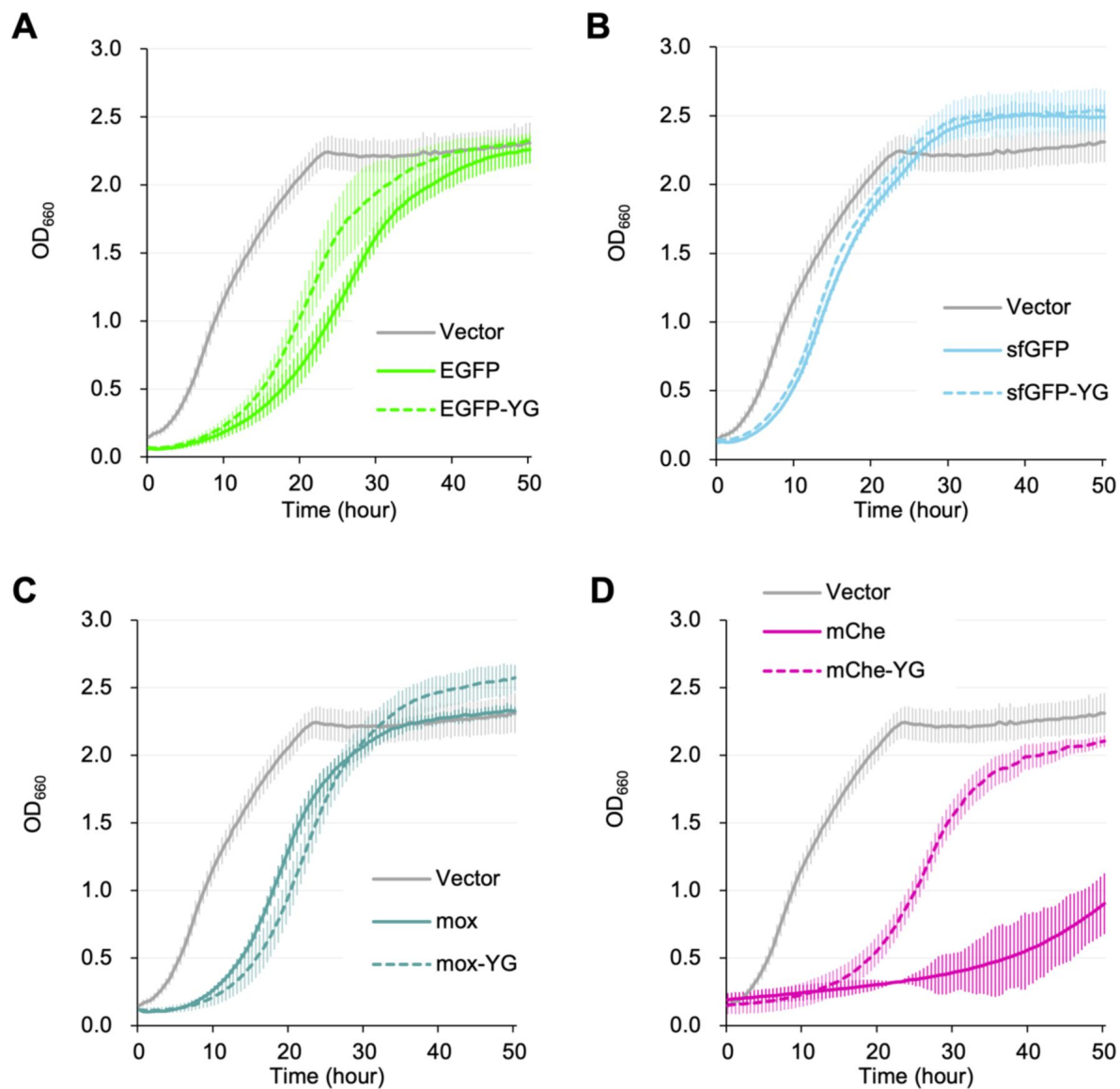


Figure 2—figure supplement 2.

Comparison of growth of fluorescent protein overexpressing strains Growth curves of the cells with the vector (pTOW40836) and the cells overexpressing fluorescent proteins and their non-fluorescent YG mutants in synthetic medium (–LeuUra).

The solid or dotted lines show the average calculated from three biological replicates. Shaded zones indicate their standard deviations. A: EGFP and EGFP-YG, B: sfGFP and sfGFP-YG, C: moxGFP (mox) and mox-YG, D: mCherry (mChe) and mChe-YG.

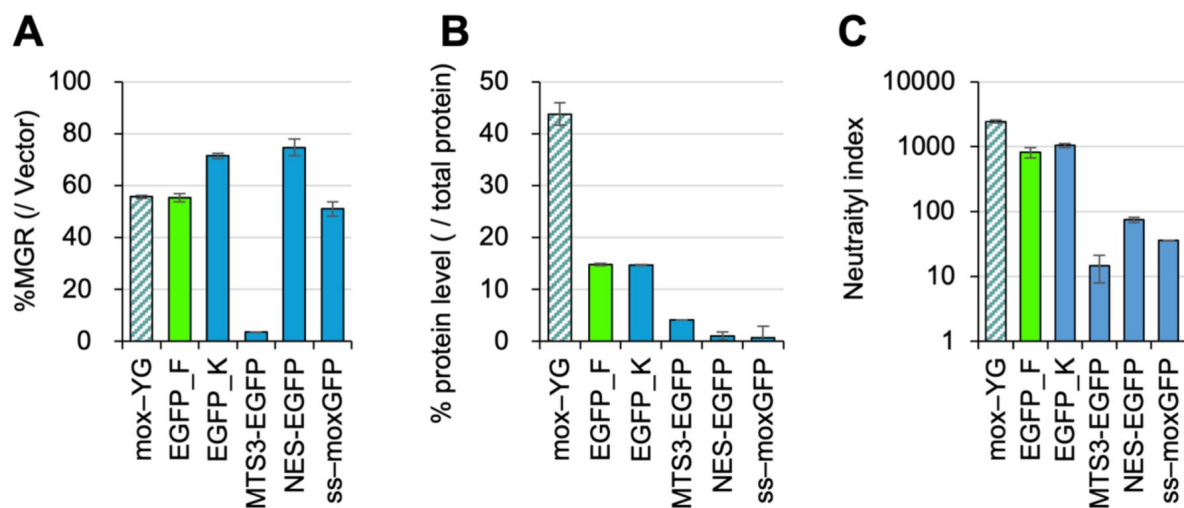


Figure 2—figure supplement 3.

Neutrality indexes of constrained proteins

A) MGRs of cells overexpressing indicated proteins. B) Protein levels of the indicated target proteins (percent over the total proteins). C) The neutrality index for the indicated proteins. MTS3-EGFP: EGFP targeted to mitochondria; cultured in SC-LeuUra; expressed by the PYK1 promoter. NES-EGFP: EGFP with nuclear export signal; cultured in SC-Ura; expressed by the HXT7 promoter. ss-moxGFP: moxGFP targeted to the endoplasmic reticulum; cultured in SC-Ura; expressed by the PYK1 promoter. EGFP_K: the EGFP result of Kintaka et al. (Kintaka et al., 2016); cultured in SC-LeuUra; expressed by the PYK1 promoter. EGFP_F: the EGFP result of this paper. In A and B, the data was obtained from Kintaka et al. (Kintaka et al., 2016), except mox-YG and EGFP_F (same as Figure 2 in this paper). The bars and error bars show the means and SDs calculated from three biological replicates. moxGFP

moxGFP

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1 M S K G E E L F T G V V P I L V E L D G 20
1 ATGTCTAAGGTGAAGAATTATCTACTGGTGTGTCCCAATTTGGTTGAATTAGATGGT 60

21 D V N G H K F S V R G E G E G D A T N G 40
61 GATGTTAATGGTCACAAATTTCTGTCcgtGGTGAAGGTGAAGGTGATGCTACTaAtGGT 120

41 K L T L K F I S T T G K L P V P W P T L 60
121 AAATTGACCTTAAATTTATTTCTACTACTGGTAAATTGCCAGTTCATGGCCACCTTA 180

61 V T T L T Y G V Q S F S R Y P D H M K R 80
181 GTCACTACTTTAACTTATGGTGTTCaATcTTTTCTAGATACCCAGATCATATGAAACgt 240

81 H D F F K S A M P E G Y V Q E R T I S F 100
241 CATGACTTTTTCAAGTCTGCCATGCCAGAAGGTTATGTTCAAGAAAGAACTATTTcTTTC 300

101 K D D G T Y K T R A E V K F E G D T L V 120
301 AAAGATGACGGTActTACAAGACCAGAGCTGAAGTCAAGTTTGAAGGTGATACCTTAGTT 360

121 N R I E L K G I D F K E D G N I L G H K 140
361 AATAGAATCGAATTAAAGGTATTGATTTTAAAGAAGATGGTAACATTTTAGGTCACAAA 420

141 L E Y N F N S H N V Y I T A D K Q K N G 160
421 TTGGAATACAACTtCAACTCTCACAATGTTTACATCActGCTGACAAACAAAAGAAATGGT 480

161 I K A N F K I R H N V E D G S V Q L A D 180
481 ATCAAAGcTAactTCAAATTAGACACAACgtTGAAGATGGTTCTGTTCAATTAGCTGAC 540

181 H Y Q Q N T P I G D G P V L L P D N H Y 200
541 CATTATCAACAAATACTCCAATTGGTGATGGTCCAGTCTTGTTACCAGACAACCATTAC 600

201 L S T Q S R L S K D P N E K R D H M V L 220
601 TTATCCACTCAATCTcgtTTATCCAAAGATCCAAACGAAAAGAGAGACCACATGGTCTTG 660

221 L E F V T A A G I T H G M D E L Y K * 239
661 TTAGAATTTGTTACTGCTGCTGGTATTACCCATGGTATGGATGAATTGTACAAATAA 717
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sfGFP

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1 M S K G E E L F T G V V P I L V E L D G 20
1 ATGTCTAAGGTGAAGAATTATCTACTGGTGTGTCCCAATTTGGTTGAATTAGATGGT 60

21 D V N G H K F S V R G E G E G D A T N G 40
61 GATGTTAATGGTCACAAATTTCTGTCcgtGGTGAAGGTGAAGGTGATGCTACTaAtGGT 120

41 K L T L K F I C T T G K L P V P W P T L 60
121 AAATTGACCTTAAATTTATTTgTACTACTGGTAAATTGCCAGTTCATGGCCACCTTA 180

61 V T T L T Y G V Q C F S R Y P D H M K R 80
181 GTCACTACTTTAACTTATGGTGTTCaATgTTTTCTAGATACCCAGATCATATGAAACgt 240

81 H D F F K S A M P E G Y V Q E R T I S F 100
241 CATGACTTTTTCAAGTCTGCCATGCCAGAAGGTTATGTTCAAGAAAGAACTATTTcTTTC 300

101 K D D G T Y K T R A E V K F E G D T L V 120
301 AAAGATGACGGTActTACAAGACCAGAGCTGAAGTCAAGTTTGAAGGTGATACCTTAGTT 360

121 N R I E L K G I D F K E D G N I L G H K 140
361 AATAGAATCGAATTAAAGGTATTGATTTTAAAGAAGATGGTAACATTTTAGGTCACAAA 420

141 L E Y N F N S H N V Y I T A D K Q K N G 160
421 TTGGAATACAACTtCAACTCTCACAATGTTTACATCActGCTGACAAACAAAAGAAATGGT 480

161 I K A N F K I R H N V E D G S V Q L A D 180
481 ATCAAAGcTAactTCAAATTAGACACAACgtTGAAGATGGTTCTGTTCAATTAGCTGAC 540

181 H Y Q Q N T P I G D G P V L L P D N H Y 200
541 CATTATCAACAAATACTCCAATTGGTGATGGTCCAGTCTTGTTACCAGACAACCATTAC 600

201 L S T Q S R L S K D P N E K R D H M V L 220
601 TTATCCACTCAATCTcgtTTATCCAAAGATCCAAACGAAAAGAGAGACCACATGGTCTTG 660

221 L E F V T A A G I T H G M D E L Y K * 239
661 TTAGAATTTGTTACTGCTGCTGGTATTACCCATGGTATGGATGAATTGTACAAATAA 717
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Figure 2—figure supplement 4.

Base and amino acid sequences of moxGFP and sfGFP.

Amino acids relevant to the text are shown in red.

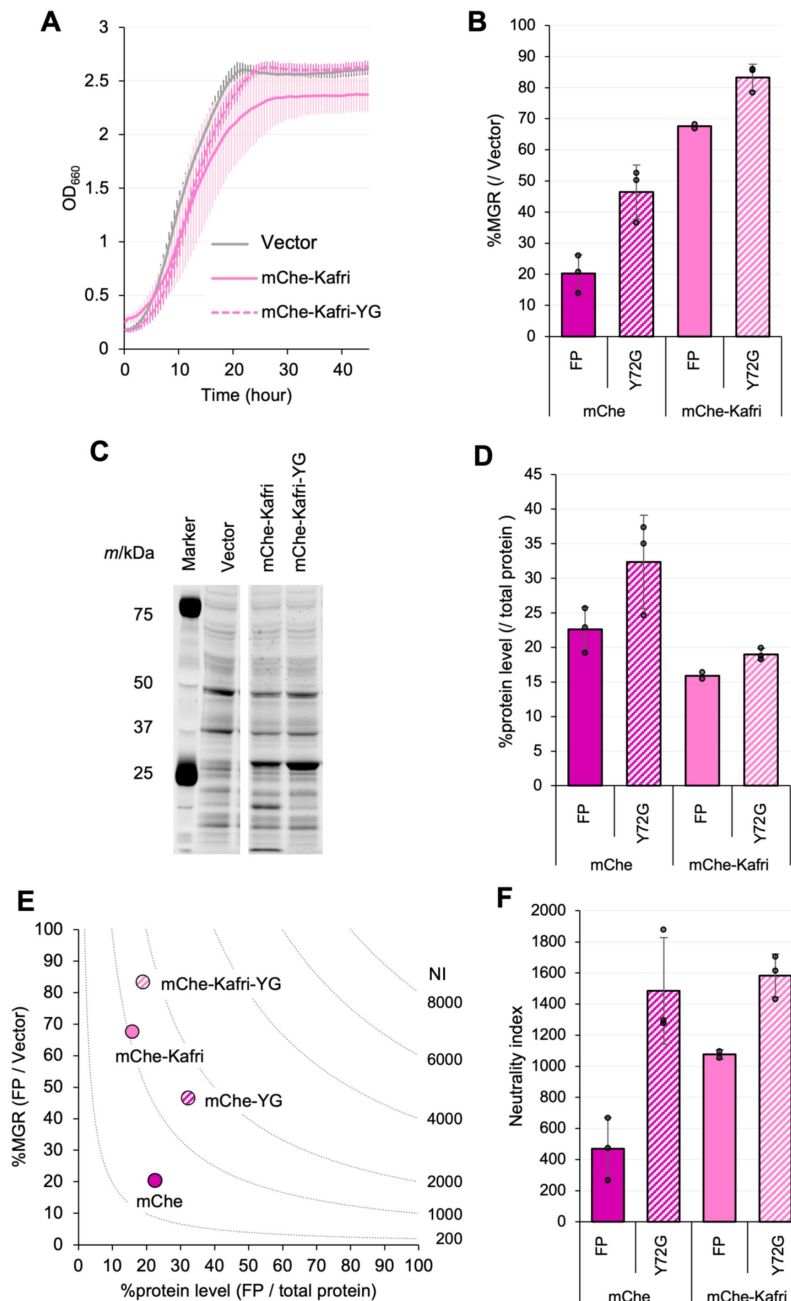


Figure 2—figure supplement 5.

Comparative evaluation of mCherry (mCherry-Kafri) used in previous studies and this study.

A) Growth curves of cells overexpressing WT and YG mutants of mCherry used in Kafri et al. (Kafri et al. 2016) in synthetic medium (–LeuUra). The solid or dotted lines show the average calculated from three biological replicates. B) Ratio of MGR of cells overexpressing FPs and their mutants over the vector control. C) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing FPs. All proteins were separated after staining with a fluorescent dye. D) Protein level of the target protein (percent over the total protein). The amount was calculated from the intensity of the target bands separated by SDS-PAGE of D. Note that mChe may not be accurately measured due to molecular cleavage of the protein. E) Relationship between MGR and the protein level. MGR data are the same as in B, and the protein level data are the same as in D. The neutrality indexes (NIs), calculated from the products of MGR and %protein levels, are indicated by the lines. If the investigated proteins have the same NI, they are located on the same dotted line segment. F) The neutrality index for the FPs and mutants. The neutrality index is the product of %MGR (/Vector) and %protein level (/total protein). The bars and error bars in B, D, and F show the means and SDs calculated from three biological replicates. The raw data is shown with dot plots.

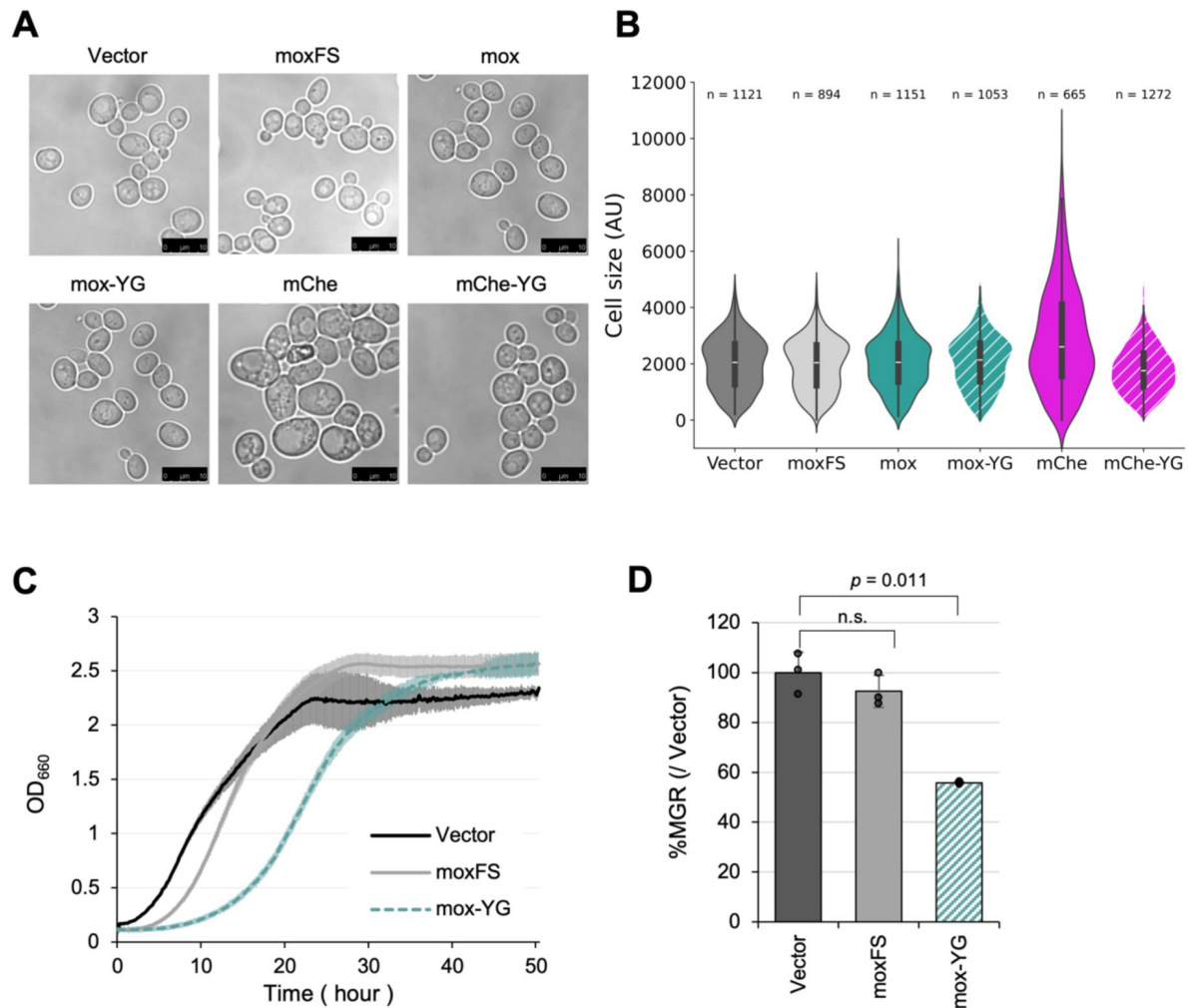


Figure 2—figure supplement 6.

Overexpression of mox-YG does not cause cell enlargement or burden on transcription.

A) Representative microscopic images of cells overexpressing indicated target proteins or moxFS (a frameshift mutant of mox). B) Violin plots showing the cell size quantified from microscopic images. The number of cells used for the quantification is shown. C) Growth curves of the vector control and cells overexpressing moxFS and mox-YG in synthetic medium (–LeuUra). The solid or dotted lines show the average calculated from three biological replicates. D) MGR of cells overexpressing moxFS and mox-YG (percent over the vector control). The bars and error bars show the means and SDs of the MGR calculated from three biological replicates. The raw data is shown with dot plots. The p-value of Welch’s t-test is shown. n.s.: not significant.

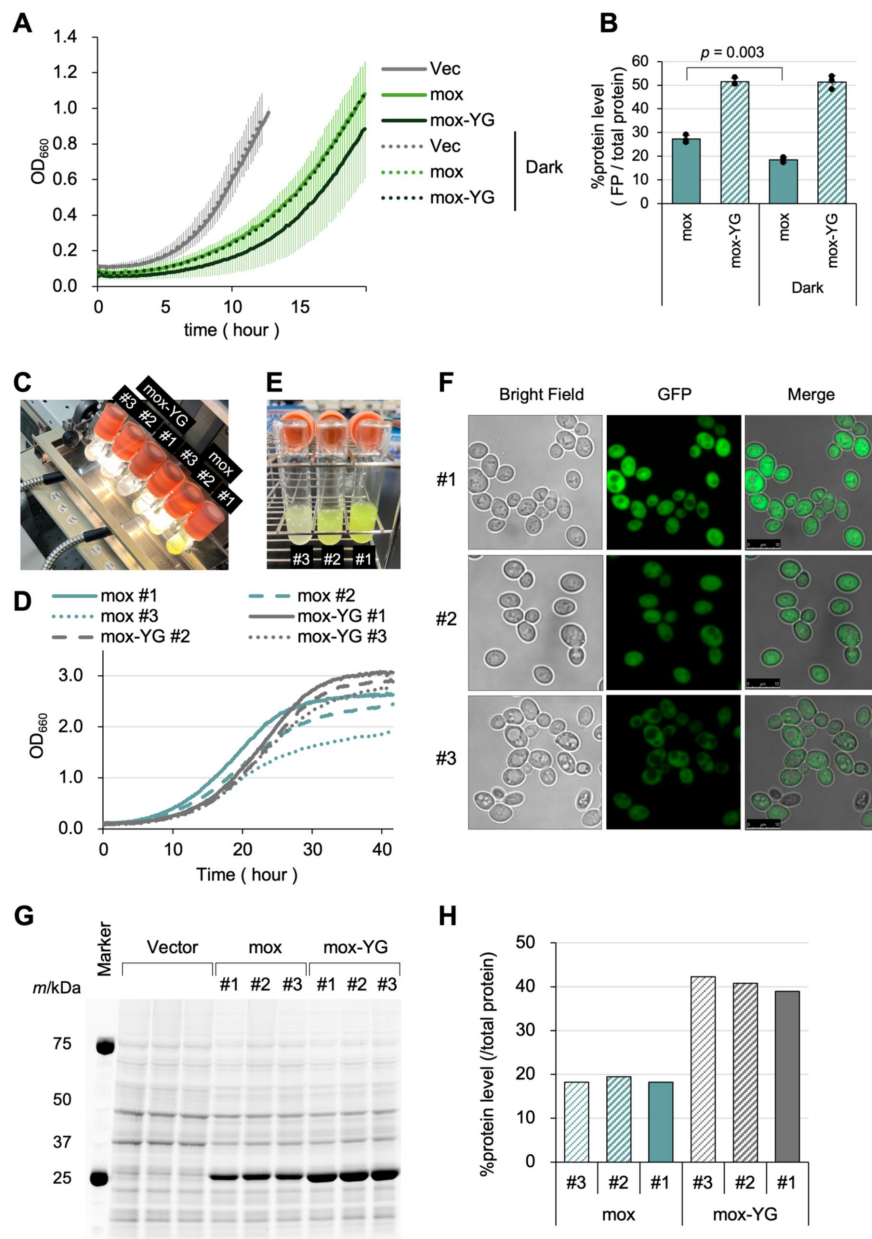


Figure 2—figure supplement 7.

Phototoxicity is not the constraint of mox expression.

A) Growth curves of the vector control and cells overexpressing mox and mox-YG grown under room light or in the dark (Dark) in synthetic medium (-LeuUra). The solid or dotted lines show the average calculated from three biological replicates. B) Protein level of mox and mox-YG (percent over the total protein) under room light or in the dark (Dark) in synthetic medium (-LeuUra). C) Experimental setup for the cultivation of cells under strong light. Strong light was irradiated using a halogen lighting source (Megalight 100, HOYA-SCHOTT). Among the three cultures of cells overexpressing mox and mox-YG, culture #3 was closest to the light, followed by #2 and #1 in decreasing order. Cells were cultured in SC-LeuUra medium. Note that the light does not affect the OD660 measurement itself in this setup. D) Growth curves of the cells overexpressing mox and mox-YG grown under the strong light. Growth curves of cells at the positions (#1, #2, and #3) in C are shown. E) Color of the culture of cells overexpressing mox cultivated at the positions (#1, #2, and #3) in C. The photo was taken after 40 hrs cultivation. F) Fluorescence microscopy images of the cells cultivated under strong light. The cells were cultivated at the position in C for 40 hrs. G) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing mox and mox-YG cultured under the light. Cells were cultured at the position in C to OD660 = 1 and analyzed. All proteins were separated after staining with a fluorescent dye. H) Protein level of mox and mox-YG level in G (percent over the total protein).

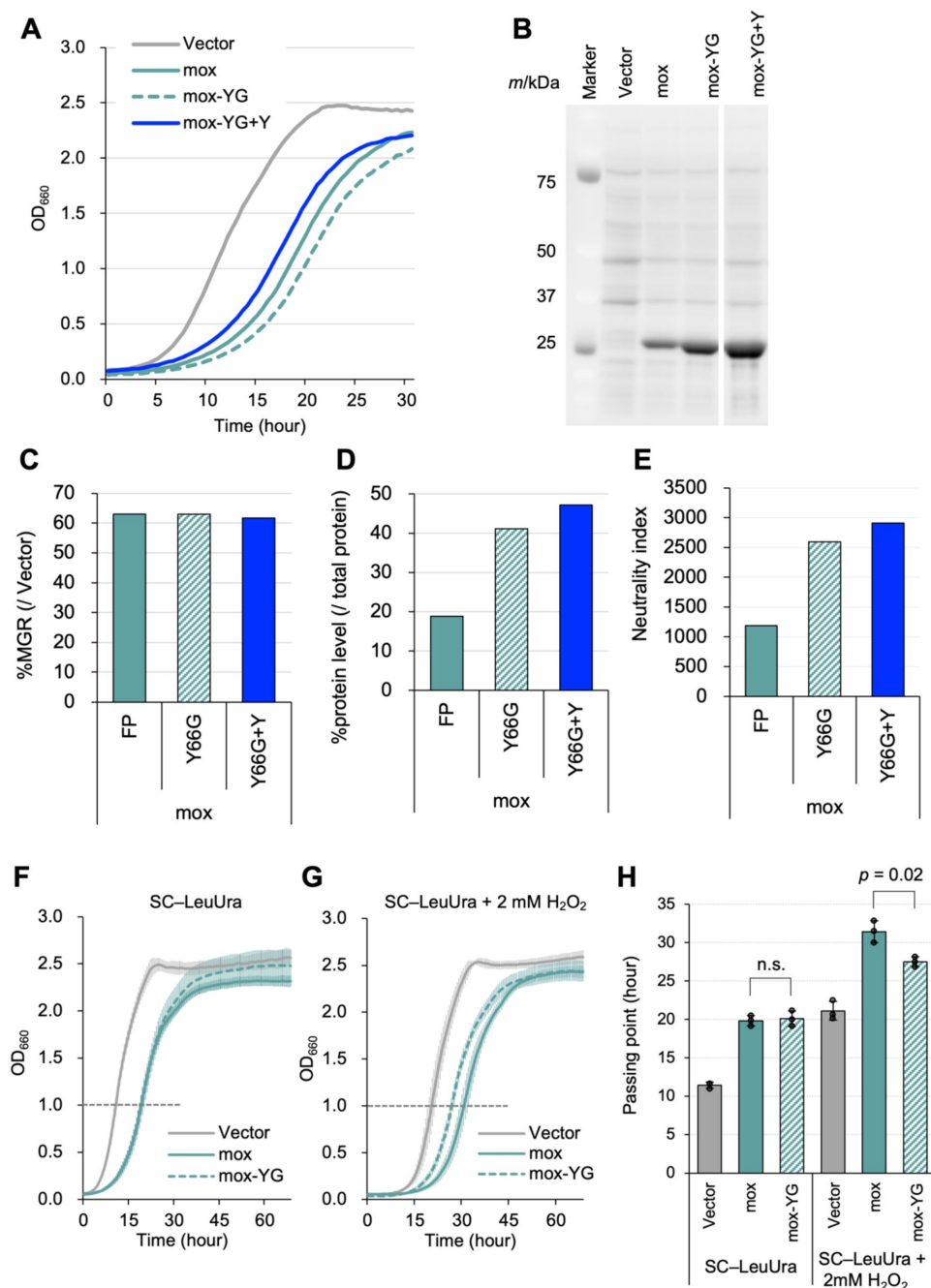


Figure 2—figure supplement 8.

Effect of Tyr and oxidative stress for the expression limits of mox and mox-YG.

A) Growth curves of cells with the vector (Vector) and overexpressing mox, mox-YG, and mox-YG with one Tyr added to the C-terminal (mox-YG+Y) (n = 1 for each). B) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing FPs. All proteins were separated after staining with a fluorescent dye. C-E) Quantification of MGR (C), the protein level (D), and the neutrality index (E) of cells overexpressing mox (FP), mox-YG (Y66G), and mox-YG+Y (Y66G+Y). The MGR was calculated from the growth curve in A. The protein level was calculated from the band in B. The neutrality index was calculated as the product of %MGR in C and the %protein level in D. The results for n = 1 are shown. F, G) Growth curves of the vector control (Vector) and cells overexpressing mox and mox-YG in synthetic medium (F) and medium with 2 mM H₂O₂ (G). Solid or dotted lines indicate mean values calculated from three biological replicates. H) Passing point when the time for Vector, mox, and mox-YG overexpressing cell cultures to reach OD₆₆₀ = 1 (the intersection of the dotted line in F and G with the respective growth curves).

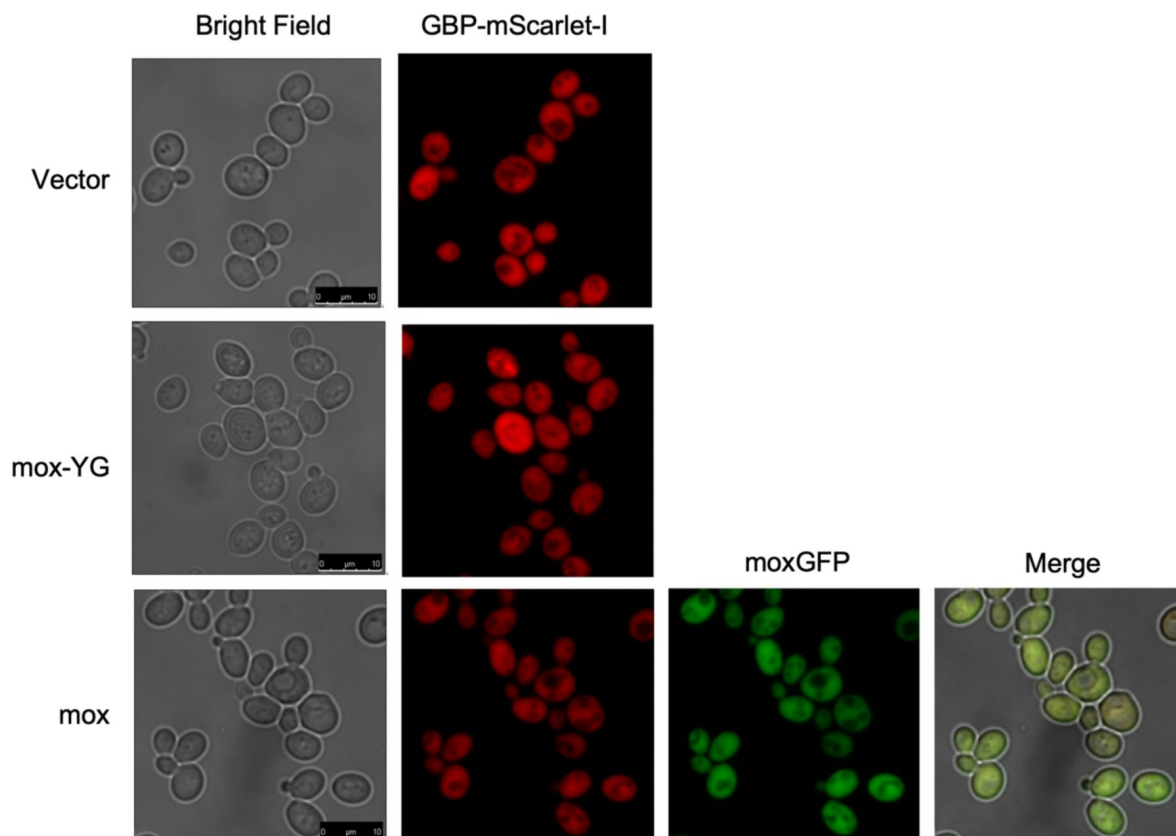


Figure 2—figure supplement 9.

Overexpressed mox-YG does not show any specific localization.

Fluorescence microscopy images of the cells with the vector (Vector) and overexpressing mox and mox-YG in a strain expressing a GFP-binding protein (GBP) fused to mScarlet-I under the control of ACT1 promoter.

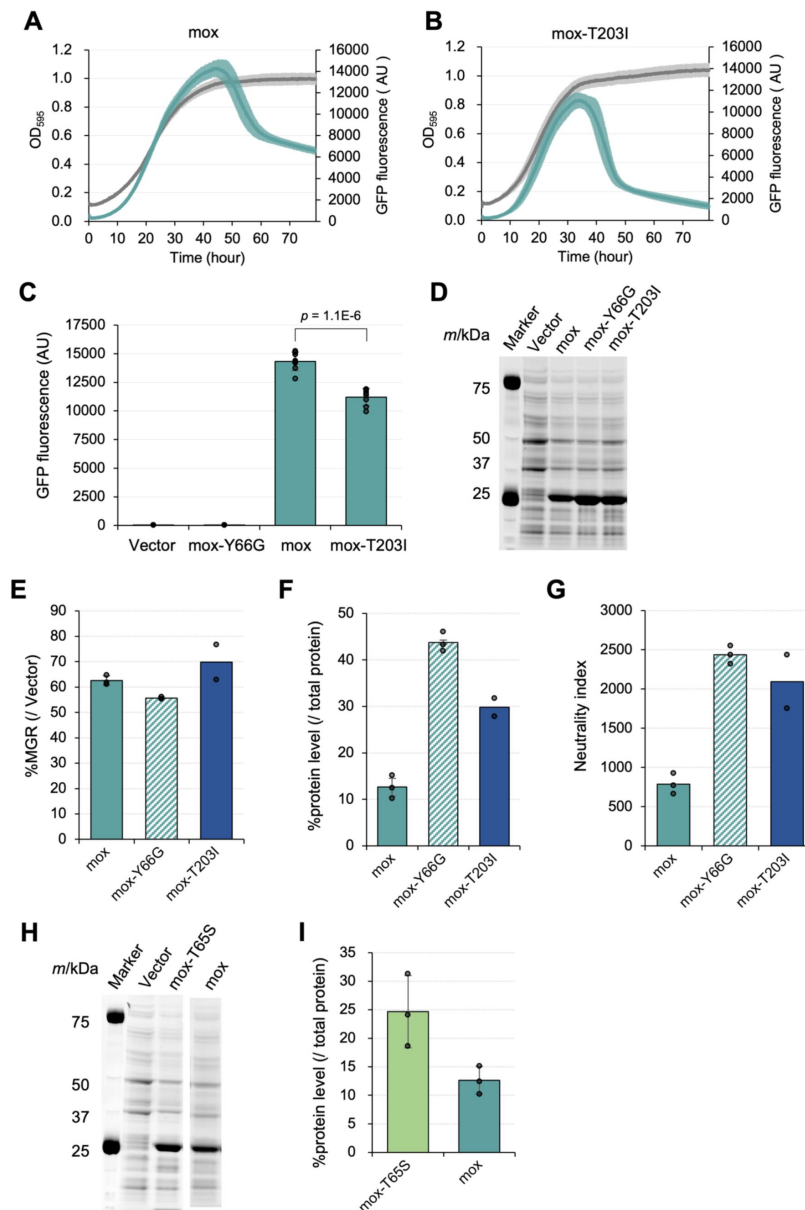


Figure 2—figure supplement 10.

Mutational analysis of *mox*.

A, B) Curves representing time series data of cell growth (gray, OD₅₉₅) and GFP fluorescence (green) overexpressing *mox* and *mox-T203I* mutant in the synthetic medium (–LeuUra). Solid lines are means calculated from eight biological replicates. C) Bar graph comparing the maximum value of GFP fluorescence obtained from the GFP fluorescence curves shown in A and B. GFP fluorescence values for the vector control and *mox-YG* overexpressing cells were also shown as controls. Average values, SD, and p-values of Welch's t-test were calculated from eight biological replicates. D) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing modified GFP. All proteins were separated after staining with a fluorescent dye. E) MGR of cells overexpressing modified *moxGFP* and their mutants (percent over the vector control). F) Protein levels of the *mox* variants (percent over the total protein). The amount was calculated from the intensity of the target bands separated by SDS-PAGE of D. G) The neutrality index for the modified GFP. The neutrality index was calculated as the product of the %MGR in E and the %protein level in F. H) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing *mox-T65S* mutant and *mox* of WT. All proteins were separated after staining with a fluorescent dye. I) Protein levels of *mox-T65S* mutant and *mox* (percent over the total protein). The amount was calculated from the intensity of the target bands separated by SDS-PAGE of H. The bars and error bars in E, F, G, and I show the means and SDs calculated from three biological replicates, except the T203I mutant (n = 2). The raw data is shown with dot plots.

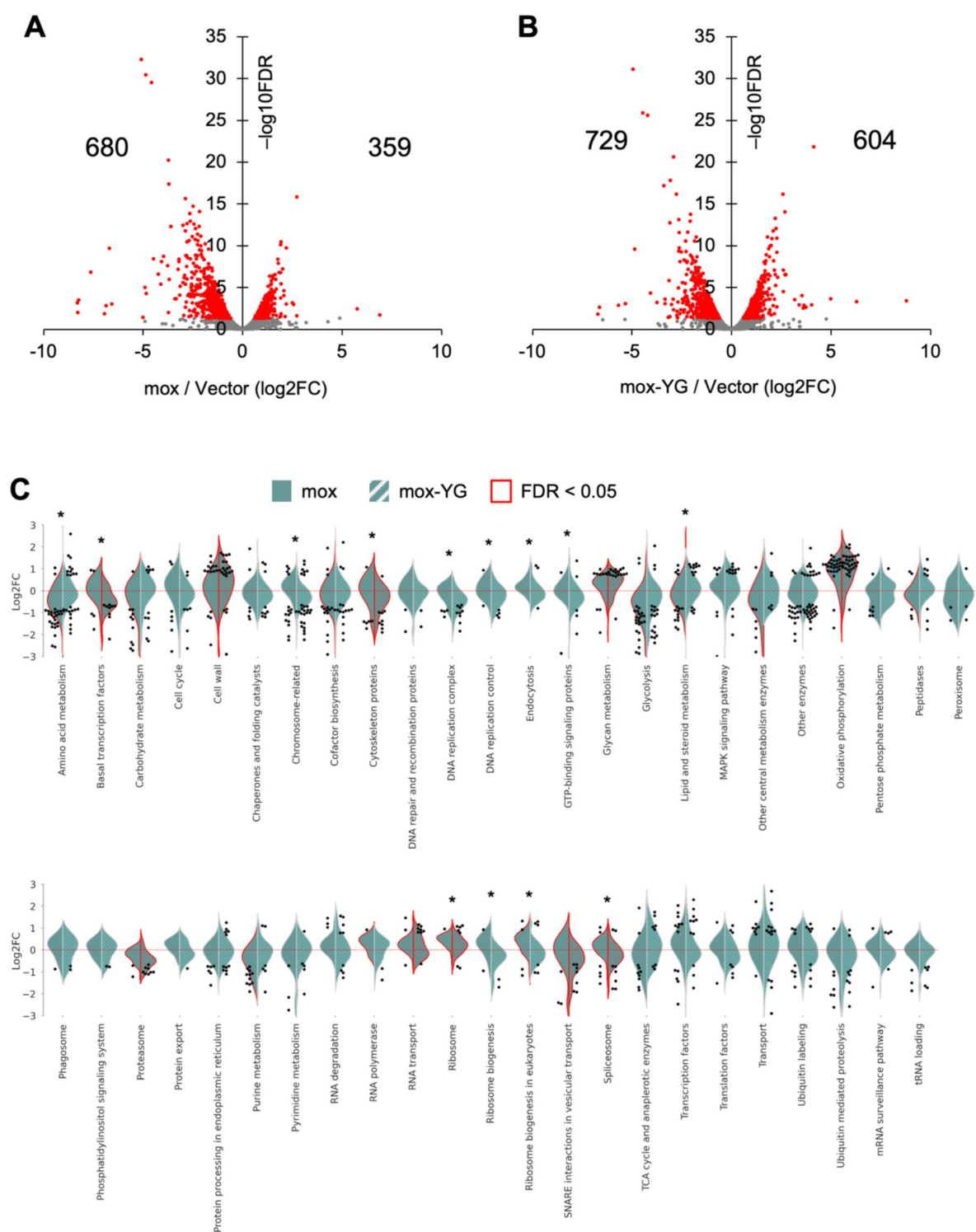


Figure 3—figure supplement 1.

Comparison of the transcriptional response of mox and mox-YG overexpressing cells.

A, B) Transcriptome changes in mox (A) and mox-YG (B) overexpressing cells. Significantly altered ORFs (FDR < 0.05) compared to the vector control are shown in red. The number of ORFs that were significantly increased and decreased are shown. C) Violin plots showing transcriptome results for each category of KEGG orthology level 3. Categories that were significantly up or down-regulated (FDR < 0.05) are indicated by red violin plots. Genes that showed significant expression changes (FDR < 0.05) within the same category are indicated by swarm plots. Asterisks indicate significant differences (FDR < 0.05) between mox and mox-YG comparisons for groups of genes in the same category.

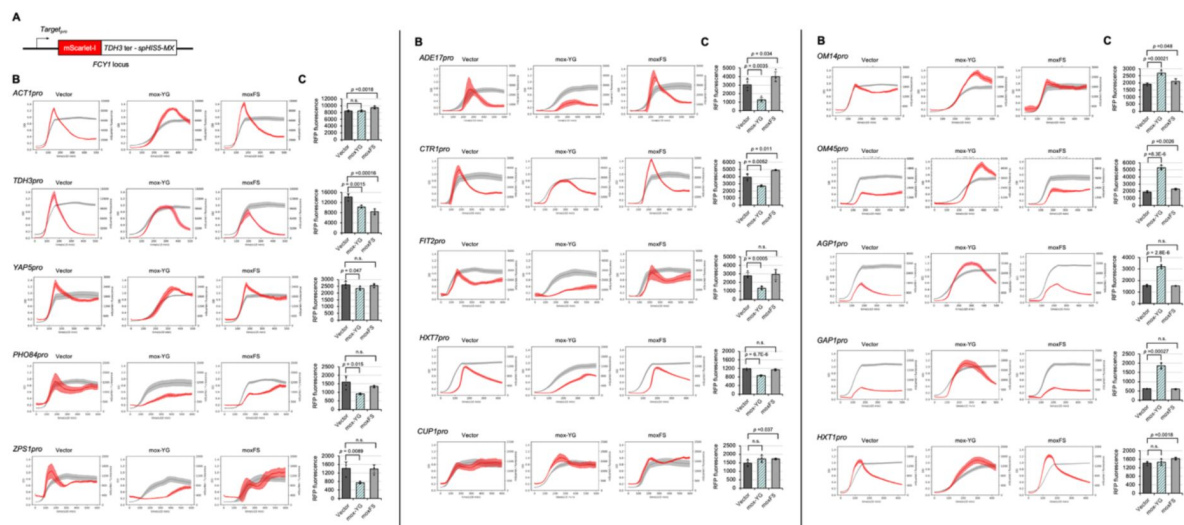


Figure 3—figure supplement 2.

Verification of promoter activity by the reporter assay.

A) The constructs used for promoter analysis by the transcriptional reporter included a sequence of RFP (mScarlet-I) behind each promoter, a terminator for TDH3, and a sequence incorporating spHIS5MX as a selection marker, which was integrated into the genomic FCY1 locus. "moxFS" means the frameshift mutant of mox-YG is overexpressed. B) Curves representing time series data of cell growth (gray, OD595) and mScarlet-I fluorescence (red, arbitrary unit) overexpressing mox-YG and moxFS in SC-LeuUra medium for each promoter. Solid lines are averages calculated from four biological replicates. C) Quantitative results of mScarlet-I fluorescence values for each promoter. Bars and error bars indicate the mean and SD of maximum fluorescence values (arbitrary unit) for mScarlet-I calculated from four biological replicates. Raw data are shown as dot plots. The p-values were calculated by performing Welch's t-test and applying the Bonferroni correction. "n.s." means $p > 0.05$.

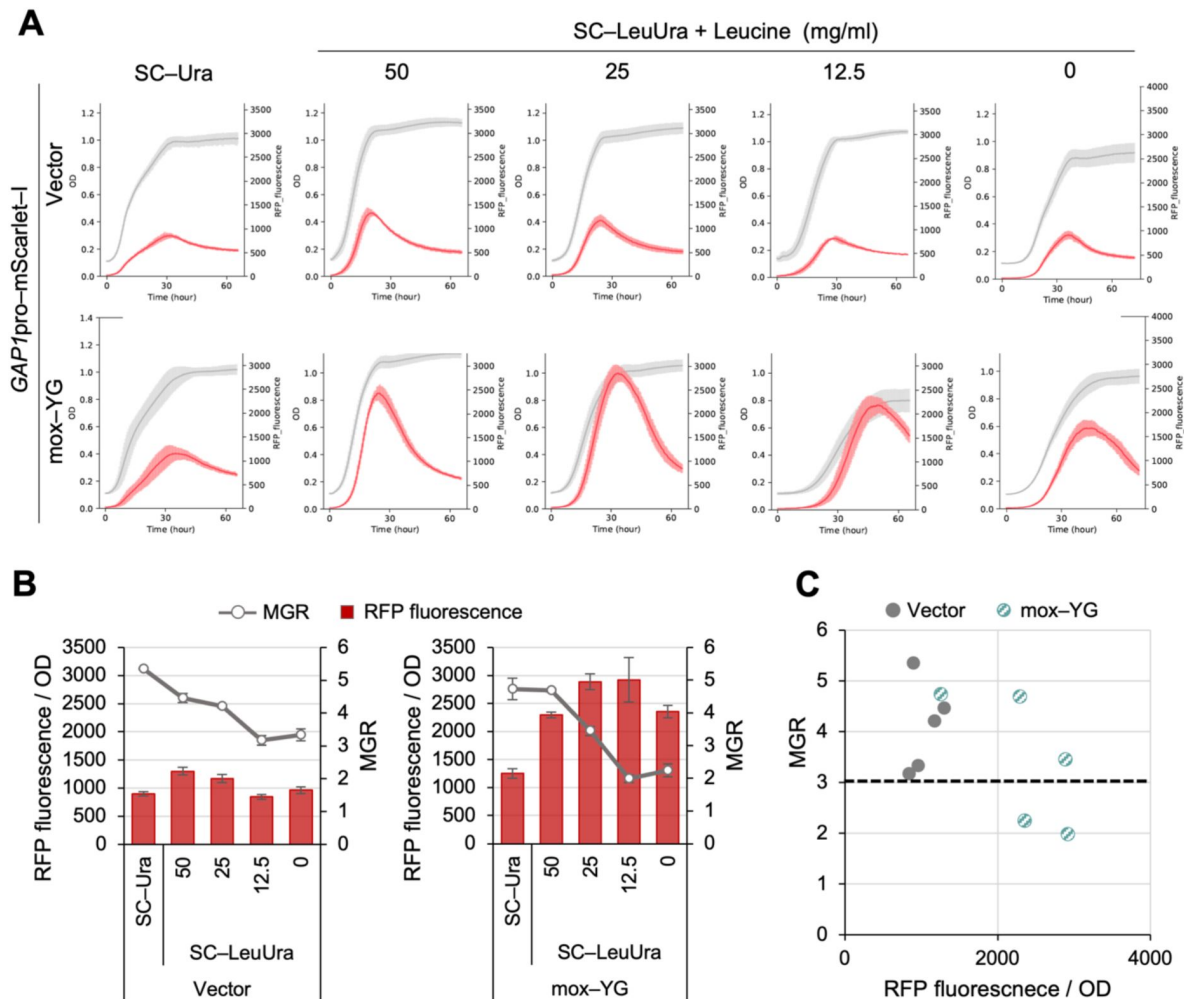


Figure 3—figure supplement 3.

Depletion of leucine in the gTOW experiment does not induce the GAP1 promoter.

A) Reporter assay of the GAP1 promoter in media with different concentrations of leucine. Curves representing time series data of cell growth (gray, OD595) and mScarlet-I fluorescence (red, arbitrary unit) overexpressing mox-YG. Solid lines are averages calculated from four biological replicates. B) Quantification of the maximum growth rate (MGR) and maximum RFP fluorescence derived from the results in A. Bar graphs indicate the mean maximum RFP fluorescence, and error bars represent the SD. Line graphs indicate the MGR, with error bars representing SD. Means and SDs were calculated from four biological replicates. C) Scatter plots of maximum fluorescence values and MGRs created using the values in B. Note that GAP1 induction was observed only in mox-YG overexpression, even under conditions where the MGR was reduced to around three due to leucine depletion (the dotted line).

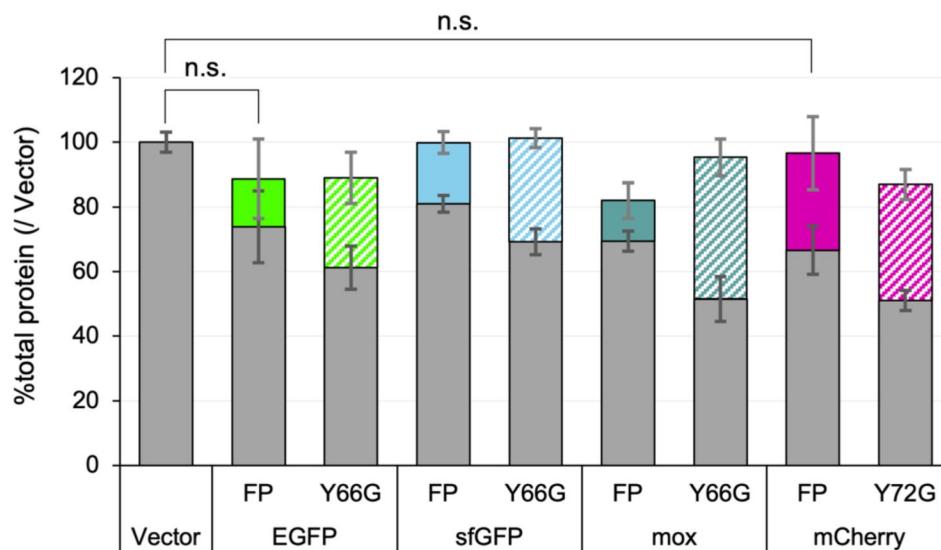


Figure 4—figure supplement 1.

Change in total protein levels upon overexpression of target proteins.

Ratio of total protein content of cells overexpressing FPs over the vector control. Gray bars and error bars indicate expression levels and SDs for proteins other than FPs; non-gray bars and error bars indicate expression levels and SDs for each overexpressed protein. The mean and SD are calculated from three biological replicates. The p-values were calculated by performing Welch's t-test and applying Bonferroni correction. "n.s." means $p > 0.05$.

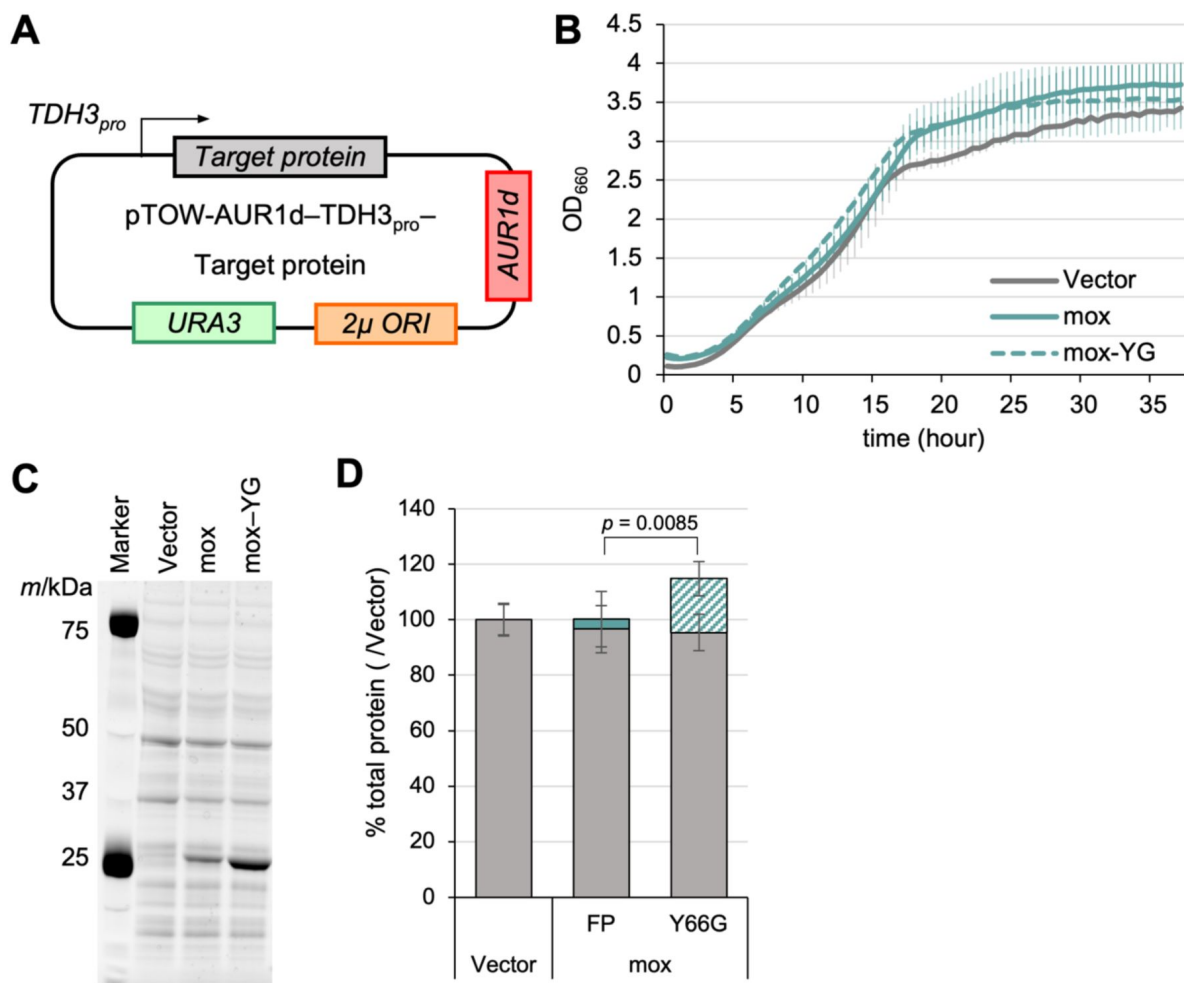


Figure 4—figure supplement 2.

Evaluation of expression constraint of fluorescent proteins under nutrient-rich medium.

A) Experimental setup of the analysis. Target proteins were expressed under the control of the TDH3 promoter (TDH3_{pro}) using the multicopy plasmid pTOW-AUR1d. This plasmid uses the aureobasidin resistance gene AUR1d, which has a short promoter region, replacing the leu2-89 used in the plasmid shown in Figure 2A. B) Growth curves of the vector control and cells overexpressing FPs in YPD included 500 ng/mL aureobasidin A. The solid or dotted lines show the average calculated from three biological replicates. Shaded zones indicate their standard deviations. C) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing FPs. All proteins were separated after staining with a fluorescent dye. D) Ratio of total protein content of cells overexpressing FPs over the vector control in YPD added aureobasidin A and SC-LeuUra. Gray bars and error bars indicate expression levels and SDs for proteins other than; non-gray bars and error bars indicate expression levels and SDs for each overexpressed protein. The mean and SD are calculated from three biological replicates. The p-values were calculated by performing Welch's t-test.

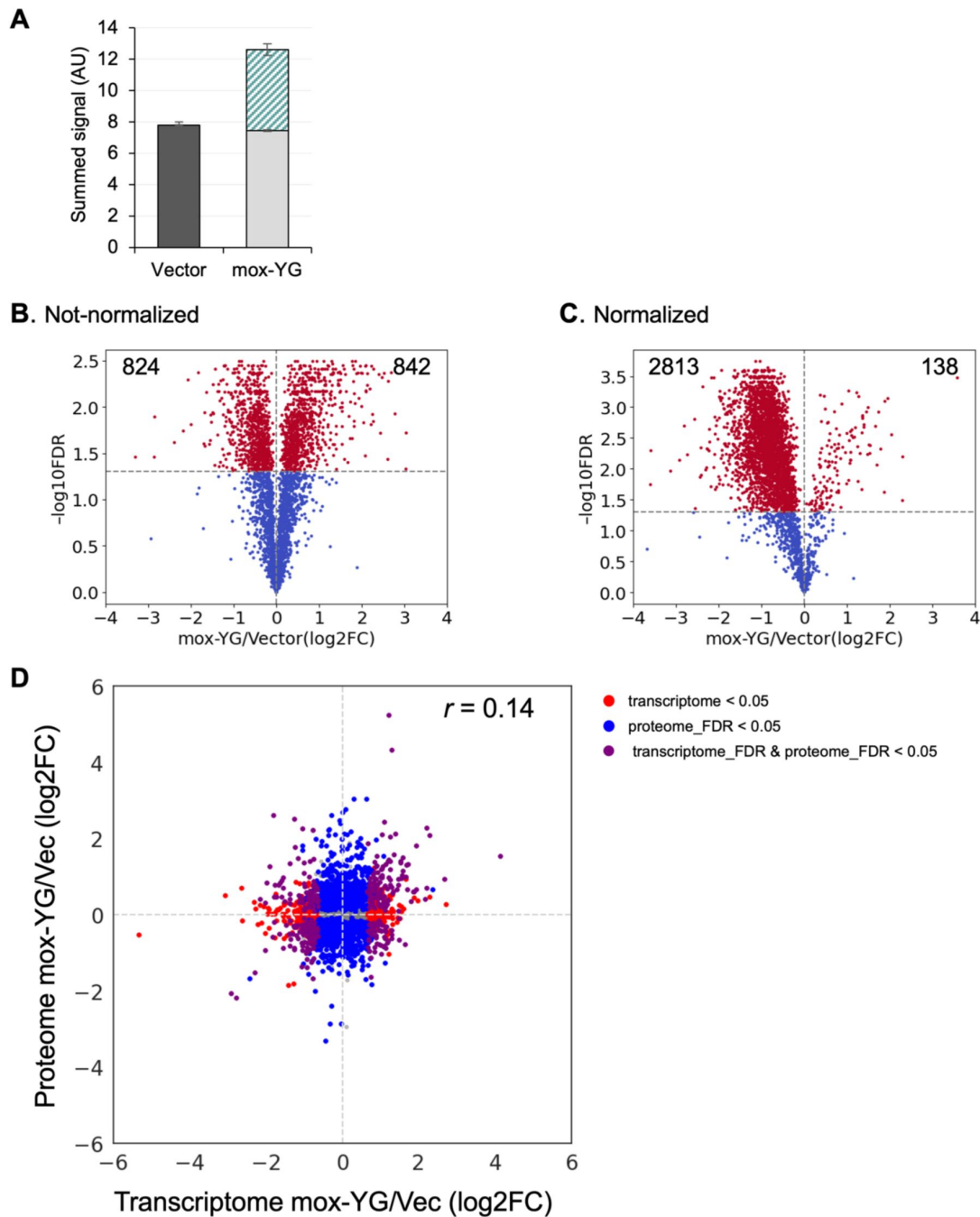


Figure 4—figure supplement 3.

Comparison of normalized or not-normalized proteome results.

A) Bar graph showing the sum of signals for each gene detected by proteome analysis. Black or light gray bars indicate the sum of signals other than mox-YG, and green-shaded bars indicate mox-YG signals. The error bar indicates the SD calculated from three biological replicates. B) Volcano plot of proteome results excluding proteins not detected in either condition. C) Volcano plot of B's results normalized so that the sum of mox-YG signals equals the sum of the vector control signals. Significantly changed proteins (FDR < 0.05) are shown in red. The number of genes whose expression decreased or increased during mox-YG overexpression compared to the vector control is indicated in the upper left or upper right of the graph. D) Comparison of proteome and transcriptome changes in cells overexpressing mox-YG. The transcriptome data used are the same as those in Figure 3—figure supplement 1B, and the proteome data are the same as those in Figure 4—figure supplement 3B. The Pearson correlation coefficient (r) is shown.

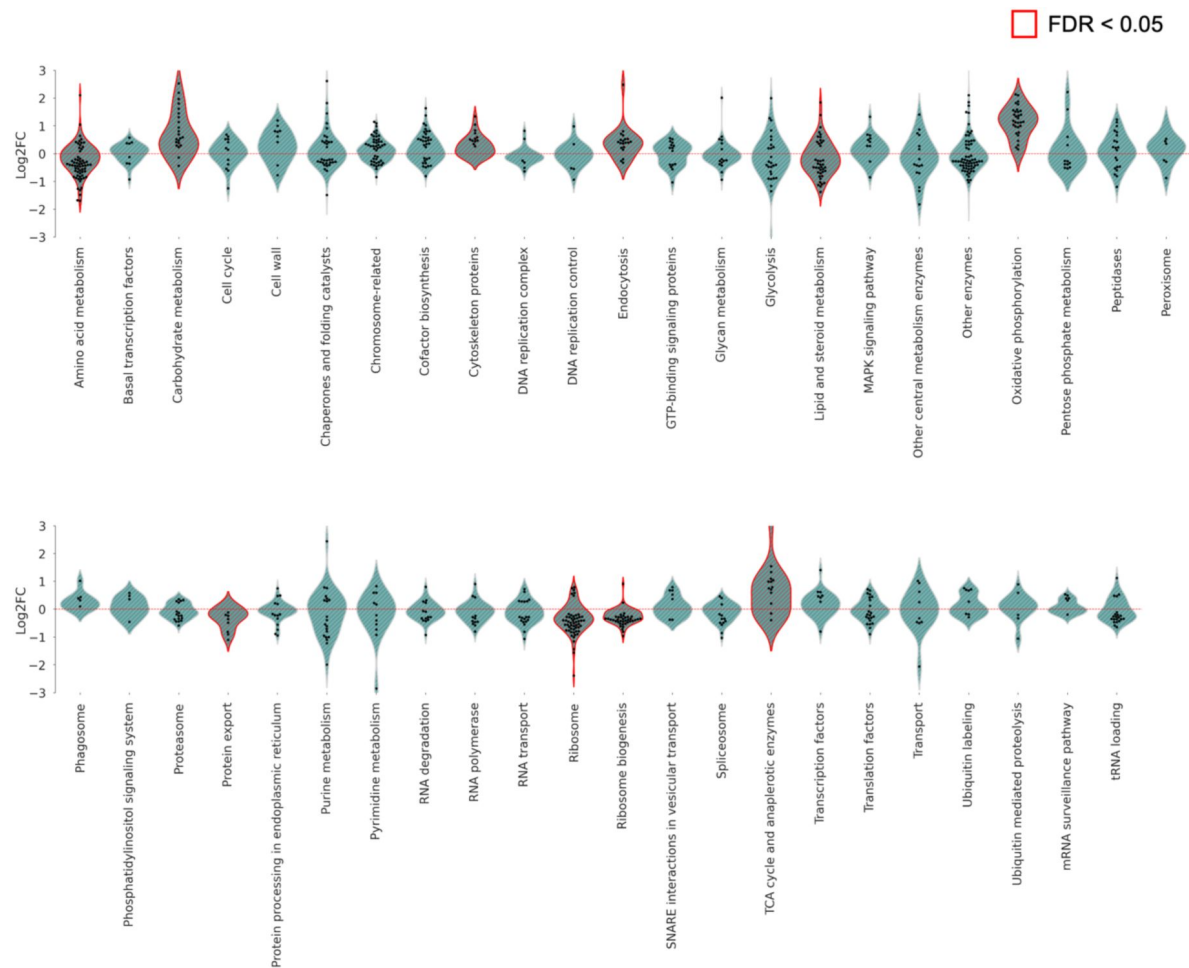


Figure 4—figure supplement 4.

Violin plots showing groups of proteins whose expression was changed by mox-YG overexpression.

Protein groups with significantly altered expression ($FDR < 0.05$) upon mox-YG overexpression over the vector control are shown in red violin plots. Proteins with significant expression changes ($FDR < 0.05$) within the same category are shown by swarm plots.

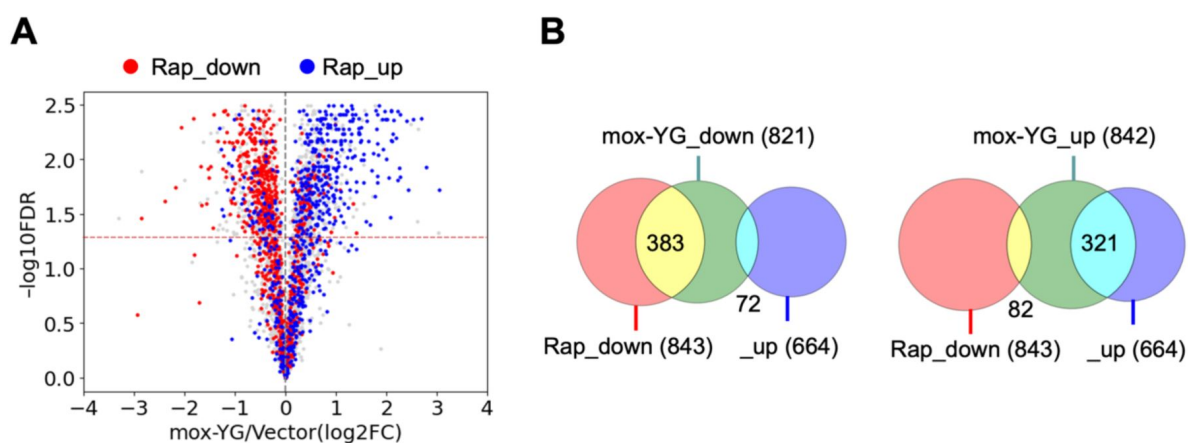


Figure 4—figure supplement 5.

Cells overexpressing mox-YG exhibit proteomic changes that overlap with those observed in rapamycin-treated cells.

A) Volcano plots showing genes whose expression levels were altered by rapamycin treatment in the proteome results of mox-YG overexpressing cells. B) Venn diagram representing the overlap of the proteome during mox-YG overexpression and rapamycin treatment. The number of genes in each area is shown. Published data (Gowans et al., 2018) was used for comparison with the rapamycin response genes.

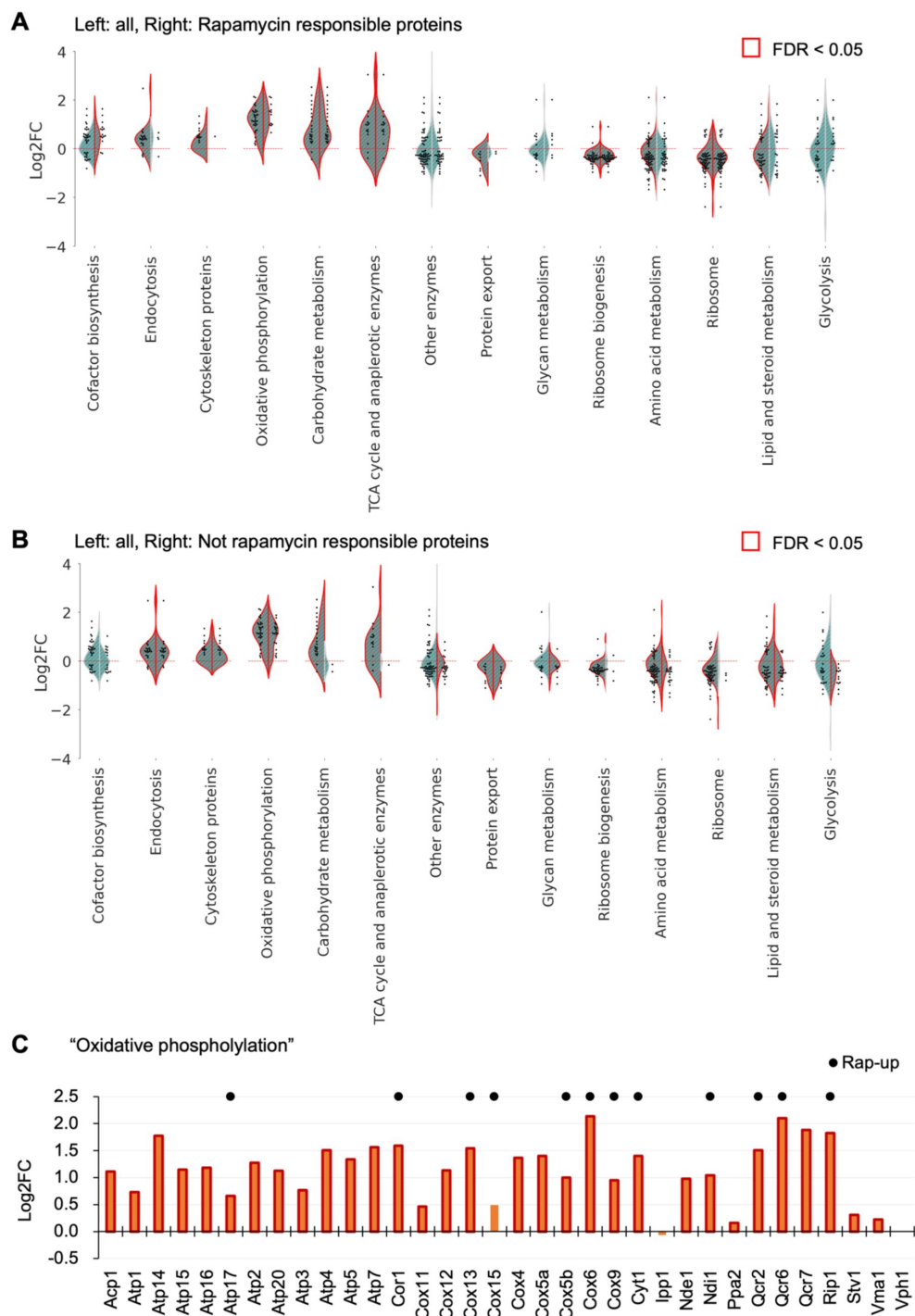


Figure 4—figure supplement 6.

Comparison of groups of proteins whose expression levels are changed by mox-YG overexpression and rapamycin treatment.

A, B) Violin plots showing protein expression changes upon mox-YG overexpression for each KEGG ontology level 3 category. Plots with red borders represent categories that changed significantly (FDR < 0.05). Proteins with significantly (FDR < 0.05) altered expression within a category are shown in the swarm plot. The left side of the violin plot shows the analysis with all the data. The right side of the plot shows the analysis with only rapamycin-responsive protein (A) and rapamycin-non-responsive protein (B). C) Bar graph showing the results for proteins in the "Oxidative phosphorylation" category. Proteins whose expression is also increased during rapamycin treatment (Rap-up) are marked.

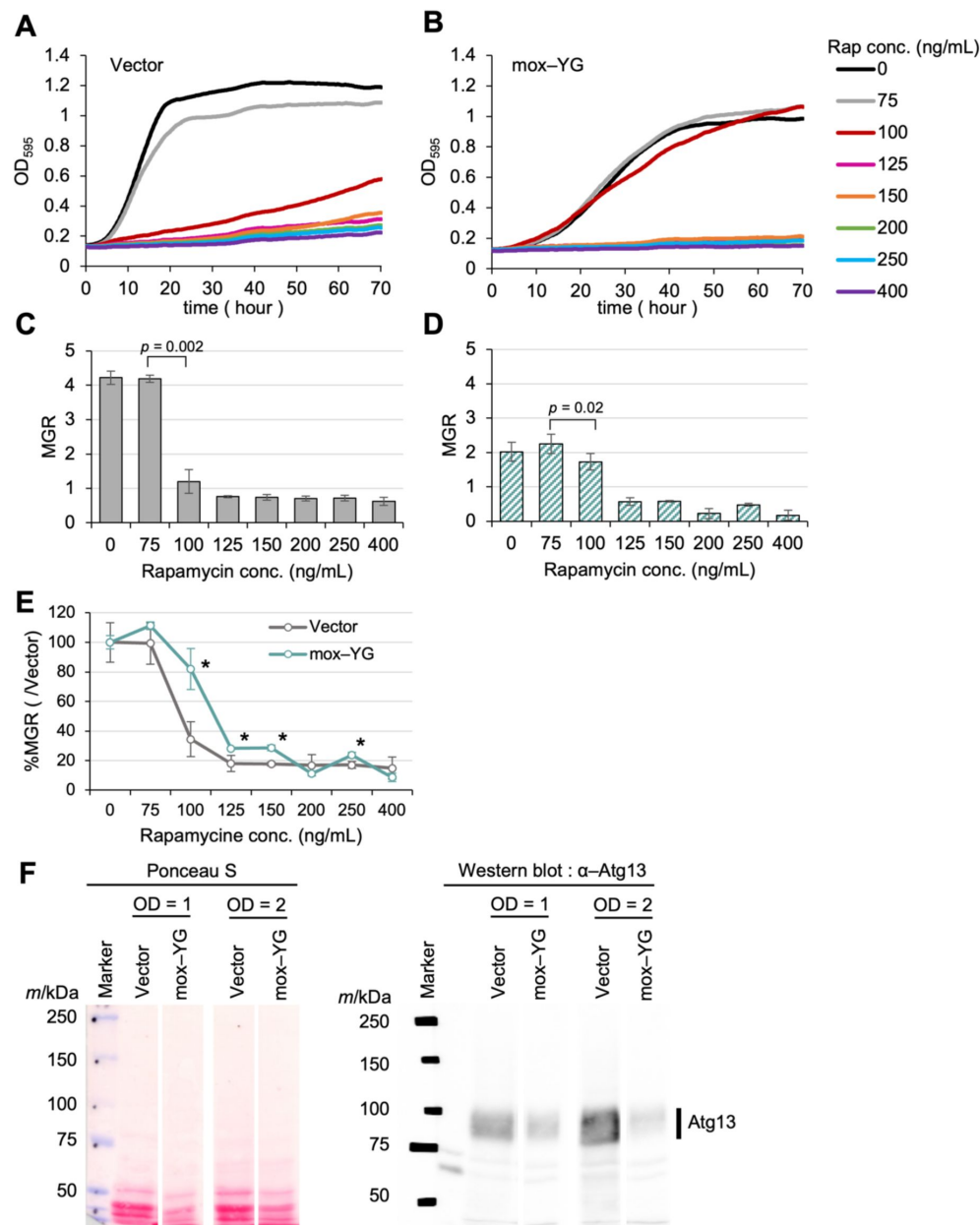


Figure 4—figure supplement 7.

Assessment of TORC1 pathway activity in mox-YG overexpressing cells

A, B) Growth curves of the vector control (A) or mox-YG overexpressing (B) cells under different rapamycin concentrations. The solid lines show the average calculated from four biological replicates. C, D) Max growth rate of the vector control (C) or mox-YG overexpressing (D) cells under different rapamycin concentrations. The bars and error bars show the means and SDs of the MGR calculated from four biological replicates. The p-values were calculated by performing Welch's t-test. E) Line graph showing the change in MGR with increasing rapamycin concentration, using 0 ng/mL of rapamycin concentration as reference. Rapamycin concentrations with significant differences ($p < 0.05$) between the vector control and mox-YG are marked with * above the mox-YG marker. The p-values were calculated by performing Welch's t-test. In the vector control, the addition of 100 ng/mL rapamycin caused significant growth inhibition, whereas in mox-YG overexpressing cells, the inhibitory effect was limited. This highlights a decreased sensitivity to rapamycin in mox-YG overexpressing cells. F) Detection of Atg13 phosphorylation in the vector control and mox-YG overexpressing cells cultured in SC-LeuUra medium to the indicated optical density (OD₆₆₀). Total protein was visualized by ponceau S staining (left), and Atg13 was detected by Western blotting using an anti-Atg13 antibody (right). Atg13 phosphorylated by TORC1 appears as a broad band (indicated by a line).

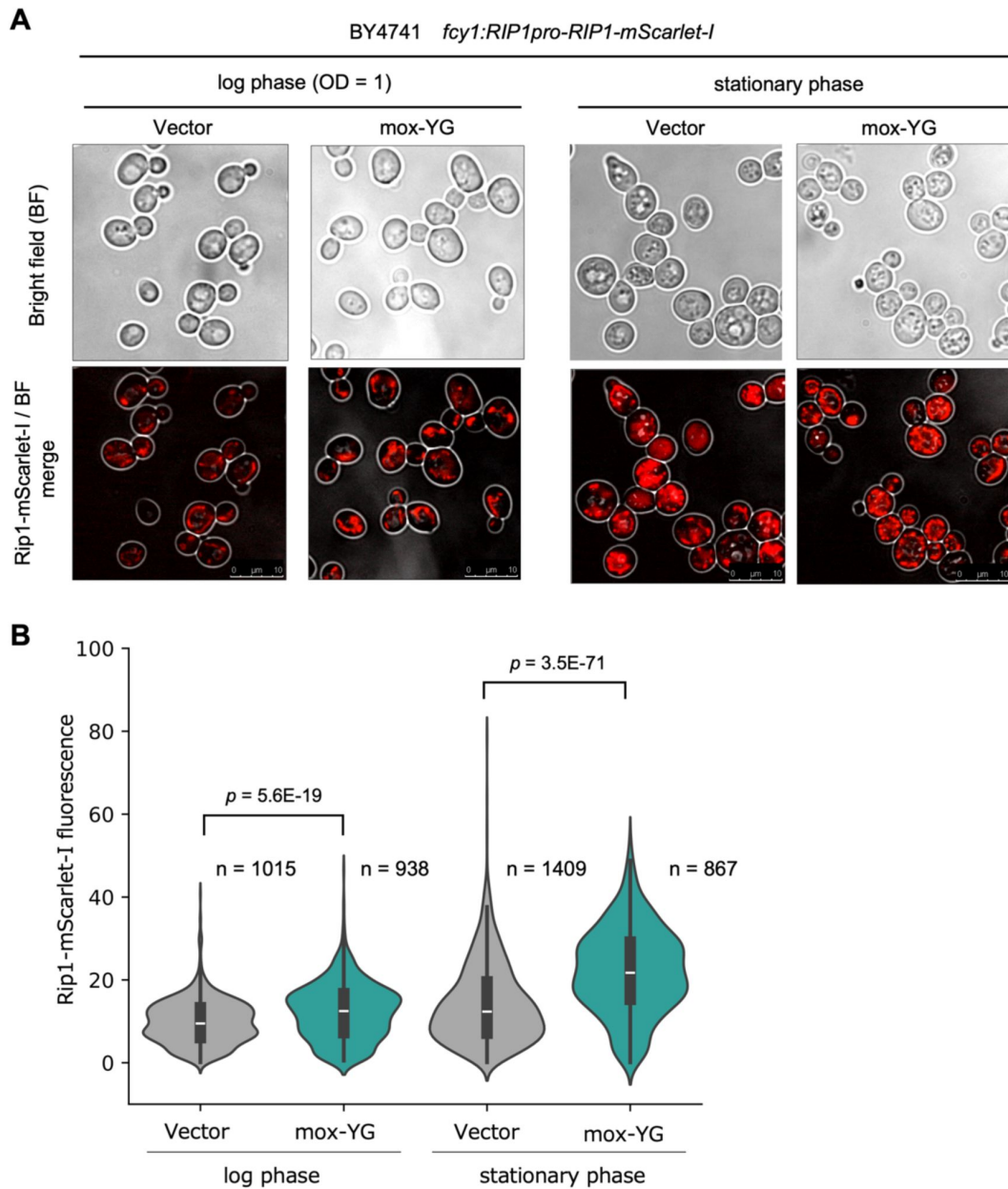


Figure 4—figure supplement 8.

Observation of mitochondria in mox-YG overexpressing cells

A) Fluorescence microscopy image showing the localization of Rip1-mScarlet-I. The vector control and mox-YG overexpressing cells were observed in the log phase and stationary phase. The upper panel is a bright-field (BF) image, and the lower panel is a merged image of the BF and fluorescence images. B) Violin plots showing the quantified fluorescence values of Rip1-mScarlet-I from fluorescence microscopy images taken in the log phase and stationary phase. The number (n) of observed cells is shown. The p-values were calculated by Welch's t-test.

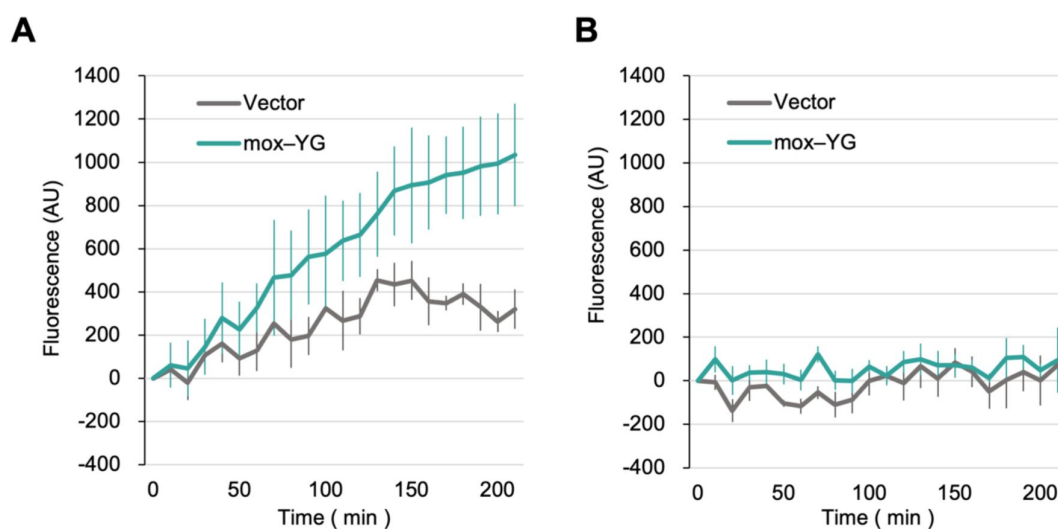


Figure 4—figure supplement 9.

Time-course analysis of oxygen consumption in mox-YG overexpressing cells.

A, B) Line graphs showing changes in the fluorescence of an oxygen-sensitive probe, measured in synthetic medium (A) and in medium containing 50 μ M antimycin A (B). Solid lines represent the mean values from three biological replicates; Error bars indicate standard deviations.

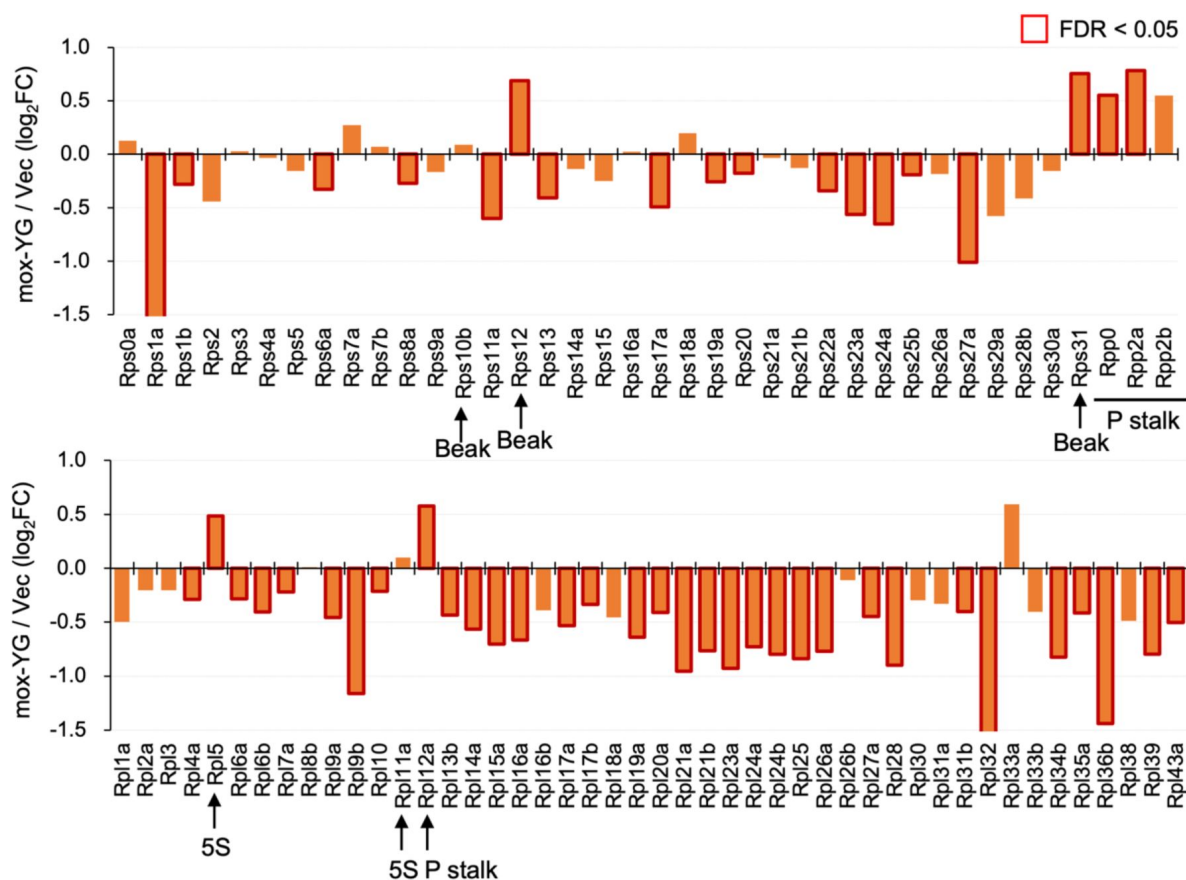


Figure 5—figure supplement 1.

Expression changes of ribosomal proteins upon mox-YG overexpression.

Bar graphs show the changes in the expression of ribosomal proteins in the proteome analysis of Figure 4. Significantly changed proteins are boxed in red. Structure names containing increased proteins are shown. Note that when the amino acid sequences of paralogs are identical, they cannot be distinguished by proteomic analysis, and the protein abundance of both members of the paralog pair is represented under the name of only one.

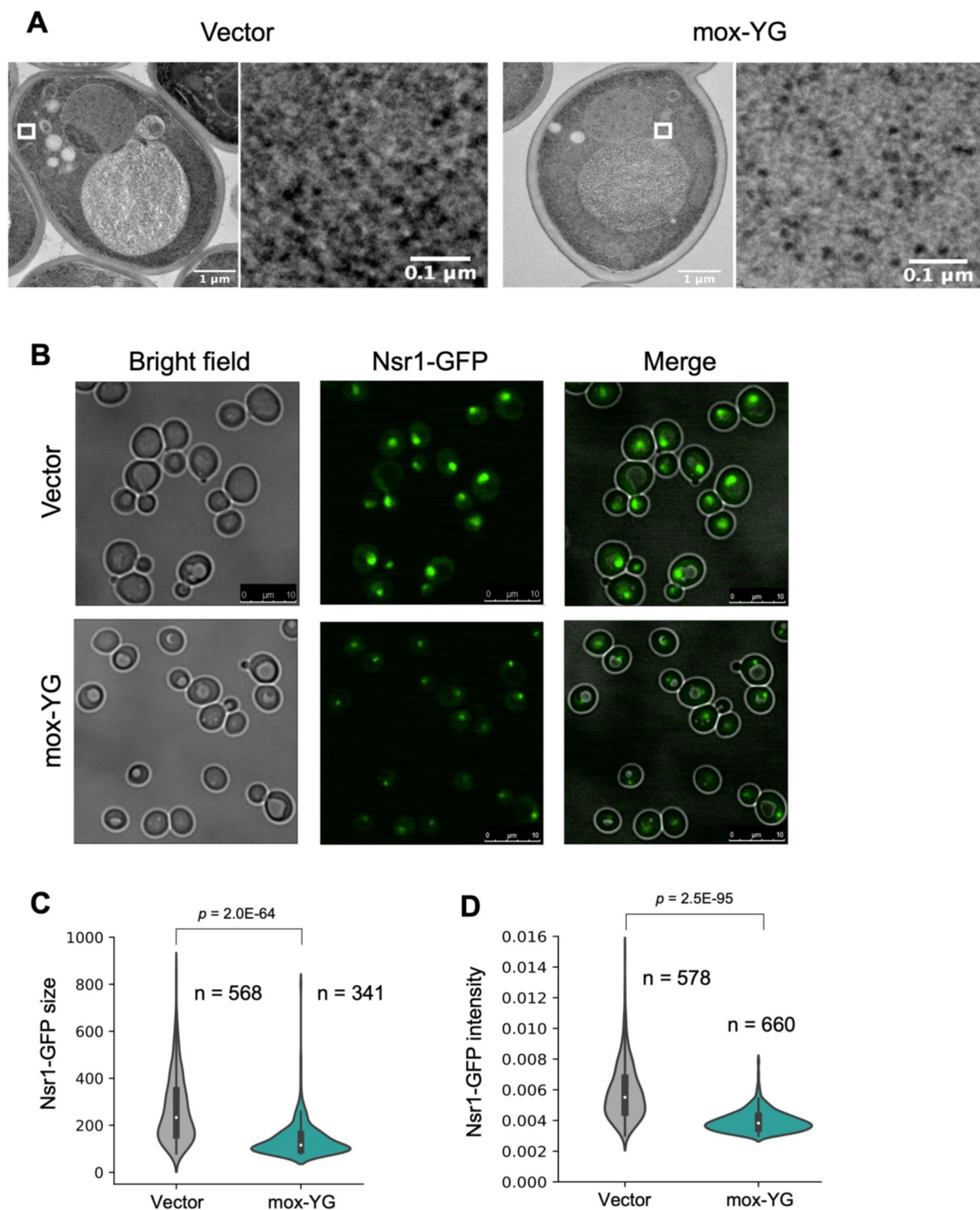
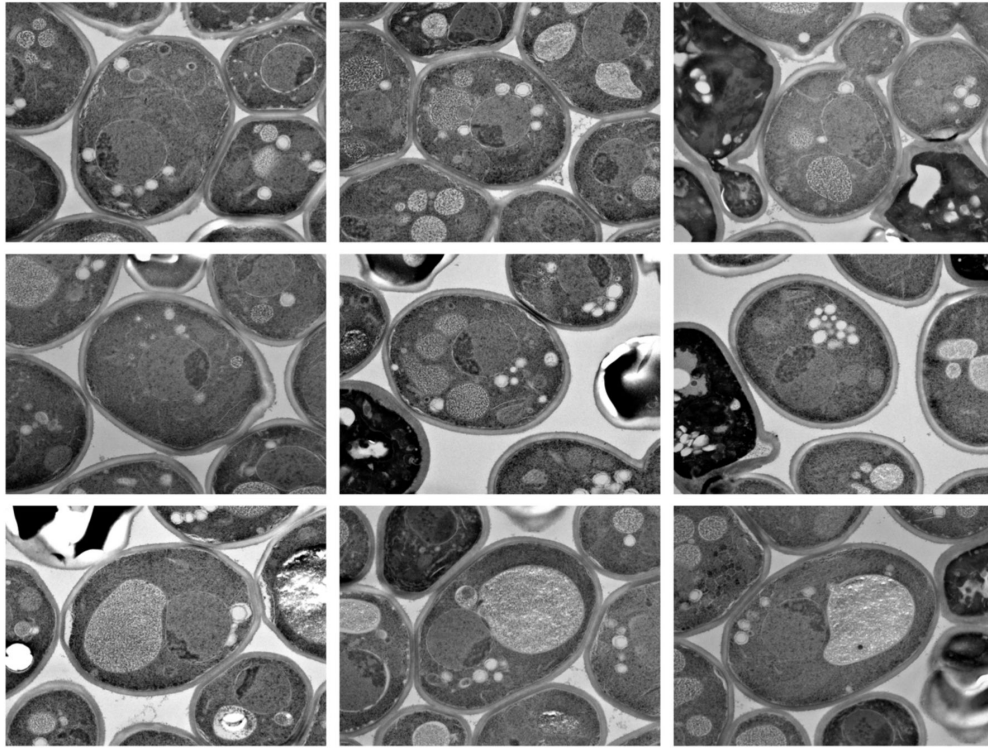


Figure 5—figure supplement 2.

Microscopic analysis of the cells under mox-YG overexpression.

A) Electron microscopic images of the cells with the vector or under mox-YG overexpression. The white rectangular area in each left image is magnified to show the cytoplasmic density (shown in each right image). B) Fluorescence microscopy image showing the size of nucleolus with Nsr1-EGFP. Log phase cells ($OD_{660} = 1.0$) with the vector or under mox-YG overexpression were observed. C, D) Quantification of the size of Nsr1-GFP fluorescent region (C) and fluorescence value of Nsr1- GFP (D) in the cells with the vector or mox-YG overexpression. The number (n) of observed cells is shown. The p-values were calculated by Welch's t-test.

A Vector



B *mox*-YG

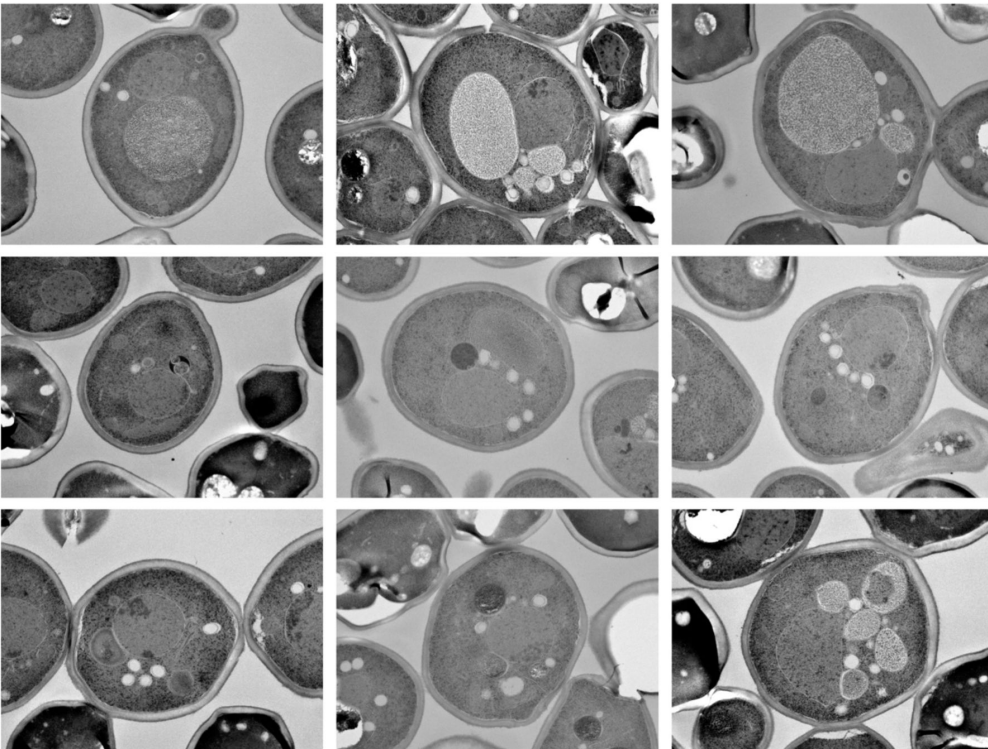


Figure 5—figure supplement 3.

Electron microscopic images of the cells with the vector (A) or under *mox*-YG overexpression (B).

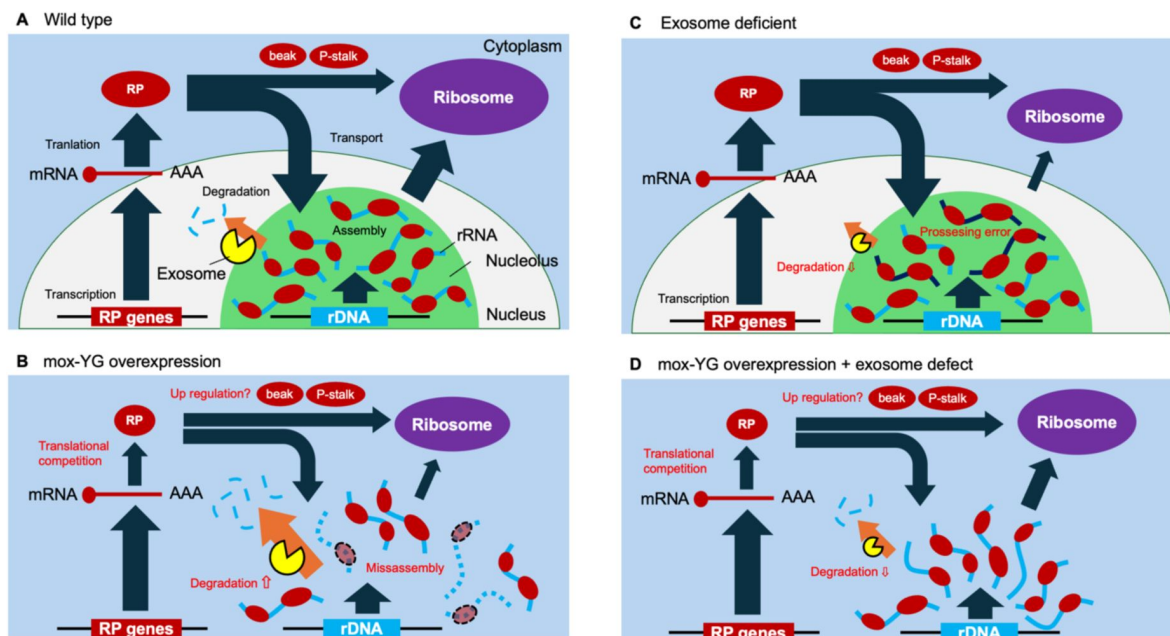


Figure 5—figure supplement 4.

Model diagrams showing the hypothetical situation of the cells.

A) In the wild type cells, enough ribosomal proteins (RPs) are produced, resulting in RP-assembled rRNAs and the formation of nucleolus with normal morphology. B) In *mox-YG* overexpressing cells, the amount of RP is reduced due to translation competition, increasing misassembled rRNA. As a result, degradation of rRNA by exosomes may be accelerated, resulting in abnormal nucleolus morphology and reduction of ribosomes. C) In exosome-deficient cells like the *mtr4-1* cells, errors in rRNA processing occur due to the reduction of degradation by exosomes, resulting in the reduction of ribosomes and growth defects. D) In *mox-YG* overexpressing, exosome-deficient cells, degradation of misassembled rRNA is reduced, and nucleolus formation and ribosome biosynthesis are partially recovered.

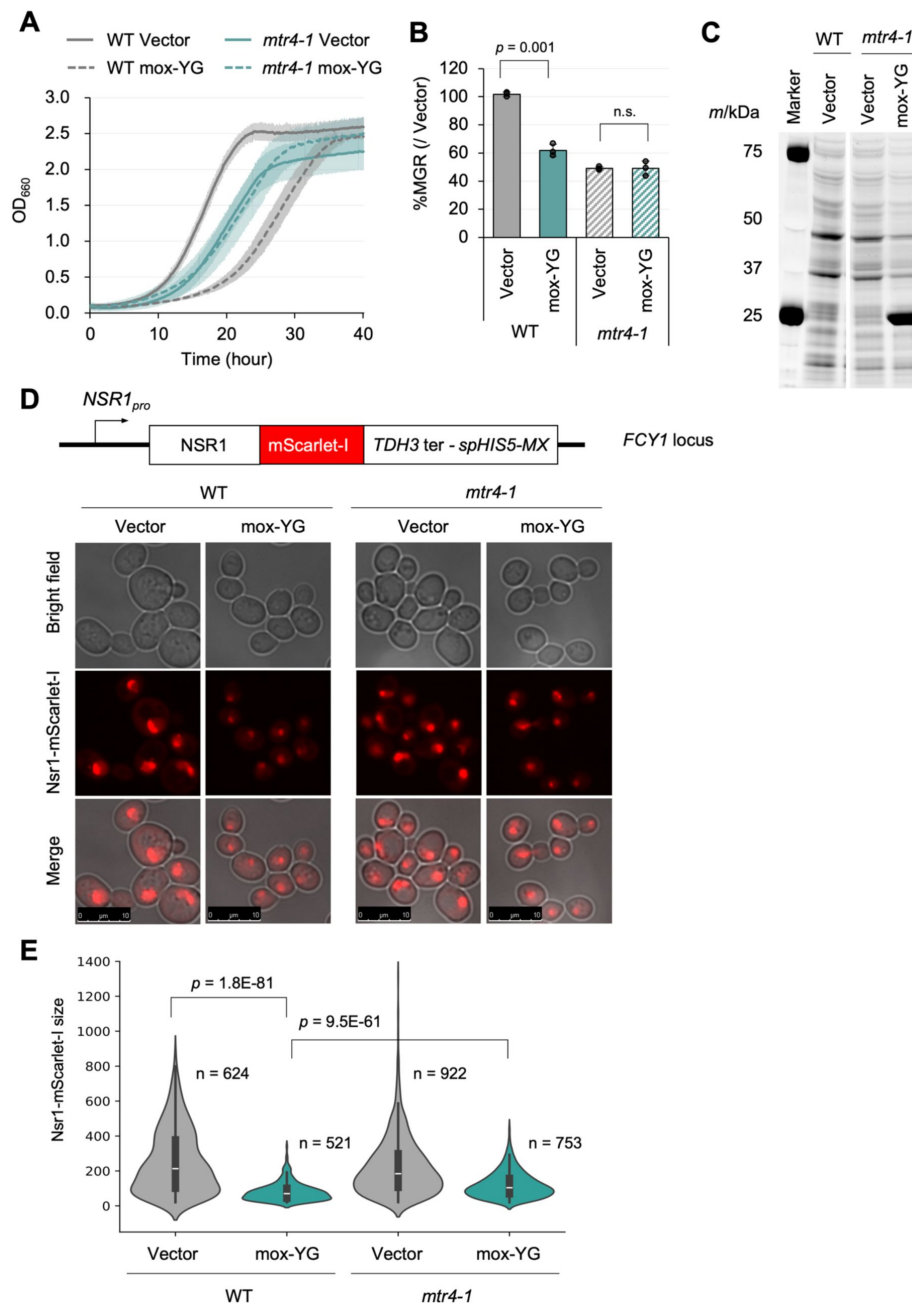


Figure 5—figure supplement 5.

Positive genetic interaction between exosome mutation (*mtr4-1*) and *mox*-YG overexpression

A) Growth curves of the wild type (WT) and *mtr4-1* mutant cells with the vector or under overexpression of *mox*-YG in SC-LeuUra medium at 30°C. The solid or dotted lines show the average calculated from three biological replicates. Shaded zones indicate their SDs. B) MGR of the wild type (WT) and *mtr4-1* mutant cells with the vector or under overexpression of *mox*-YG (percent over the vector control). The bars and error bars show the means and SDs calculated from three biological replicates. The raw data is shown with dot plots. The p-value of Welch's t-test is shown. n.s.: $p > 0.05$. C) Gel images of SDS-PAGE-separated proteins extracted from the wild type (WT) and *mtr4-1* mutant cells with the vector or under overexpression of *mox*-YG. All proteins were separated after staining with a fluorescent dye. D) Observation of nucleolus. The construct to observe nucleolus (Nsr1-mScarlet-I fusion gene) was integrated into the genomic *FCY1* locus. Fluorescence microscopy image of nucleolus-localized protein Nsr1-mScarlet-I of the WT and *mtr4-1* mutant cells with the vector or *mox*-YG overexpression. E) Quantification of the size of Nsr1-mScarlet-I fluorescent region in WT and *mtr4-1* cells with the vector or *mox*-YG overexpression. The number (n) of observed cells is shown. The p-values were calculated by Welch's t-test.

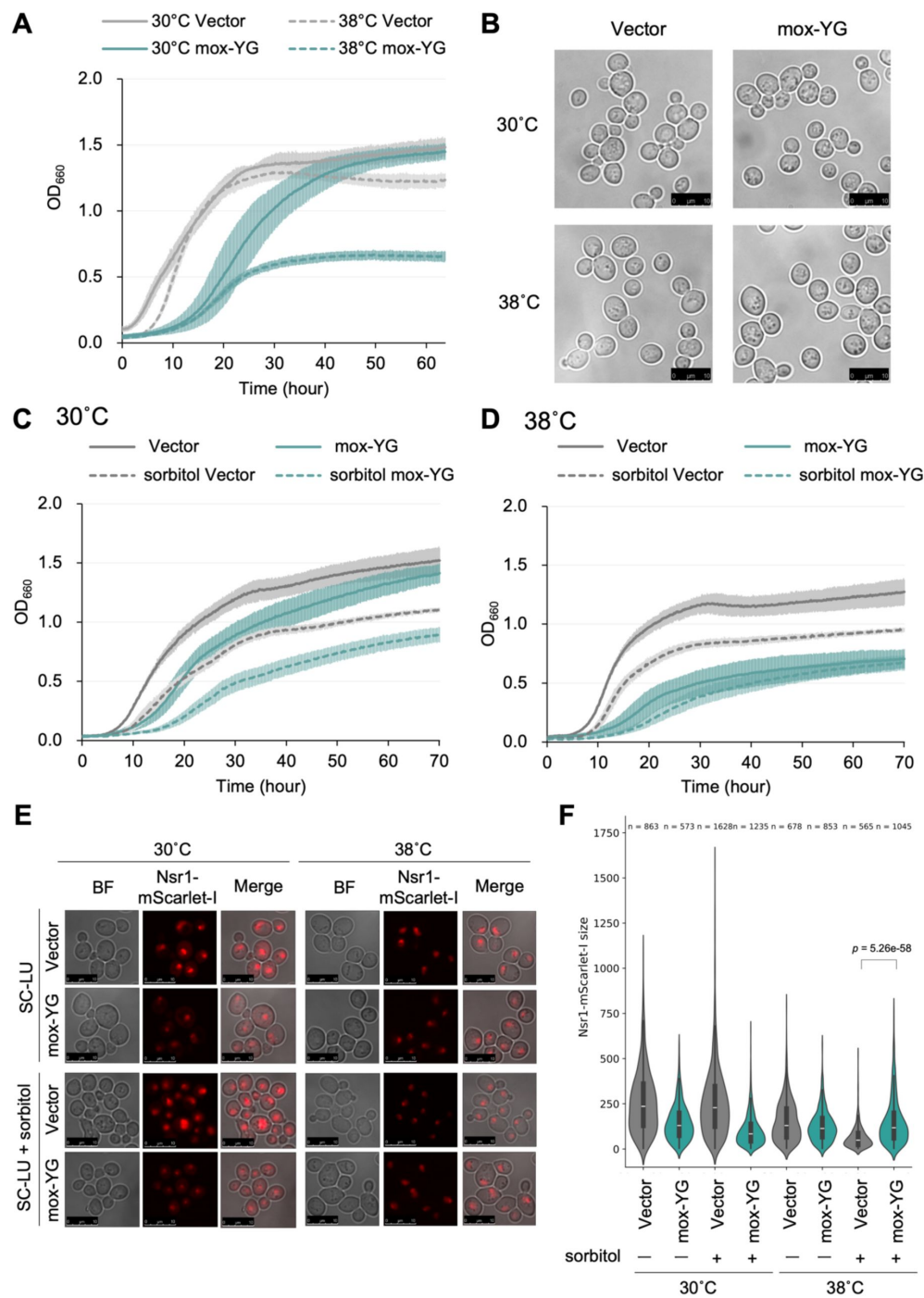


Figure 5—figure supplement 6.

Comparison of growth of glycolytic enzymes overexpressing strains.

A, C, D) Growth curves of cells with the vector or under mox-YG overexpression in SC-LeuUra medium at the indicated temperature. In the indicated culture, sorbitol (1 M) is added. The solid or dotted lines show the average calculated from three biological replicates. Shaded zones indicate their standard deviations. B) Bright-field microscopic images of cells with the vector or under mox-YG overexpression cultured at the indicated temperature. E) Fluorescence microscopy image showing the size of the Nsr1-mScarlet-I nucleolus, observed in the vector control or mox-YG over-expressing cells (OD660 = 1.0) when cultured at 38°C or in sorbitol- enriched medium. F) Quantification results of the Nsr1-mScarlet-I regions observed in the eight conditions indicated in E. Number of cells observed (n) is shown. The p-value was calculated by the Mann-Whitney U-test.

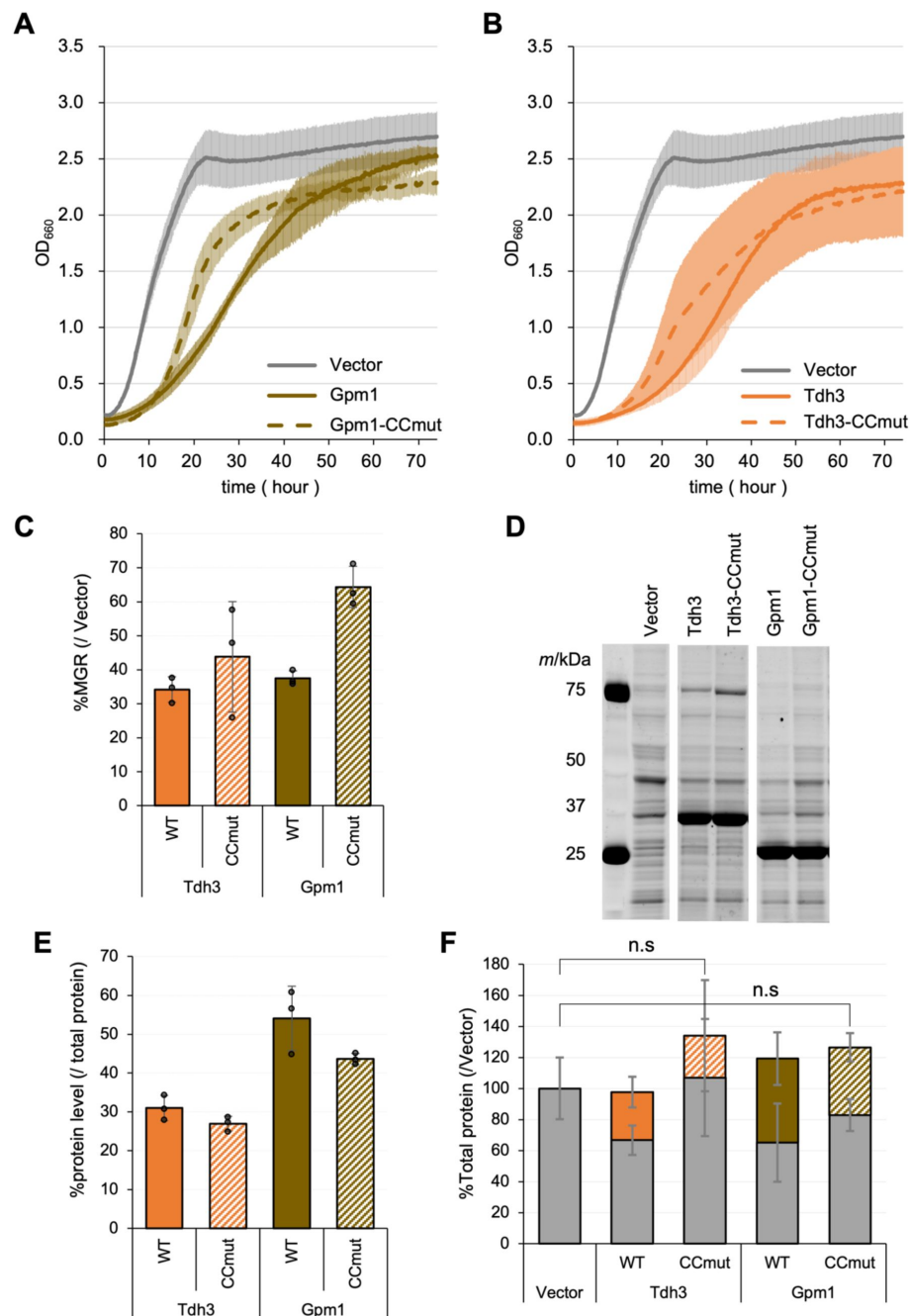


Figure 6—figure supplement 1.

Evaluation of the neutrality of Tdh3 and Gpm1.

A, B) Growth curves of cells with the vector (Vector) and overexpressing Gpm1 and Tdp3 and their catalytic center mutants (CCmut) in synthetic medium (–LeuUra). The solid or dotted lines show the average calculated from three biological replicates. Shaded zones indicate their SD. C) MGR of cells overexpressing indicated proteins (percent over the vector control). D) Gel images of SDS-PAGE-separated proteins extracted from cells overexpressing indicated proteins. All proteins were separated after staining with a fluorescent dye. E) Protein levels of the target protein (percent over the total protein). The amount was calculated from the intensity of the target bands separated by SDS-PAGE of D. F) Ratio of total protein content of cells overexpressing glycolytic enzymes over the vector control. Gray bars and error bars indicate expression levels and SDs for proteins other than glycolytic enzymes; non-gray bars and error bars indicate expression levels and SDs for each overexpressed protein. The bars and error bars in C, E, and F show the means and SDs calculated from three biological replicates. The raw data is shown with dot plots.

Gpm1

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1 M P K L V L V R H G Q S E W N E K N L F 20
959 ATGCCAAAGTAGTTTGTAGTACACGGTCAATCCGAATGGAACGAAAGAACTTATTC 1018

21 T G W V D V K L S A K G Q Q E A A R A G 40
1019 ACCGGTTGGGTGTGTTAAATTGTCTGCCAAGGGTCAACAAGAAGCCGCTAGAGCCGGT 1078

41 E L L K E K K V Y P D V L Y T S K L S R 60
1079 GAATTGTTGAAGGAAAAGAGTCTACCCAGACGCTTGTACACTTCCAAGTTGTCCAGA 1138

61 A I Q T A N I A L E K A D R L W I P V N 80
1139 GCTATCCAACTGCTAACATTGCTTTGAAAAGGCTGACAGATTATGGATTCCAGTCAAC 1198

81 R S W R L N E R H Y G D L Q G K D K A E 100
1199 AGATCCTGGAGATTGAACGAAAGACATTACGGTGACTTACAAGGTAAGGACAAGGCTGAA 1258

101 T L K K F G E E K F N T Y R R S F D V P 120
1259 ACTTTGAAGAAGTTCGGTGAAGAAAATTCAACACCTACAGAAGATCCTTCGATGTTCCA 1318

121 P P P I D A S S P F S Q K G D E R Y K Y 140
1319 CCTCCCCAATCGACGCTTCTTCCATTCTCTCAAAGGGTGATGAAAGATACAAGTAC 1378

141 V D P N V L P E T E S L A L V I D R L L 160
1379 GTTGACCCAAATGCTTGGCAGAACTGAATCTTTGGCTTGGTCATTGACAGATTGTTG 1438

161 P Y W Q D V I A K D L L S G K T V M I A 180
1439 CCATCTGGCAAGATGTCATTGCCAAGGACTTGTGAGTGGTAAGACCGTCATGATCGCC 1498

181 A A G N S L R G L V K H L E G I S D A D 200
1499 GCTgCGGTAAGTCTTGAAGGTTGGTTAAGCACTTGAAGGTATCTCTGATGCTGAC 1558

201 I A K L N I P T G I P L V F E L D E N L 220
1559 ATTGCTAAGTTGAACATCCCACTGGTATTCATTGGTCTTCAATTTGACGAAACTTG 1618

221 K P S K P S Y Y L D P E A A A A G A A A 240
1619 AAGCCATCTAAGCCATCTTACTACTTGGACCCAGAAGCTGCCGCTGCTGGTCCGCTGCT 1678

241 V A N Q G K K * 248
1679 GTTGCCAACCAAGGTAAGAAATAA 1702

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Tdh3

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1 M V R V A I N G F G R I G R L V M R I A 20
959 ATGGTTAGAGTTGCTATTACGGTTTCGGTAGAATCGGTAGATTGGTCATGAGAATTGCT 1018

21 L S R P N V E V V A L N D P F I T N D Y 40
1019 TTGCTAGACCAAACGTGGAAGTTGTGCTTTGAACGACCATTCATCACAACGACTAC 1078

41 A A Y M F K Y D S T H G R Y A G E V S H 60
1079 GCTGCTTACATGTTCAAGTACGACTCCACTCAGGTAGATACGCTGGTGAAGTTTCCAC 1138

61 D D K H I I V D G K K I A T Y Q E R D P 80
1139 GATGACAAGCACATCATGTCGATGGTAAGAAGATTGCTACTTACCAAGAAAGAGACCCA 1198

81 A N L P W G S S N V D I A I D S T G V F 100
1199 GCTAACTTGCATGGGGTCTTCCAACGTTGACATCGCCATTGACTCCACTGGTGTTC 1258

101 K E L D T A Q K H I D A G A K K V V I T 120
1259 AAGGAATTAGACACTGCTCAAAAGCATTGACGCTGGTGCCAAGAAGTTGTTATCACT 1318

121 A P S S T A P M F V M G V N E E K Y T S 140
1319 GCTCATCTTCCACCGCCCAATGTCGTCATGGGTGTTAACGAAGAAAAATACACTTCT 1378

141 D L K I V S N A S S T T N C L A P L A K 160
1379 GACTTGAAGATTGTTCCAACGCTTCTcTACCACCACTGTTGGCTCCATTGGCCAAG 1438

161 V I N D A F G I E E G L M T T V H S L T 180
1439 GTTATCAACGATGCTTTTCGTATTGAAGAAGGTTTGATGACCACTGTCCACTCTTTGACT 1498

181 A T Q K K T V D G P S H K D W R G G R T A 200
1499 GCTACTCAAAAGACTGTTGACGGTCCATCCACAAGGACTGGAGAGTGGTAGAACCGCT 1558

201 S G N I I P S S T G A A K A V G K V L P 220
1559 TCCGGTAACATCATCCATCTCCACCGGTGCTGTAAGGCTGCGTAAGGTTAGGCTTGGCA 1618

221 E L Q G K L T G M A F R V P T V D V S V 240
1619 GAATTGAAGTAAGTTGACCGGTATGGCTTTCAGAGTCCCAACCGTCGATGTCTCCGTT 1678

241 V D L T V K L N K E T T Y D E I K K V V 260
1679 GTTGACTTGACTGTCAAGTTGAACAAGGAAACCACTACGATGAAATCAAGAAGTTGTT 1738

261 K A A A E G K L K G V L G Y T E D A V V 280
1739 AAGGCTCCGCTGAAGGTAAGTTGAAGGTTGTTTGGGTTACACCAAGACGCTGTTGTC 1798

281 S S D F L G D S H S S I F D A S A G I Q 300
1799 TCCTCTGACTTCTGGGTGACTCTCACTCTTCCATCTTCGATGCTTCCGCTGGTATCCAA 1858

301 L S P K F V K L V S W Y D N E Y G Y S T 320
1859 TTGCTCCTAAAGTTGCTCAAGTTGGTCTCCTGGTACGACACGAATACGGTTACTCTACC 1918

321 R V V D L V E H V A K A * 333
1919 AGAGTTGTCGACTTGGTTGAACACGTTGCCAAGGCTTAA 1957

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Figure 6—figure supplement 2.

Nucleotide and amino acid sequences of Gpm1 and Tdh3.

The amino acid substitutions introduced to eliminate enzyme activity are shown in red. The H182A mutation in Gpm1 was referred to the study by White et al. (White and Fothergill-Gilmore, 1992 [\[1\]](#)), and the C150S mutation in Tdh3 was referred to the study by Peralta et al. (Peralta et al., 2015 [\[2\]](#)).

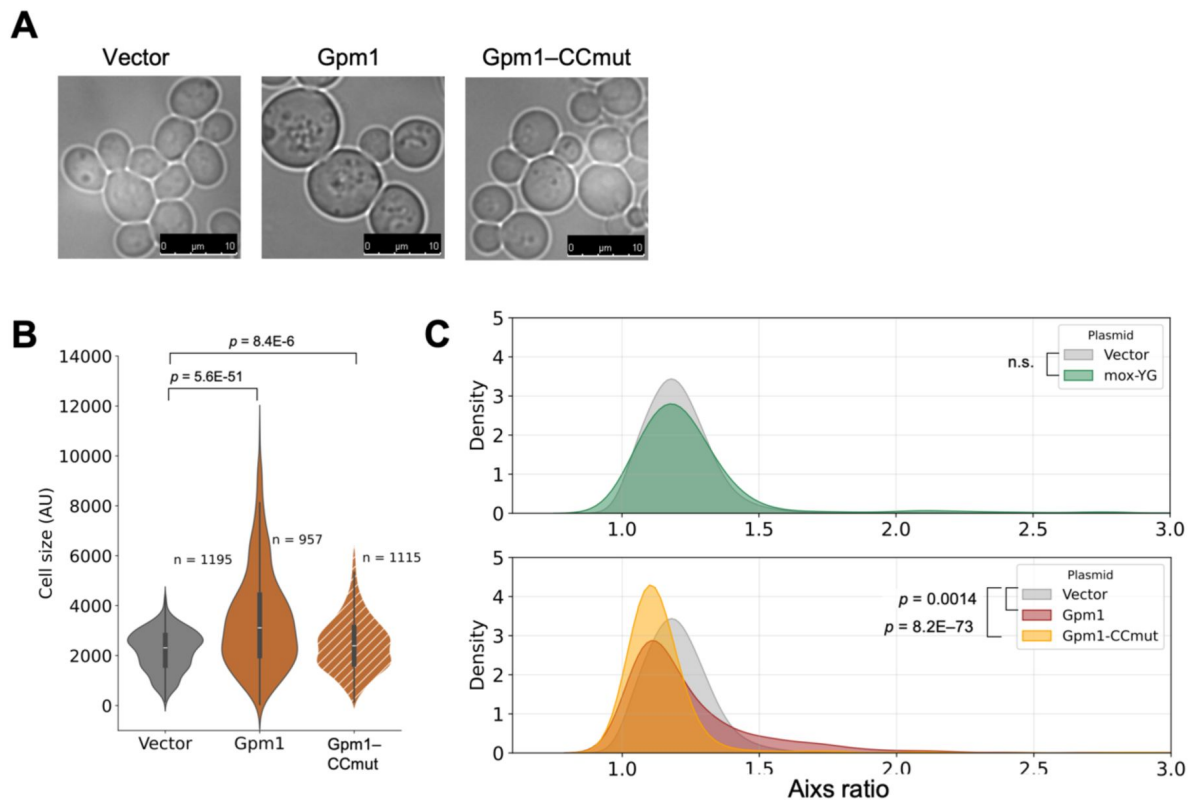


Figure 6—figure supplement 3.

Overexpression of Gpm1 and its CC-mutant induces abnormal cell morphology.

A) Representative microscopic images of cells overexpressing Gpm1 or Gpm1-CCmut. B) Violin plots showing cell size quantified from microscopic images. The number of cells used for the quantification is shown. The p-values were calculated by the Mann-Whitney U-test. C) Distribution of cell axis ratios upon overexpression of mox-YG, Gpm1, and Gpm1-CCmut, compared with the vector control. The p-values were calculated by the Mann-Whitney U-test. "n.s." means $p > 0.05$.

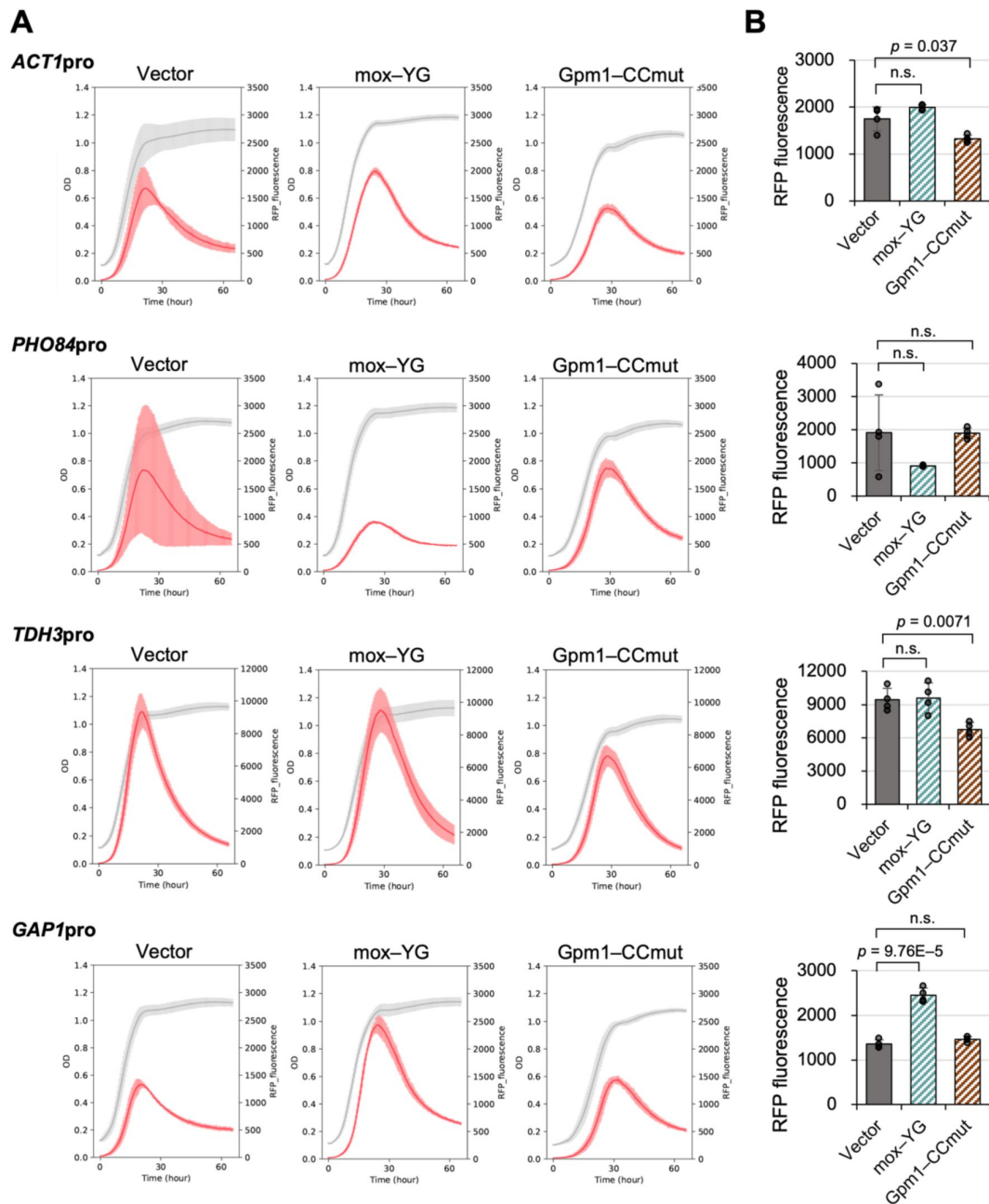


Figure 6—figure supplement 4.

Transcriptional response under mox-YG and Gpm1 CC-mutant overexpression.

A) Curves representing time series data of cell growth (gray, OD595) and mScarlet-I fluorescence (red, arbitrary unit) overexpressing mox-YG and Gpm1-CCmut in SC-LeuUra + 50 mg/mL leucine medium for each promoter. Solid lines are averages calculated from four biological replicates. B) Quantitative results of mScarlet-I fluorescence values for each promoter. Bars and error bars indicate the mean and SD of maximum fluorescence values (arbitrary unit) for mScarlet-I calculated from four biological replicates. Raw data are shown as dot plots. The p-values were calculated by performing Welch's t-test and applying the Bonferroni correction. "n.s." means $p > 0.05$.

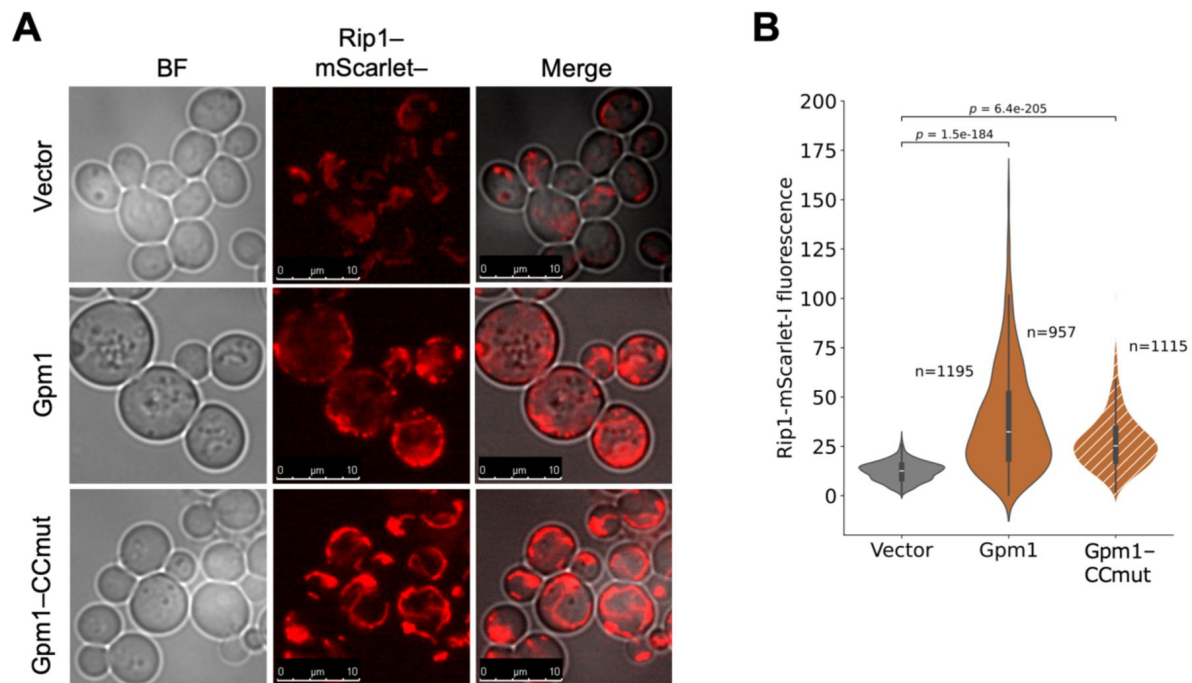


Figure 6—figure supplement 5.

Microscopic observation of mitochondria in cells overexpressing Gpm1 or Gpm1-CCmut.

A) Fluorescence microscopy image showing the size of nucleolus with Rip1-mScarlet-I. Log phase cells (OD₆₆₀ = 1.0) with the vector or under Gpm1 or Gpm1-CCmut overexpression were observed. B) Quantification of the fluorescence value of Rip1-mScarlet-I in the cells with the vector or Gpm1 or Gpm1-CCmut overexpression. The number of cells used for the quantification is shown. The p-values were calculated by Mann-Whitney U test.

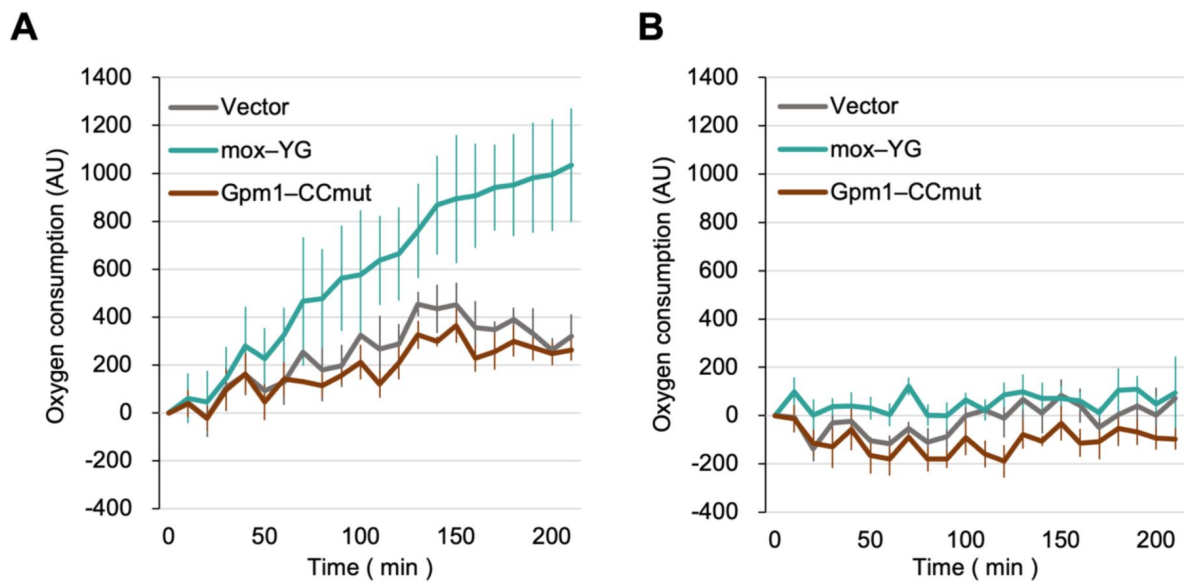


Figure 6—figure supplement 6.

Time-course analysis of oxygen consumption in Gpm1-CCmut overexpressing cells.

A, B) Line graphs showing changes in the fluorescence of an oxygen-sensitive probe, measured in synthetic medium (A) and in medium containing 50 μ M antimycin A (B). Solid lines represent the mean values from three biological replicates; error bars indicate standard deviations.

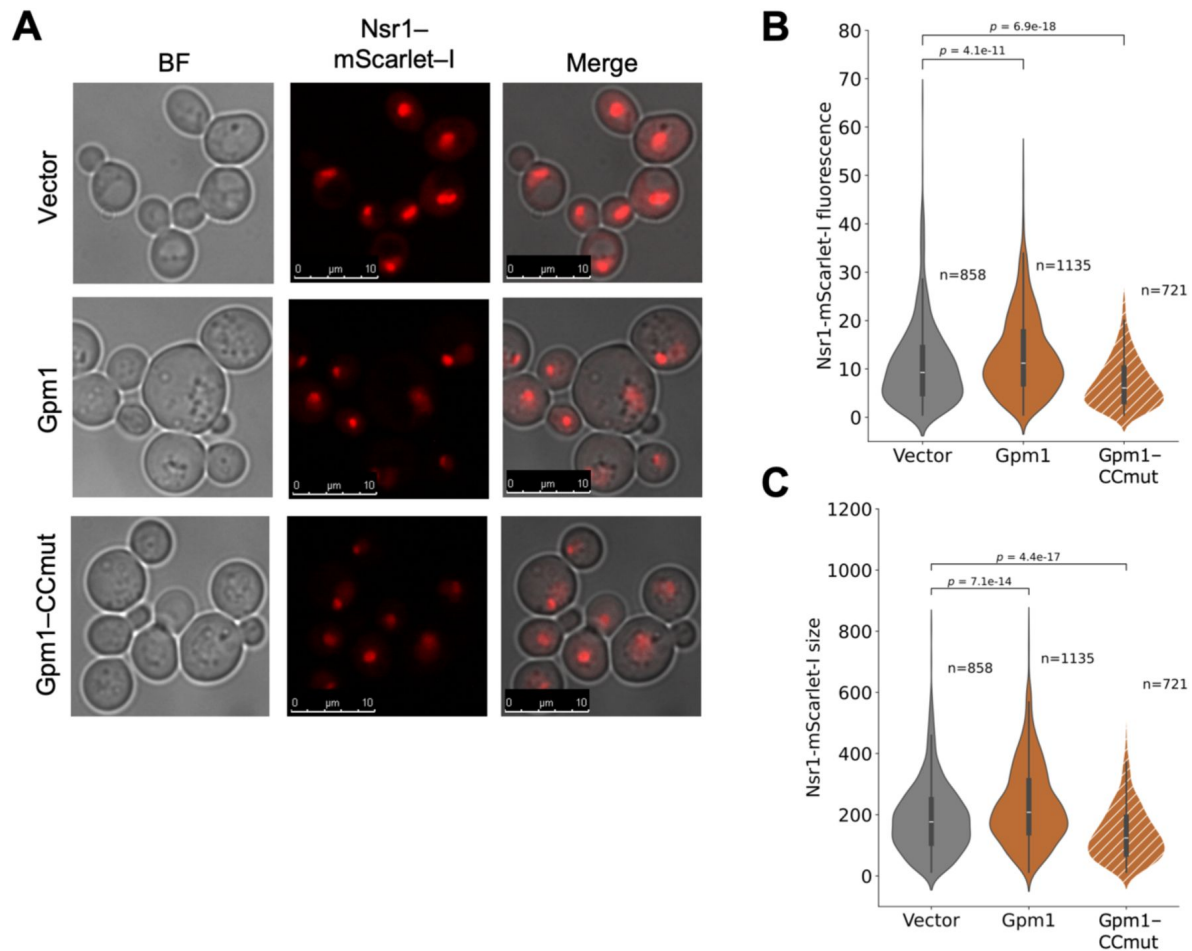


Figure 6—figure supplement 7.

Microscopic observation of nucleoli in cells overexpressing Gpm1 or Gpm1-CCmut.

A) Fluorescence microscopy images showing the size of nucleolus with Nsr1-mScarlet-I. Log phase cells (OD660 = 1.0) with the vector or under Gpm1 or Gpm1-CCmut overexpression were observed. B, C) Quantification of the size of Nsr1-mScarlet-I fluorescent region (B) and fluorescence value of Nsr1-mScarlet-I (C) in the cells with the vector or Gpm1 or Gpm1-CCmut overexpression. The number of cells used for the quantification is shown. The p-values were calculated by Mann-Whitney U test.

Acknowledgements

We thank the members of the Moriya laboratory (Okayama University) for their helpful discussions.

Additional information

Funding

This work was partly supported by the Nagase Science and Technology Foundation, Institute for Fermentation, Osaka (IFO) General Research Funding, JSPS KAKENHI Grant Numbers 22K19294, and 24K02313, JST SPRING Japan Grant Number JPMJSP2126.

Author contribution

YF and HM conceived and designed the experiments, YF and YK performed experiments, YF, SN, and HM analyzed the data, and YF and HM wrote the paper.

Funding

Nagase Science Technology Foundation

Institute for Fermentation

Japan Society for the Promotion of Science (22K19294)

Japan Society for the Promotion of Science (24K02313)

Japan Science and Technology Agency (JPMJSP2126)

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Reviewer #1 (Public Review):

Summary:

The study "Impact of Maximal Overexpression of a Non-toxic Protein on Yeast Cell Physiology" by Fujita et al. aims to elucidate the physiological impacts of overexpressing non-toxic proteins in yeast cells. By identifying model proteins with minimal cytotoxicity, the authors claim to provide insights into cellular stress responses and metabolic shifts induced by protein overexpression.

Strengths:

The study introduces a neutrality index to quantify cytotoxicity and investigates the effects of protein burden on yeast cell physiology. The study identifies mox-YG (a non-fluorescent fluorescent protein) and Gpm1-CCmut (an inactive glycolytic enzyme) as proteins with the lowest cytotoxicity, capable of being overexpressed to more than 40% of total cellular protein while maintaining yeast growth. Overexpression of mox-YG leads to a state resembling nitrogen starvation probably due to TORC1 inactivation, increased mitochondrial function, and decreased ribosomal abundance, indicating a metabolic shift towards more energy-efficient respiration and defects in nucleolar formation.

Weaknesses:

While the introduction of the neutrality index seems useful to differentiate between cytotoxicity and protein burden, the biological relevance of the effects of overexpression of the model proteins is unclear.

<https://doi.org/10.7554/eLife.99572.2.sa3>

Reviewer #2 (Public Review):

Summary:

In this manuscript, Fujita et al. characterized the neutrality indexes of several protein mutants in *S. cerevisiae* and uncovered that mox-YG and Gpm1-CCmut can be expressed as abundant as 40% of total proteins without causing severe growth defects. The authors then looked at the transcriptome and proteome of cells expressing excess mox-YG to investigate how protein burden affects yeast cells. Based on RNA-seq and mass-spectrometry results, the authors uncover that cells with excess mox-YG exhibit nitrogen starvation, respiration increase, inactivated TORC1 response, and decreased ribosomal abundance. The authors further showed that the decreased ribosomal amount is likely due to nucleoli defects, which can be partially rescued by nuclear exosome mutations.

Strengths:

Overall, this is a well-written manuscript that provides many valuable resources for the field, including the neutrality analysis on various fluorescent proteins and glycolytic enzymes, as well as the RNA-seq and proteomics results of cells overexpressing mox-YG. Their model on how mox-YG overexpression impairs the nucleolus and thus leads to ribosomal abundance decline will also raise many interesting questions for the field.

Weaknesses:

The authors concluded from their RNA-seq and proteomics results that cells with excess mox-YG expression showed increased respiration and TORC1 inactivation. I think it will be more convincing if the authors can show some characterization of mitochondrial respiration/membrane potential and the TOR responses to further verify their -omic results.

In addition, the authors only investigated how overexpression of mox-YG affects cells. It would be interesting to see whether overexpressing other non-toxic proteins causes similar

effects, or if there are protein-specific effects. It would be good if the authors could at least discuss this point considering the workload of doing another RNA-seq or mass-spectrum analysis might be too heavy.

<https://doi.org/10.7554/eLife.99572.2.sa2>

Reviewer #3 (Public Review):

Summary:

Protein overexpression is widely used in experimental systems to study the function of the protein, assess its (beneficial or detrimental) effects in disease models, or challenge cellular systems involved in synthesis, folding, transport, or degradation of proteins in general. Especially at very high expression levels, protein-specific effects and general effects of a high protein load can be hard to distinguish. To overcome this issue, Fujita et al. use the previously established genetic tug-of-war system to identify proteins that can be expressed at extremely high levels in yeast cells with minimal protein-specific cytotoxicity (high 'neutrality'). They focus on two versions of the protein *mox*-GFP, the fluorescent version and a point mutation that is non-fluorescent (*mox*-YG) and is the most 'neutral' protein on their screen. They find that massive protein expression (up to 40% of the total proteome) results in a nitrogen starvation phenotype, likely inactivation of the TORC1 pathway, and defects in ribosome biogenesis in the nucleolus.

Strengths:

This work uses an elegant approach and succeeds in identifying proteins that can be expressed at surprisingly high levels with little cytotoxicity. Many of the changes they see have been observed before under protein burden conditions, but some are new and interesting. This work solidifies previous hypotheses about the general effects of protein overexpression and provides a set of interesting observations about the toxicity of fluorescent proteins (that is alleviated by mutations that render them non-fluorescent) and metabolic enzymes (that are less toxic when mutated into inactive versions).

Weaknesses:

The data are generally convincing, however in order to back up the major claim of this work - that the observed changes are due to general protein burden and not to the specific protein or condition - a broader analysis of different conditions would be highly beneficial.

Major points:

(1) The authors identify several proteins with high neutrality scores but only analyze the effects of *mox*/*mox*-YG overexpression in depth. Hence, it remains unclear which molecular phenotypes they observe are general effects of protein burden or more specific effects of these specific proteins. To address this point, a proteome (and/or transcriptome) of at least a *Gpm1-CCmut* expressing strain should be obtained and compared to the *mox*-YG proteome. Ideally, this analysis should be done simultaneously on all strains to achieve a good comparability of samples, e.g. using TMT multiplexing (for a proteome) or multiplexed sequencing (for a transcriptome). If feasible, the more strains that can be included in this comparison, the more powerful this analysis will be and can be prioritized over depth of sequencing/proteome coverage.

(2) The genetic tug-of-war system is elegant but comes at the cost of requiring specific media conditions (synthetic minimal media lacking uracil and leucine), which could be a potential confound, given that metabolic rewiring, and especially nitrogen starvation are among the observed phenotypes. I wonder if some of the changes might be specific to these conditions.

The authors should corroborate their findings under different conditions. Ideally, this would be done using an orthogonal expression system that does not rely on auxotrophy (e.g. using antibiotic resistance instead) and can be used in rich, complex mediums like YPD. Minimally, using different conditions (media with excess or more limited nitrogen source, amino acids, different carbon source, etc.) would be useful to test the robustness of the findings towards changes in media composition.

(3) The authors suggest that the TORC1 pathway is involved in regulating some of the changes they observed. This is likely true, but it would be great if the hypothesis could be directly tested using an established TORC1 assay.

(4) The finding that the nucleolus appears to be virtually missing in *mox-YG*-expressing cells (Figure 6B) is surprising and interesting. The authors suggest possible mechanisms to explain this and partially rescue the phenotype by a reduction-of-function mutation in an exosome subunit. I wonder if this is specific to the *mox-YG* protein or a general protein burden effect, which the experiments suggested in point 1 should address. Additionally, could a *mox-YG* variant with a nuclear export signal be expressed that stays exclusively in the cytosol to rule out that *mox-YG* itself interferes with phase separation in the nucleus?

Minor points:

(5) It would be great if the authors could directly compare the changes they observed at the transcriptome and proteome levels. This can help distinguish between changes that are transcriptionally regulated versus more downstream processes (like protein degradation, as proposed for ribosome components).

<https://doi.org/10.7554/eLife.99572.2.sa1>

Author Response:

The following is the authors' response to the original reviews.

General response

(1) Evaluation of mitochondrial activity in *mox-YG* overexpression cells

To determine whether the observed “mitochondrial development” seen in transcriptomic, proteomic, and microscopic analyses corresponds to an actual phenotypic shift toward respiration, we measured oxygen consumption in *mox-YG* overexpression cells. The results showed that oxygen consumption rates were indeed elevated in these cells, suggesting a metabolic shift from fermentation toward respiration. These findings have been incorporated into the revised manuscript as new Figure 4E and Figure 4—figure supplement 9, along with the corresponding descriptions in the Results section.

(2) Evaluation of TORC1 Pathway Inactivation in *mox-YG* Overexpression Cells

While the proteomic response in *mox-YG* overexpression cells overlapped with known responses to TORC1 pathway inactivation, we had not obtained direct evidence that TORC1 activity was indeed reduced. To address this, we assessed TORC1 activity by testing the effect of rapamycin, a TORC1 inhibitor, and by attempting to detect the phosphorylation state of known TORC1 targets. Our results showed that *mox-YG* overexpressing cells exhibited reduced sensitivity to rapamycin compared to vector control cells, supporting the idea that TORC1 is already inactivated in the *mox-YG* overexpression condition.

In parallel, we attempted to detect phosphorylation of TORC1 targets Sch9 and Atg13 by Western blotting. Specifically, we tested several approaches: detecting phospho-Sch9 using a phospho-specific antibody, assessing the band shift of HA-tagged Sch9, and monitoring Atg13

band shift using an anti-Atg13 antibody. While we were unable to detect Sch9 phosphorylation, likely due to technical limitations, we finally succeeded in detecting Atg13 with the help of our new co-author, Dr. Kamada. However, we observed a marked reduction in Atg13 protein levels in *mox-YG* overexpression cells, making it difficult to interpret the biological significance of any apparent decrease in phosphorylation. Therefore, we decided not to pursue further experiments on TORC1 phosphorylation within the current revision period.

These findings have been summarized in new Figure 4—figure supplement 7, and the relevant description has been added to the Results section.

(3) Phenotypes of *Gpm1-CCmut*

We focused our initial analysis on the phenotypes of cells overexpressing *mox-YG*, the protein with the lowest Neutrality Index (NI) in our dataset, as a model of protein burden. However, it remained unclear to what extent the phenotypes observed in *mox-YG* overexpression cells are generalizable to protein burden as a whole. We agree with the reviewers' suggestion that it is important to examine whether similar phenotypes are also observed in cells overexpressing *Gpm1-CCmut*, which was newly identified in this study as having a similarly low NI. We therefore performed validation experiments using *Gpm1-CCmut* overexpression cells to assess whether they exhibit the characteristic phenotypes observed in *mox-YG* overexpression cells. These phenotypes included: transcriptional responses, mitochondrial development, metabolic shift toward respiration, and nucleolar shrinkage.

As a result, mitochondrial development and nucleolar shrinkage were also observed in *Gpm1-CCmut* overexpression cells, consistent with *mox-YG*. In contrast, the transcriptional response associated with amino acid starvation and the metabolic shift toward respiration were not observed. Furthermore, an abnormal rounding of cell morphology—absent in *mox-YG* overexpression cells—was uniquely observed in *Gpm1-CCmut* cells. These results suggest that the phenotypes observed under *mox-YG* overexpression may comprise both general effects of protein burden and effects specific to the *mox-YG* protein. Alternatively, it is possible that *Gpm1-CCmut* imposes a different kind of constraint or toxicity not shared with *mox-YG*. In any case, these findings highlight that the full range of phenotypes associated with protein burden cannot yet be clearly defined and underscore the need for future analyses using a variety of “non-toxic” proteins.

Given that these results form a coherent set, we have relocated original Figure 3—which previously presented the NI values of *Gpm1* and *Tdh3* in the original version—to new Figure 6, which now includes all related phenotypic analyses. Correspondingly, we have added new Figures 6—figure supplement 1 through 6—figure supplement 7. The associated results have been incorporated into the Results section, and we have expanded the Discussion to address this point

As a result of these revisions, the order of figures has changed from the original version. The correspondence between the original and revised versions is as follows:

original → Revised

Figure 1 → Figure 1
 Figure 2 → Figure 2
 Figure 3 → Figure 6
 Figure 4 → Figure 3
 Figure 5 → Figure 4
 Figure 6 → Figure 5

Public Reviews:

Reviewer #1 (Public Review):

Weaknesses:

While the introduction of the neutrality index seems useful to differentiate between cytotoxicity and protein burden, the biological relevance of the effects of overexpression of the model proteins is unclear.

Thank you for your comment. This point is in fact the core message we wished to convey in this study. We believe that every protein possesses some degree of what can be described as “cytotoxicity,” and that this should be defined by the expression limit—specifically, the threshold level at which growth inhibition occurs. This index corresponds to what we term the neutrality index. We further argue that protein cytotoxicity arises from a variety of constraints inherent to each protein. These constraints act in a stepwise manner to determine the expression limit (i.e., the neutrality) of a given protein (Figure 1A). To demonstrate the real existence of such constraints, there are two complementary approaches: an inductive one that involves large-scale, systematic investigation of naturally occurring proteins, and a deductive one that tests hypotheses using selected model proteins. Our current study follows the latter approach. In addition, we define protein burden as a phenomenon that can only be elicited by proteins that are ultimately harmless (Figure 1B). We assume that such burden results in a shared physiological state, such as depletion of cellular resources. Through continued efforts to identify a protein suitable for investigating this phenomenon, we eventually arrived at *mox*-YG. As the reviewer rightly pointed out, examining only *mox*-YG does not reveal the full picture of protein burden. In fact, in response to the reviewer’s suggestion, we investigated the physiological consequences of overexpressing a mutant glycolytic protein, *Gpm1-CCmut* (General Response 3). We found that the resulting phenotype was notably different from that observed in cells overexpressing *mox*-YG. Going forward, we believe that our study provides a foundation for further systematic exploration of “harmless proteins” and the cellular impacts of their overexpression.

Reviewer #2 (Public Review):

Weaknesses:

*The authors concluded from their RNA-seq and proteomics results that cells with excess *mox*-YG expression showed increased respiration and TORC1 inactivation. I think it will be more convincing if the authors can show some characterization of mitochondrial respiration/membrane potential and the TOR responses to further verify their -omic results.*

These points are addressed in General Response 1 and 2.

*In addition, the authors only investigated how overexpression of *mox*-YG affects cells. It would be interesting to see whether overexpressing other non-toxic proteins causes similar effects, or if there are protein-specific effects. It would be good if the authors could at least discuss this point considering the workload of doing another RNA-seq or mass-spectrum analysis might be too heavy.*

These points are addressed in General Response 3.

Reviewer #3 (Public Review):

Weaknesses:

The data are generally convincing, however in order to back up the major claim of this work - that the observed changes are due to general protein burden and not to the specific protein or condition - a broader analysis of different conditions would be highly beneficial.

These points are addressed in General Response 3.

Major points:

(1) The authors identify several proteins with high neutrality scores but only analyze the effects of mox/mox-YG overexpression in depth. Hence, it remains unclear which molecular phenotypes they observe are general effects of protein burden or more specific effects of these specific proteins. To address this point, a proteome (and/or transcriptome) of at least a Gpm1-CCmut expressing strain should be obtained and compared to the mox-YG proteome. Ideally, this analysis should be done simultaneously on all strains to achieve a good comparability of samples, e.g. using TMT multiplexing (for a proteome) or multiplexed sequencing (for a transcriptome). If feasible, the more strains that can be included in this comparison, the more powerful this analysis will be and can be prioritized over depth of sequencing/proteome coverage.

This comment has been addressed in General Response 3. Gpm1-CCmut overexpression cells exhibited both phenotypes that were shared with, and distinct from, those observed in mox-YG overexpression cells. To define a unified set of phenotypes associated with "protein burden," we believe that extensive omics analyses targeting multiple "non-toxic" protein overexpression strains will be necessary. However, such an effort goes beyond the scope of the current study, and we would like to leave it as an important subject for future investigation.

(2) The genetic tug-of-war system is elegant but comes at the cost of requiring specific media conditions (synthetic minimal media lacking uracil and leucine), which could be a potential confound, given that metabolic rewiring, and especially nitrogen starvation are among the observed phenotypes. I wonder if some of the changes might be specific to these conditions. The authors should corroborate their findings under different conditions. Ideally, this would be done using an orthogonal expression system that does not rely on auxotrophy (e.g. using antibiotic resistance instead) and can be used in rich, complex mediums like YPD. Minimally, using different conditions (media with excess or more limited nitrogen source, amino acids, different carbon source, etc.) would be useful to test the robustness of the findings towards changes in media composition.

We appreciate the reviewer's clear understanding of both the advantages and limitations of the gTOW system. As rightly pointed out, since our system relies on leucine depletion, it is essential to carefully consider the potential impact this may have on cellular metabolism. Another limitation—though it also serves as one of the strengths—of the gTOW system is its reliance on copy number variation to achieve protein overexpression. This feature limits the possibility of observing rapid responses, as immediate induction is not feasible. To address this issue, we have recently developed a strong and inducible promoter that minimizes effects on other metabolic systems (Higuchi et al., 2024), and we believe this tool will be essential in future experiments.

In response to the reviewer's comments, we conducted two additional sets of experiments. First, we established a new overexpression system in nutrient-rich conditions (YPD medium) that is conceptually similar to gTOW but uses aureobasidin A and the *AUR1d* resistance gene to promote gene amplification (new Figure 4—figure supplement 2). Using this system, we observed that non-fluorescent YG mutants led to increased expression of mox. Total protein

levels appeared to rise correspondingly, suggesting that the overall synthetic capacity of cells might be higher in YPD compared to SC medium. However, the degree of overexpression achieved in this system was insufficient to strongly inhibit growth, meaning we could not replicate the stress conditions observed with the original gTOW system. Further studies will be needed to determine whether stronger induction under these nutrient-rich conditions will yield comparable responses.

Second, we performed a control experiment to examine whether the amino acid starvation response observed in *mox-YG* overexpressing cells could be attributed to leucine depletion from the medium (new Figure 3—figure supplement 3). By titrating leucine concentrations in SC medium, we confirmed that lower leucine levels reduced the growth rate of vector control cells, indicating leucine limitation. However, *GAP1* induction was not observed under these conditions. In contrast, *mox-YG* overexpression led to strong *GAP1* induction under similar growth-inhibitory conditions, suggesting that the amino acid starvation response is not simply due to environmental leucine depletion, but rather a consequence of the cellular burden imposed by *mox-YG* overexpression.

These findings have been incorporated into the manuscript, along with the corresponding figures (new Figure 4—figure supplement 2, Figure 3—figure supplement 3), and relevant descriptions have been added to the Results and Discussion sections.

(3) The authors suggest that the TORC1 pathway is involved in regulating some of the changes they observed. This is likely true, but it would be great if the hypothesis could be directly tested using an established TORC1 assay.

This comment has been addressed in General Response 2. We assessed the rapamycin sensitivity of *mox-YG* overexpression cells—which was found to be reduced—and attempted to detect phosphorylation of the TORC1 target Atg13, although the latter was only partially successful. These findings have been incorporated into the Results section.

*(4) The finding that the nucleolus appears to be virtually missing in *mox-YG*-expressing cells (Figure 6B) is surprising and interesting. The authors suggest possible mechanisms to explain this and partially rescue the phenotype by a reduction-of-function mutation in an exosome subunit. I wonder if this is specific to the *mox-YG* protein or a general protein burden effect, which the experiments suggested in point 1 should address. Additionally, could a *mox-YG* variant with a nuclear export signal be expressed that stays exclusively in the cytosol to rule out that *mox-YG* itself interferes with phase separation in the nucleus?*

As also described in our General Response 3, we observed nucleolar shrinkage upon Gpm1-CCmut overexpression as well (new Figure 6E and 6—figure supplement 7), suggesting that this phenomenon may represent a general feature of protein burden. The reviewer's suggestion to test whether this effect persists when *mox-YG* is excluded from the nucleus is indeed intriguing. However, based on our previous work, we have shown that overexpression of NES-tagged proteins (e.g., NES-EGFP) causes severe growth inhibition due to depletion of nuclear export factors (Kintaka et al., 2020). Unfortunately, this technical limitation makes it difficult for us to carry out the proposed experiment as suggested.

Minor points:

(5) It would be great if the authors could directly compare the changes they observed at the transcriptome and proteome levels. This can help distinguish between changes that are transcriptionally regulated versus more downstream processes (like protein degradation, as proposed for ribosome components).

We also considered this point to be important, and therefore compared the transcriptomic and proteomic changes associated with mox-YG overexpression. However, somewhat unexpectedly, we found little correlation between these two layers of response. As shown in new Figure 3 and 4 (original Figures 4 and 5), while genes related to oxidative phosphorylation were consistently upregulated at both the mRNA and protein levels in mox-YG overexpressing cells, ribosomal proteins showed a discordant pattern: their mRNA levels were significantly increased, whereas their protein levels were significantly decreased.

Several factors may explain this discrepancy: (1) differences in analytical methods between transcriptomics and proteomics; (2) temporal mismatches arising from the dynamic changes in mRNA and protein expression during batch culture; and (3) the possibility that, under protein burden conditions, specific regulatory mechanisms may govern the selective translation or targeted degradation of certain proteins. However, at this point, we were unable to clearly determine which of these factors account for the observed differences.

For this reason, we did not originally include a global transcriptome–proteome comparison in the manuscript. In response to the reviewer’s comment, however, we have now included the comparison data (new Figure 4—figure supplement 3D).

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

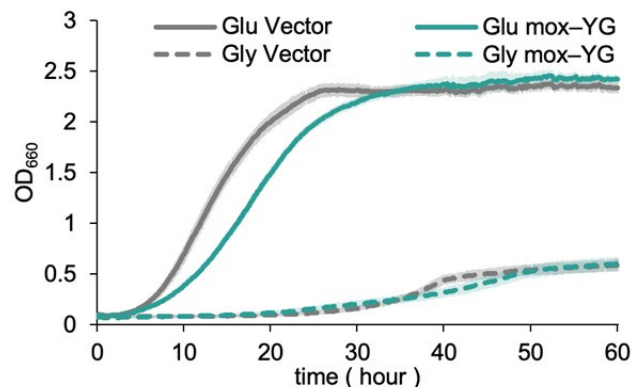
Major points:

(1) While the study provides a detailed description of physiological changes, the underlying mechanisms remain speculative. For example, the exact reasons for nitrogen source depletion or increased respiration are unclear. The transcriptomic and proteomic data should be complemented by basic growth assay tests on rapamycin or glycerol to strengthen these observations.

This comment has been addressed in General Responses 1 and 2. We conducted oxygen consumption assays and growth assays in the presence of rapamycin, and incorporated these results into the revised version of the manuscript.

We also performed culture experiments using glycerol as a carbon source. However, both the vector control and mox-YG overexpression cells showed extremely poor growth. Although there was a slight difference between the two, we judged that it would be difficult to draw any meaningful conclusions from these results. Therefore, we have chosen not to include them in the main text (the data are attached below for reference).

Author response image 1.



(2) The study mainly focuses on two proteins, *mox-YG*/FP proteins and *Gpm1-CCmut*. Did the authors look also at a broader range of proteins with varying degrees of cytotoxicity to validate the neutrality index and generalize their findings? Such as known cytotoxic proteins.

In our calculation of the Neutrality Index (NI), we use two parameters: the maximum growth rate (expressed as %MGR relative to the control) and the protein expression level. For the latter, we measure the abundance of the overexpressed protein as a percentage of total cellular protein, based on the assumption that the protein is expressed at a sufficiently high level to be detectable by SDS-PAGE. In our view, proteins typically regarded as “cytotoxic” cannot be overexpressed to levels detectable by SDS-PAGE without the use of more sensitive techniques such as Western blotting. This limitation in expression itself is an indication of their high cytotoxicity. Consequently, for such proteins, NI is determined solely by the MGR value, and will inherently fall below 100.

To test whether this interpretation is valid, we re-evaluated a group of EGFP variants previously reported by us to exhibit higher cytotoxicity than EGFP (Kintaka et al., 2016), due to overloading of specific cellular transport pathways. These include EGFPs tagged with localization signals. At the time of the original study, we had not calculated their NI values. Upon re-analysis, we found that all of these localization-tagged EGFP variants indeed have NI values below 100.

This result has been included as a new Figure 2—figure supplement 3, and the relevant descriptions have been added to the Results section.

(3) The partial rescue of ribosomal biosynthesis defects by a mutation in the nuclear exosome is intriguing but not fully explored. The specific role of the nuclear exosome in managing protein burden remains unclear. This result could be supported by alternative experiments. For example, would *tom1* deletion or proteasome inhibition (degradation of ribosomal proteins in the nucleus) partially rescue the nuclear formation?

As described in the main text, our interest in exosome mutants was prompted by our previous SGA (Synthetic Genetic Array) analysis, in which these mutants exhibited positive genetic interactions with GFP overexpression—namely, they acted in a rescuing manner (Kintaka et al., 2020). In contrast, proteasome mutants did not show such positive interactions in the same screening. On the contrary, proteasome mutants that displayed negative genetic interactions have been identified, such as the *pre7ts* mutant. Furthermore, the proteasome is

involved in various aspects of proteostasis beyond just orphan ribosomal proteins, making the interpretation of its effects potentially quite complex.

Regarding the *TOM1* mutant raised by the reviewer, we attempted to observe nucleolar morphology using the *NSR1-mScarlet-I* marker in the *tom1Δ* deletion strain. However, we were unsuccessful in constructing the strain. This failure may be due to the strong detrimental effects of this perturbation in the *tom1Δ* background. As we were unable to complete this experiment within the revision period, we would like to address this issue in future work.

Minor comments:

(1) *It would be interesting to include long-term cellular and evolutionary responses to protein overexpression to understand how cells adapt to chronic protein burden.*

Thank you for the suggestion. We are currently conducting experiments related to these points. However, as they fall outside the scope of the present study, we would like to refrain from including the data in this manuscript.

(2) *The microscopy of Nsr1 in Figure 6G does not clearly demonstrate the restored formation of the nucleolus in the mrt4-1 mutant. Electron microscopy images would be a better demonstration.*

The restoration of nucleolar size in the *mtr4-1* mutant, as shown in Figure 5—figure supplement 5 (original Figure 6_S5), is statistically significant. However, as described in the main text, the degree of rescue by the mutation is partial, and, as the reviewer notes, not clearly distinguishable by eye. It becomes apparent only when analyzing a large number of cells, allowing for detection as a statistically significant difference. Given that electron microscopy images are inherently limited in the number of cells that can be analyzed and pose challenges for statistical evaluation, we believe it would be difficult to detect such a subtle difference using this method. Therefore, we respectfully ask for your understanding that we will not include additional EM experiments in this revision.

(3) *On page 24, line 451 it says that of the 84 ribosomal proteins... latest reviews and structures described/ identified 79 ribosomal proteins in budding yeast of which the majority are incorporated into the pre-ribosomal particles in the nucleolus. We could not find this information in the provided reference. Please align with the literature.*

Thank you for the comment. In *S. cerevisiae*, many ribosomal protein genes are duplicated due to gene duplication events, resulting in a total of 136 ribosomal proteins (<http://ribosome.med.miyazaki-u.ac.jp/rpg.cgi?mode=genetable>). However, not all of them are duplicated, and among the duplicated pairs, some can be distinguished by proteomic analysis based on differences in amino acid sequences, while others cannot. As a result, we report that 84 ribosomal proteins were “detected” in our proteomic analysis. To avoid confusion, we have added the following explanation to the legend of Figure 5—figure supplement 1 (original Figure 6_S1), as follows.

“Note that when the amino acid sequences of paralogs are identical, they cannot be distinguished by proteomic analysis, and the protein abundance of both members of the paralog pair is represented under the name of only one.”

Reviewer #2 (Recommendations for the authors):

(1) *The authors mentioned that based on their proteomics results, overexpressing mox-YG appears to increase respiration. I think it is worth doing some quick verification, such*

as oxygen consumption experiments or mitochondrial membrane potential staining to provide some verification on that.

This comment has been addressed in General Response 1. We measured oxygen consumption in mox-YG overexpression cells and found that it was indeed elevated, suggesting a metabolic shift from fermentation toward aerobic respiration.

(2) Similar to point 1, the authors concluded from their proteomics data that the mox-YG overexpression induced responses that are similar to TORC1 inactivation. It might be worth testing whether there is any actual TORC1 inactivation, e.g. by detecting whether there is reduced Sch9 phosphorylation by western blot.

This comment has been addressed in General Response 2. We assessed the rapamycin sensitivity of mox-YG overexpression cells—which was found to be reduced—and attempted to detect phosphorylation of the TORC1 target Atg13, although the latter was only partially successful. These findings have been incorporated into the Results section.

(3) The authors showed that overexpressing excess mox-YG caused downregulated glycolysis pathways. It is worth discussing whether overexpressing glycolysis-related non-toxic proteins such as Gpm1-CCmut will also lead to similar results.

This comment has been addressed in General Response 3. Gpm1-CCmut overexpression cells exhibited both phenotypes shared with mox-YG overexpression and distinct ones. These findings suggest that a unified set of phenotypes associated with "protein burden" has yet to be clearly defined, and further investigation will be necessary to elucidate this.

Reviewer #3 (Recommendations for the authors):

(1) The authors identify several proteins with high neutrality scores but only analyze the effects of mox/mox-YG overexpression in depth. Hence, it remains unclear which molecular phenotypes they observe are general effects of protein burden or more specific effects of these specific proteins. To address this point, a proteome (and/or transcriptome) of at least a Gpm1-CCmut expressing strain should be obtained and compared to the mox-YG proteome. Ideally, this analysis should be done simultaneously on all strains to achieve a good comparability of samples, e.g. using TMT multiplexing (for a proteome) or multiplexed sequencing (for a transcriptome). If feasible, the more strains that can be included in this comparison, the more powerful this analysis will be and can be prioritized over depth of sequencing/proteome coverage.

This comment has been addressed in General Response 3. Gpm1-CCmut overexpression cells exhibited both phenotypes that were shared with, and distinct from, those observed in mox-YG overexpression cells. To define a unified set of phenotypes associated with "protein burden," we believe that extensive omics analyses targeting multiple "non-toxic" protein overexpression strains will be necessary. However, such an effort goes beyond the scope of the current study, and we would like to leave it as an important subject for future investigation.

(2) The genetic tug-of-war system is elegant but comes at the cost of requiring specific media conditions (synthetic minimal media lacking uracil and leucine), which could be a potential confound, given that metabolic rewiring, and especially nitrogen starvation are among the observed phenotypes. I wonder if some of the changes might be specific to these conditions. The authors should corroborate their findings under different conditions. Ideally, this would be done using an orthogonal expression system that does not rely on auxotrophy (e.g. using antibiotic resistance instead) and can be used in rich,

complex mediums like YPD. Minimally, using different conditions (media with excess or more limited nitrogen source, amino acids, different carbon source, etc.) would be useful to test the robustness of the findings towards changes in media composition.

We appreciate the reviewer's clear understanding of both the advantages and limitations of the gTOW system. As rightly pointed out, since our system relies on leucine depletion, it is essential to carefully consider the potential impact this may have on cellular metabolism. Another limitation—though it also serves as one of the strengths—of the gTOW system is its reliance on copy number variation to achieve protein overexpression. This feature limits the possibility of observing rapid responses, as immediate induction is not feasible. To address this issue, we have recently developed a strong and inducible promoter that minimizes effects on other metabolic systems (Higuchi et al., 2024), and we believe this tool will be essential in future experiments.

In response to the reviewer's comments, we conducted two additional sets of experiments. First, we established a new overexpression system in nutrient-rich conditions (YPD medium) that is conceptually similar to gTOW but uses aureobasidin A and the *AUR1d* resistance gene to promote gene amplification (new Figure 4—figure supplement 2). Using this system, we observed that non-fluorescent YG mutants led to increased expression of *mox*. Total protein levels appeared to rise correspondingly, suggesting that the overall synthetic capacity of cells might be higher in YPD compared to SC medium. However, the degree of overexpression achieved in this system was insufficient to strongly inhibit growth, meaning we could not replicate the stress conditions observed with the original gTOW system. Further studies will be needed to determine whether stronger induction under these nutrient-rich conditions will yield comparable responses.

Second, we performed a control experiment to examine whether the amino acid starvation response observed in *mox*-YG overexpressing cells could be attributed to leucine depletion from the medium (new Figure 3—figure supplement 3). By titrating leucine concentrations in SC medium, we confirmed that lower leucine levels reduced the growth rate of vector control cells, indicating leucine limitation. However, *GAP1* induction was not observed under these conditions. In contrast, *mox*-YG overexpression led to strong *GAP1* induction under similar growth-inhibitory conditions, suggesting that the amino acid starvation response is not simply due to environmental leucine depletion, but rather a consequence of the cellular burden imposed by *mox*-YG overexpression.

These findings have been incorporated into the manuscript, along with the corresponding figures (new Figure 4—figure supplement 2, Figure 3—figure supplement 3), and relevant descriptions have been added to the Results and Discussion sections.

(3) The authors suggest that the TORC1 pathway is involved in regulating some of the changes they observed. This is likely true, but it would be great if the hypothesis could be directly tested using an established TORC1 assay.

This comment has been addressed in General Response 2. We assessed the rapamycin sensitivity of *mox*-YG overexpression cells—which was found to be reduced—and attempted to detect phosphorylation of the TORC1 target Atg13, although the latter was only partially successful. These findings have been incorporated into the Results section.

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exclusively in the cytosol to rule out that mox-YG itself interferes with phase separation in the nucleus?

As also described in our General Response 3, we observed nucleolar shrinkage upon Gpm1-CCmut overexpression as well (new Figure 6E and 6—figure supplement 7), suggesting that this phenomenon may represent a general feature of protein burden. The reviewer's suggestion to test whether this effect persists when mox-YG is excluded from the nucleus is indeed intriguing. However, based on our previous work, we have shown that overexpression of NES-tagged proteins (e.g., NES-EGFP) causes severe growth inhibition due to depletion of nuclear export factors (Kintaka et al., 2020). Unfortunately, this technical limitation makes it difficult for us to carry out the proposed experiment as suggested.

(5) It would be great if the authors could directly compare the changes they observed at the transcriptome and proteome levels. This can help distinguish between changes that are transcriptionally regulated versus more downstream processes (like protein degradation, as proposed for ribosome components).

We also considered this point to be important, and therefore compared the transcriptomic and proteomic changes associated with mox-YG overexpression. However, somewhat unexpectedly, we found little correlation between these two layers of response. As shown in new Figure 3 and 4 (original Figures 4 and 5), while genes related to oxidative phosphorylation were consistently upregulated at both the mRNA and protein levels in mox-YG overexpressing cells, ribosomal proteins showed a discordant pattern: their mRNA levels were significantly increased, whereas their protein levels were significantly decreased.

Several factors may explain this discrepancy: (1) differences in analytical methods between transcriptomics and proteomics; (2) temporal mismatches arising from the dynamic changes in mRNA and protein expression during batch culture; and (3) the possibility that, under protein burden conditions, specific regulatory mechanisms may govern the selective translation or targeted degradation of certain proteins. However, at this point, we were unable to clearly determine which of these factors account for the observed differences.

For this reason, we did not originally include a global transcriptome–proteome comparison in the manuscript. In response to the reviewer's comment, however, we have now included the comparison data (new Figure 4—figure supplement 3D).

Minor points:

(1) The authors repeatedly state that 'mitochondrial function' is increased. This is inaccurate in two ways: first, mitochondria have multiple functions, and it should be specified which one is referred to (probably mitochondrial respiration); second, the claim is based solely on the abundance of transcripts/proteins, which may or may not reflect increased activity.

The authors should either perform functional tests (e.g. measure oxygen consumption or extracellular acidification), or change their wording to more accurately reflect the findings.

To more directly reflect our findings, we revised two instances of the phrase “mitochondrial function” to “mitochondrial proteins” in the manuscript. Furthermore, as described in General Response 1, we confirmed that oxygen consumption is elevated in mox-YG overexpression cells. This observation suggests that mitochondrial respiratory activity is indeed enhanced under these conditions.

(2) Similarly, the authors state that FPs are 'not localized' (e.g. line 137). This should be specified (e.g. 'not actively sorted into cellular compartments other than the cytosol').

As pointed out by the reviewer, we have revised the relevant sections accordingly.

(3) In Figure 4D, some of the reporter assays don't fully recapitulate the RNAseq findings (e.g. for PHO84 and ZPS1, where mox-FS and mox-YG behave differently in the reporter assay, but not in the RNAseq data). This may stem from technical limitations given that the reporter assay relies on RFP expression which could generally be affected by protein overexpression (cf. ACT1pro in mox-FS), but it should be mentioned in the text.

We apologize for the confusion caused by our insufficient explanation of "moxFS" in new Figure 3D (original Figure 4D). As clarified here, "moxFS" refers to a frameshift mutant in which the mRNA is transcribed but the protein is not translated due to an early frameshift mutation. This is not a functional mox protein. The behavior of this mutant is nearly identical to that of the vector control, indicating that the transcriptional response observed in this assay is not triggered by mRNA expression itself, but rather by events occurring after protein synthesis begins. Importantly, the transcriptional responses identified by RNA-seq in mox-YG overexpression cells are largely recapitulated by this reporter assay, supporting the reliability of our experimental design.

We appreciate the reviewer's comment, which helped us recognize the lack of clarity in our original description. In response, we have added an explanation of the FS mutation to the figure legend (new Figure 3D), and we have also expanded the description of the moxFS experimental results in the Results section.

<https://doi.org/10.7554/eLife.99572.2.sa0>